ALL SOFTWARE TEST RESULTS - COMPRESSION RATIO AND SPEED FOR EVERY TOOL TESTED
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HX is Patent Pending (U.S. App. 64/192,843)
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Built only from results already recorded (no new benchmarks were run for this document).

HOW TO READ
  Ratio = original bytes / compressed bytes (higher is better), written x.xx×. Speeds are given both as seconds (as recorded) and as MB/s, where MB = 1,000,000 bytes of ORIGINAL data per second (original size / recorded time). "n/r" = not recorded. A "~" before a byte count means the source recorded only a size relative to the original; the ratio is exact, the byte count is derived from it and the known original size (rounding error under 0.0001%). Every listed result decoded back exactly unless a note says otherwise.
  Group column g: blank = original third-party program, * = our program, m = modified tool (theirs plus our changes; the stock decoder reads it), r = mutant or research build (needs our decoder, uses a research-only part, or is a test control).
  Machines: "cloud" = 2-core cloud machine (busy, noisy times); "desktop" = Michael's Ryzen 5 7500X3D, 6 cores / 12 threads. Compare speeds only inside one table (same machine, same session).
  Section order: 1 tool list by group; 2 results by corpus; 3 HX; 4 modified tools; 5 mutants and research builds; 6 task-level tests; 7 our programs outside the corpus tables; 8 cmix / paq8px; 9 final rankings; 10 gaps.

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1. EVERY TOOL TESTED, BY GROUP
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  (1) ORIGINAL THIRD-PARTY PROGRAMS - 140 entries
      7-Zip -mx9; 7-Zip -mx9 (LZMA2); 7-Zip PPMd (-mx9 -m0=PPMd); 7-Zip Ultra -mx9; adaptive arithmetic (ac0); ALAC (ffmpeg); Apache ORC C++; Apache Tika 3.3.3; bgzip (htslib 1.21); binwalk 3.1.0; brotli 11; Brotli 11; Brotli 5; Brotli 8; BSC; BSC -b64e2; bsc 0; bzip2 -9; bzip3; bzip3 0; ClamAV 1.5.1; cmix v20 (published); cmix v21 (lab only); difference+shuffle, xz -9; difference+shuffle, zstd -19; DuckDB Parquet writer; FFV1 (level 3); FLAC (xiph) -8; flac -8; flac -8 -e -p (max); FSE (tANS) 1 MB; FSE (tANS) 128 KB; gcc/make (builds); Ghostscript 10; GNU grep 3.11 / ripgrep 14.1; gzip -1; gzip -6 (zlib 1.3.1 minigzip); gzip -9; huff0 (Huffman) 128 KB; JPEG XL -d 0 -e 7; JPEG XL -d 0 -e 7 + xz header; JPEG XL -d 0 -e 9; JPEG XL -d 0 -e 9 + xz header; JPEG XL lossless (e7); JPEG XL lossless (e9); JPEG-LS (CharLS); JPEG-LS (CharLS) + xz header; JPEG-LS (ffmpeg); jq 1.7; kanzi -l 9; kanzi 9; Lepton; lpaq1; lpaq1 -6; lpaq1 -9; lpaq1 8; lpaq1 9; LZ4 (default); lz4 -1; lz4 -12; LZ4 1.10.0 default; LZ4 default; LZFSE; LZMA -9 (7-Zip SDK, king); lzma 9; MagicYUV -pred gradient (FFmpeg); MagicYUV -pred median (FFmpeg); meshoptimizer; Monkey's Audio -c4000; Monkey's Audio -c5000; mzip; mzip (Cranot); OptimFROG --preset 2; OptimFROG --preset max; packJPG; packMP3; paq8px -1 single stream; paq8px v217 (lab / wrapper only); paq8px v217 -8 (full); pdftoppm (render check); pigz; PNG (level 9); PNG (Pillow / optipng-style); precomp (lzma); PrusaSlicer; pugixml; pzstd -3 -p 12; qpdf 11.9 (PDF baseline); rANS 1 MB; rapidgzip 0.16.1; RAR 7.30 beta 2; RAR 7.30b2 -m5 -ma5; RecastNavigation; river (HalfSpaceTrees); Snappy; Snort 3.12.4; Steim2 in miniSEED (archive standard); Tesseract 5; TTA (ffmpeg); Ut Video; WavPack (ffmpeg, level 8); WavPack -hh (official); WavPack -hhx6 (official, max); WebP lossless; WebP lossless (z9); whisper.cpp; Wireshark tshark 4.6.9; x264 lossless (veryslow); x265 lossless (slow); xdelta3 3.0.11; xz -0; xz -9; xz -9 on raw samples; xz -9e; xz 9; YARA; Zeek 8.0.4; zlib 9; ZPAQ -m4; ZPAQ -m5; zpaq 5; ZPAQ 7.15 -m4; ZPAQ 7.15 -m5; zstd -1; zstd -19; zstd -19 --long; zstd -19 -T2; zstd -19 on raw samples; zstd -22; zstd -3; zstd -3 (1 thread); zstd -3 --long=27 -T12; zstd -3 -T12; zstd -3, one thread; zstd -9; zstd 1.5.6 default; zstd 19; zstd 22; zstd 3; zstd default (-3)

  (2) OUR PROGRAMS - 94 entries
      cascade v1.1 (zstd) and HCM cascade v3; cascade v1.1 -19; cascade v1.1 -19, 16 MB blocks; cascade v1.1 -3; difference+shuffle, HCM v3.6.1; H-Deflate + HX Complete + image routes; H-Deflate v1; Harrington's Sets (HCM sets mode); HCM fast; HCM max; HCM Sets fast; HCM Sets max; HCM Sets max + cascade v3; HCM v2; HCM v2, v3, v3.1, v3.2, v3.3, v3.4, v3.5, v3.6, v3.6.1 (fast and max); HCM v3 fast; HCM v3 max; HCM v3.1 fast; HCM v3.1 max; HCM v3.2 fast; HCM v3.2 max; HCM v3.3 fast; HCM v3.3 fast + x86 E8/E9 filter; HCM v3.3 max; HCM v3.3 max + x86 E8/E9 filter; HCM v3.4 fast; HCM v3.4 max; HCM v3.6 fast single core; HCM v3.6 fast, Harrington's Sets 16 MB; HCM v3.6 fast, Harrington's Sets 4 MB; HCM v3.6 fast, parallel 16 MB blocks; HCM v3.6 fast, single core; HCM v3.6.1 fast; HCM v3.6.1 fast (one core); HCM v3.6.1 Harrington's Sets 16 MB, 2 cores; HCM v3.6.1 max; HCM v3.6.1 on raw samples; HCM-Bin v1 max (hcmx); hcmx / hcmx-s; hcmx fast; hcmx fast (trit2); hcmx fast best-of; hcmx fast best-of incl. HCM-Bin; hcmx max; hcmx max best-of; hcmx-s fast (x86 inside format); hcmx-s trit2 lite; hs_build; hs_clam / hs_binwalk; hs_grep 1.0 / 1.1; hs_gs; hs_lines (jq / river); hs_ocr; hs_saveload and the game-test wrappers; hs_scan; hs_shark 1.1 / 1.2; hs_whisper; hs_wrap 1.3 / 1.4 / 1.5; hs_wrap 1.4 + gzip -6; hs_wrap+paq8px -1 1 MB 1 job; hs_wrap+paq8px -1 1 MB 12 jobs; hs_wrap+paq8px -1 1 MB 3 jobs; hs_wrap+paq8px -1 1 MB 6 jobs; hs_wrap+paq8px -1 16 MB Sets 12 jobs; hs_wrap+paq8px -1 2 MB 12 jobs; hs_wrap+paq8px -6 1 MB 6 jobs; hs_zeek (pcap split + merge);

hs_zstd -19 Harrington's Sets 16 MB; hs_zstd -19 Harrington's Sets 16 MB, 2 jobs; hs_zstd -19 Harrington's Sets 4 MB; hs_zstd -19 plain blocks 16 MB; hs_zstd -19 plain blocks 16 MB, 2 jobs; hs_zstd -19 plain blocks 4 MB; hs_zstd 1.0 / 1.1 / 1.2 / 1.3; hs_zstd 1.1, 16 MB blocks; hs_zstd 1.1, 4 MB blocks; hs_zstd 1.2 rule, 16 MB; hs_zstd 1.3 (Sets, raw off, safe), 16 MB; hs_zstd 1.3 (Sets, raw off, safe), 4 MB; hs_zstd 1.3 Sets -19; hs_zstd 1.3 Sets -19, 16 MB blocks; hs_zstd 1.3 Sets -3; hs_zstd plain-blocks control 16 MB; hs_zstd plain-blocks control 4 MB; HsTika; HX Complete; HX Complete + audio/colour lines; HX Complete + specialists; HX Complete, every route (router build); HX Fast; HX Fast + audio/colour lines; HX lines; HX lines + Sets; HX lines, HX Fast, HX Complete

(3) MODIFIED TOOLS (stock decoder reads the output) - 29 entries
7z-hi; 7z-hi (Delta 2); best of lzma-hi / xz-hi per file; Brotli-hi; Brotli-hi 11; Brotli-hi 5; Brotli-hi 8; bzip2-hi; bzip2-hi (100k blocks); flac-hi (tuned FLAC); JPEG-LS tuned (CharLS); jxl-hi (cjxl settings); lzma-hi; PDF stream rewrite (zlib + PNG predictor); RAR no-pain search; RAR no-pain search (rar-hi); xz-hi; xz-hi (delta 13); xz-hi (king race); zip-hi (results kept, baseline unclear); zpaq-hi; zpaq-hi (one file per archive); zpaq-hi -m5; zstd-hi; zstd-hi -1; zstd-hi -19; zstd-hi -22; zstd-hi -3; zstd-hi -9

(4) MUTANTS AND RESEARCH BUILDS - 53 entries
bzip2 rebuild + candidate escape flag; bzip2-style rebuild + arithmetic coding; change map (activity map) + gzip / lz4 / Snappy / LZFSE; cmix low-memory build; gzip -1 + record delta; gzip -9 + change map; gzip -9 + record delta; gzip -9 + record delta 13; Harrington Exchange front ends; HCM v3.3 fast + cmix English dictionary; HCM v3.7 (research); HCM v3.7 max; HCM v3.7 max + Sets reorder (all files); HCM v3.7 max, file order; HCM v3.7 max, reorder kept only if smaller; HCM-Bin v1; HCM-Bin v1 (overall); hcm-trit v1; hcm-trit v1 / v1.1 / v2; hcm-trit v1.1; hcm-trit v2 (pointer) 6-order; hcm-trit v2 max; hcm-trit v2 max + cmix dictionary; HX + x264; HX Complete + mzip; HX Complete + specialists + x264; kanzi -l 9 + Sets reorder; lpaq1-h; lpaq1-h (lpaq1 mutant); lpaq1-h 9; lz4 -12 + record delta 13; LZFSE + record delta 13; paq8px lite model set; paq8px lite model set (lab); Parquet / ORC with record columns; precomp -cn + HX Complete; precomp deflate-only + HX Complete (ctl); precomp routes + HX Complete; record delta + gzip / zstd / lz4 / xz / Snappy / LZFSE / RAR; record delta + xz -9; record delta + zstd -19; Sets + mzip; Sets + mzip 12 jobs 4 MB; Sets + mzip 16 MB blocks; Sets + mzip 4 MB blocks; Sets reorder (setsort); Snappy + record delta 13; xz -0 + record delta; xz -9e + record delta; zstd -1 + record delta; zstd -19 + record delta; zstd -19 + record delta 2 (mutant); zstd -22 + record delta 13

(5) TASK-LEVEL TESTS (no compressor name): see section 6 - 26 tests.
Entries listed above: 316 (one entry per tool and setting as named in the result files).

2. RESULTS BY CORPUS (each table = one machine, one session; best ratio first)

CORPUS: Silesia

Silesia, 12 files - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 211,938,580 bytes in 12 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		40,785,680	5.20×	107.3	1.97	96.90	2.19
mzip		43,095,821	4.92×	2,209	0.096	24.76	8.56
Sets + mzip 16 MB blocks	r	43,813,765	4.84×	1,648	0.129	16.80	12.62
Sets + mzip 4 MB blocks	r	44,854,347	4.73×	1,528	0.139	19.15	11.07
HX Complete	*	44,861,528	4.72×	279.2	0.759	8.62	24.58
bzip3		46,426,674	4.57×	25.91	8.18	28.03	7.56
BSC		46,536,796	4.55×	18.08	11.72	13.66	15.52
7-Zip PPMd (-mx9 -m0=PPMd)		47,684,720	4.44×	47.39	4.47	49.72	4.26
7-Zip -mx9 (LZMA2)		48,758,687	4.35×	72.79	2.91	3.41	62.19
Brotli 11		49,564,659	4.28×	548.5	0.386	1.20	176.2
HX Fast	*	49,775,366	4.26×	235.5	0.900	0.686	308.9
gzip -9		67,631,918	3.13×	21.43	9.89	1.56	135.4
LZ4 (default)		100,344,651	2.11×	0.680	311.7	0.439	482.8
ZPAQ -m5		28,104,227	6.11×	1,014	0.170	906.3	0.190
partial: 7 of 12 files (171,850,994 B) - not ranked							
HCM max	*	30,303,109	5.67×	968.2	0.177	277.1	0.620
partial: 7 of 12 files (171,850,994 B) - not ranked							
lpaq1 -9		30,471,222	5.64×	341.3	0.503	427.5	0.402
partial: 7 of 12 files (171,850,994 B) - not ranked							
xz -9e		34,691,188	4.95×	291.8	0.589	5.62	30.59
partial: 7 of 12 files (171,850,994 B) - not ranked							

HX Complete + mzip	r	32,559,983	4.81×	1,708	0.092	6.72	23.32
partial: 9 of 12 files (156,660,357 B) - not ranked							
zstd -19 --long		37,493,944	4.58×	234.0	0.734	0.898	191.4
partial: 7 of 12 files (171,850,994 B) - not ranked							
zstd -19		37,740,652	4.55×	196.6	0.874	0.972	176.8
partial: 7 of 12 files (171,850,994 B) - not ranked							
bzip2 -9		40,213,473	4.27×	29.32	5.86	18.26	9.41
partial: 7 of 12 files (171,850,994 B) - not ranked							

Silesia, 12 files - HX run of 3 Nov 2026 (HX Fast, HX Complete, BSC, bzip3, mzip)
machine: cloud, 2 threads, wall clock, decode best of 3
original: 211,938,580 bytes in 12 file(s)
source: HU-pretrain-results\hx-complete\HX-Complete-and-HX-Fast-9-Silesia-2026-11-03.txt + hxc_*.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
mzip		n/r	4.92×	2,209	0.096	24.76	8.56
size recorded as a ratio only in this file; bytes from the All-Units table: 43,095,821							
HX Complete	*	44,861,528	4.72×	279.2	0.759	8.62	24.58
bzip3		46,426,674	4.57×	25.87	8.19	28.09	7.54
BSC		46,536,796	4.55×	18.14	11.68	13.68	15.49
HX Fast	*	49,775,366	4.26×	235.5	0.900	0.686	308.9

Silesia, 12 files - HCM v3.6 fast on the 12-thread desktop (30 Oct 2026)
machine: desktop Ryzen 5 7500X3D 6c/12t, Windows, one run each
original: 211,938,580 bytes in 12 file(s)
source: hcm-v36-windows\HCM-v3.6-Windows-Core-Scaling-Results.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6 fast, single core	*	42,661,041	4.97×	268.9	0.788	83.30	2.54
HCM v3.6 fast, parallel 16 MB blo	*	42,883,537	4.94×	176.9	1.20	57.40	3.69
HCM v3.6 fast, Harrington's Sets	*	42,962,663	4.93×	137.7	1.54	46.50	4.56
HCM v3.6 fast, Harrington's Sets	*	43,646,375	4.86×	76.10	2.79	27.10	7.82

Silesia, 12 files - hs_zstd 1.1 corpus run on the desktop (31 Oct 2026)
machine: desktop 12 threads, summed median decode of 3
original: 211,938,580 bytes in 12 file(s)
source: HCM-tests\hszcorp-2026-10-31\results.txt
Decode = summed per-file medians; compress = summed per-file times (both recorded per file).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zstd default (-3)		66,547,664	3.18×	1.30	163.0	1.14	185.4
hs_zstd 1.1, 16 MB blocks	*	66,606,480	3.18×	0.970	218.5	0.575	368.6
hs_zstd 1.1, 4 MB blocks	*	66,746,533	3.18×	0.760	278.9	0.481	440.6
LZ4 default		100,934,427	2.10×	0.870	243.6	1.17	181.9

Silesia, 12 files - HCM v3.7 max (research build) vs kanzi -l 9, file order and Sets reorder (4 Nov)
machine: cloud, 2 cores, wall clock
original: 211,938,580 bytes in 12 file(s)
source: setsort-reorder-2026-11-04\hcm.csv, kanzi.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		40,785,680	5.20×	93.96	2.26	92.78	2.28
kanzi -l 9 + Sets reorder	r	41,275,591	5.13×	90.05	2.35	90.82	2.33
HCM v3.7 max, reorder kept only i	r	41,744,610	5.08×	n/r	n/r	n/r	n/r
per-file pick of the two rows above; times not summed for the pick							
HCM v3.7 max, file order	r	41,751,004	5.08×	779.8	0.272	198.3	1.07
HCM v3.7 max + Sets reorder (all	r	42,141,775	5.03×	802.4	0.264	206.3	1.03

Silesia, 12 files - Tier-4 race (27 Oct 2026)
machine: cloud, 2 cores, decode seconds
original: 211,938,580 bytes in 12 file(s)
source: memory /areas/hcm-benchmarks.md (race of 27 Oct); PC Documents\data compression\tools
Compression times not recorded for this race.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
ZPAQ -m5		~39,112,629	5.42×	n/r	n/r	554.0	0.383
HCM v3.3 max + x86 E8/E9 filter	*	~41,934,591	5.05×	n/r	n/r	n/r	n/r
decode total not recorded							
hcmx-s fast (x86 inside format)	*	~42,053,701	5.04×	n/r	n/r	n/r	n/r
decode total not recorded							
HCM v3.3 max	*	~42,215,198	5.02×	n/r	n/r	217.0	0.977
hcmx fast best-of incl. HCM-Bin	*	~42,329,857	5.01×	n/r	n/r	276.0	0.768
hcmx fast (trit2)	*	~42,532,258	4.98×	n/r	n/r	241.0	0.879
hcmx-s trit2 lite	*	~42,613,219	4.97×	n/r	n/r	n/r	n/r
decode total not recorded							
ZPAQ -m4		~42,627,842	4.97×	n/r	n/r	161.0	1.32
HCM v3.3 fast + x86 E8/E9 filter	*	~42,648,612	4.97×	n/r	n/r	n/r	n/r

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      decode total not recorded
  HCM v3.3 fast                     *   ~42,926,040   4.94×      n/r       n/r    186.0     1.14
  BSC -b64e2                            ~46,536,838   4.55×      n/r       n/r    12.50    16.96
  bzip3                                 ~46,929,984   4.52×      n/r       n/r    23.00     9.21
  BSC                                   ~47,428,251   4.47×      n/r       n/r     8.30    25.53

Silesia, 12 files - LZMA / 7-Zip / RAR king races and record delta (26-27 Oct 2026)
  machine: cloud, in-process decode (LZMA, xz) or CLI (7-Zip, RAR); totals in ms
  original: 211,938,580 bytes in 12 file(s)
  source: memory /areas/hcm-benchmarks.md (race of 26-27 Oct); PC Documents\data compression\tools
  Rows come from separate races; compare speeds only inside each pair (stock vs our version).
  tool                            g  compressed B   ratio   comp s comp MB/s    dec s dec MB/s
  7z-hi                           m   ~47,333,091   4.48×      n/r       n/r     3.31    63.95
  best of lzma-hi / xz-hi per file  m   ~47,488,230   4.46×      n/r       n/r     2.92    72.48
  xz -9e + record delta           r   ~48,075,300   4.41×      n/r       n/r      n/r      n/r
      decode total not recorded
  lzma-hi                         m   ~48,136,762   4.40×      n/r       n/r     2.89    73.36
  xz -9e                              ~48,456,153   4.37×      n/r       n/r      n/r      n/r
      decode total not recorded
  LZMA -9 (7-Zip SDK, king)           ~48,752,867   4.35×      n/r       n/r     2.94    72.11
  7-Zip Ultra -mx9                    ~48,757,530   4.35×      n/r       n/r     3.48    60.85
  RAR no-pain search (rar-hi)     m   ~52,171,437   4.06×      n/r       n/r     1.02    208.0
  zstd -19 + record delta         r   ~52,568,186   4.03×      n/r       n/r      n/r      n/r
      decode total not recorded
  zstd -19                            ~52,892,028   4.01×      n/r       n/r      n/r      n/r
      decode total not recorded
  RAR 7.30b2 -m5 -ma5                 ~53,129,823   3.99×      n/r       n/r     1.06    200.1
  xz -0 + record delta            r   ~61,887,973   3.42×      n/r       n/r      n/r      n/r
      decode total not recorded
  xz -0                               ~62,590,549   3.39×      n/r       n/r      n/r      n/r
      decode total not recorded
  gzip -9 + record delta          r   ~66,463,939   3.19×      n/r       n/r      n/r      n/r
      decode total not recorded
  gzip -9                             ~67,765,665   3.13×      n/r       n/r      n/r      n/r
      decode total not recorded
  zstd -1 + record delta          r   ~71,482,432   2.96×      n/r       n/r      n/r      n/r
      decode total not recorded
  zstd -1                             ~73,229,442   2.89×      n/r       n/r      n/r      n/r
      decode total not recorded
  gzip -1 + record delta          r   ~76,910,603   2.76×      n/r       n/r      n/r      n/r
      decode total not recorded
  gzip -1                             ~77,622,505   2.73×      n/r       n/r      n/r      n/r
      decode total not recorded

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CORPUS: Silesia (silesia.tar)
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silesia.tar (211,957,760 B) - TurboBench 2026.11.01 and our tools (1 Nov 2026)
  machine: cloud, 2 cores; TurboBench reports MB/s, ours wall-clock seconds
  original: 211,957,760 bytes in 1 file(s)
  source: turbobench-silesia-2026-11-01\results-turbobench.txt, results-ours.txt
  tool                            g  compressed B   ratio   comp s comp MB/s    dec s dec MB/s
  zpaq 5                              40,485,026   5.24×      n/r     0.340      n/r    0.320
  kanzi 9                             41,855,581   5.06×      n/r      1.60      n/r     1.65
  HCM v3.6.1 max                  *   43,004,910   4.93×    1,029     0.206    226.5    0.936
  HCM v3.6.1 Harrington's Sets 16 M *   43,189,460   4.91×    453.1     0.468    107.8     1.97
  HCM v3.6.1 fast                 *   43,795,425   4.84×    1,025     0.207    196.4     1.08
  lzma 9                              48,677,116   4.35×      n/r      1.20      n/r    65.82
  xz 9                                48,774,140   4.35×      n/r     0.940      n/r    71.92
  bsc 0                               49,294,928   4.30×      n/r     15.10      n/r    16.11
  brotli 11                           49,528,131   4.28×      n/r     0.350      n/r    190.6
  zstd 22                             52,305,790   4.05×      n/r      1.08      n/r    730.8
  zstd 19                             52,874,872   4.01×      n/r      1.68      n/r    562.5
  hs_zstd -19 Harrington's Sets 16  *   53,498,614   3.96×    55.20      3.84    0.340    623.4
  hs_zstd -19 plain blocks 16 MB, 2 *   53,547,274   3.96×    51.20      4.14    0.330    642.3
  bzip3 0                             59,387,051   3.57×      n/r      4.04      n/r     4.60
  zstd 3                              66,497,774   3.19×      n/r     186.0      n/r    874.7
  zlib 9                              67,643,913   3.13×      n/r      8.37      n/r    196.1

-------------------------------------------------------------------------------------------------
CORPUS: Silesia 1 MB sample
-------------------------------------------------------------------------------------------------
```

Silesia 1 MB sample - 14 tools, one fresh same-machine run (29 Oct 2026)
  machine: cloud, wall clock per file, summed
  original: 12,582,912 bytes in 12 file(s)
  source: benchmark-charts-2026-10-29\results.json (GitHub benchmarks\charts-2026-10-29)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
ZPAQ -m5		3,418,919	3.68×	52.47	0.240	54.72	0.230
HCM v3.4 max	*	3,544,604	3.55×	61.72	0.204	16.22	0.776
hcmx fast	*	3,551,524	3.54×	106.0	0.119	17.77	0.708
lpaq1-h	r	3,559,695	3.53×	98.28	0.128	16.26	0.774
HCM v3.4 fast	*	3,568,979	3.53×	48.44	0.260	13.21	0.953
lpaq1 -6		3,643,900	3.45×	14.87	0.846	15.75	0.799
xz-hi	m	4,011,280	3.14×	413.8	0.030	0.320	39.37
bzip2-hi	m	4,083,159	3.08×	370.1	0.034	0.668	18.85
bzip2 -9		4,102,930	3.07×	1.10	11.47	0.697	18.07
xz -9e		4,147,436	3.03×	6.17	2.04	0.329	38.30
zstd -19		4,443,358	2.83×	3.96	3.18	0.069	182.9
gzip -9		5,021,695	2.51×	1.17	10.72	0.129	97.92
zstd -3		5,043,561	2.49×	0.192	65.57	0.065	194.2
lz4 -1		7,103,853	1.77×	0.081	155.7	0.072	174.3

CORPUS: Silesia 3 MB heads

First 3 MB of 10 Silesia files (31,457,280 B) - paq8px through hs_wrap (30 Oct 2026)
  machine: desktop 12 threads, wall clock
  original: 31,457,280 bytes in 10 file(s)
  source: HCM-tests\paq8px-wrapper-2026-10-30\results.txt
  The only recorded paq8px runs (level -1 and -6, 3 MB heads). This is a wrapper speed test, not a full paq8px benchmark; full-strength paq8px / cmix comparisons are in section 8.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
hs_wrap+paq8px -6 1 MB 6 jobs	*	6,895,648	4.56×	2,056	0.015	2,076	0.015
paq8px -1 single stream		6,997,774	4.50×	3,889	0.008	3,867	0.008
hs_wrap+paq8px -1 16 MB Sets 12 j	*	7,011,798	4.49×	3,715	0.008	3,676	0.009
hs_wrap+paq8px -1 2 MB 12 jobs	*	7,094,294	4.43×	2,821	0.011	2,845	0.011
hs_wrap+paq8px -1 1 MB 12 jobs	*	7,163,002	4.39×	1,829	0.017	1,824	0.017
hs_wrap+paq8px -1 1 MB 6 jobs	*	7,163,002	4.39×	1,897	0.017	1,916	0.016
hs_wrap+paq8px -1 1 MB 3 jobs	*	7,163,002	4.39×	1,881	0.017	1,873	0.017
hs_wrap+paq8px -1 1 MB 1 job	*	7,163,002	4.39×	3,920	0.008	3,905	0.008

CORPUS: Silesia x-ray

Silesia x-ray (8,474,240 B, 12-bit radiograph) - every recorded tool
  machine: mixed: see notes
  original: 8,474,240 bytes in 1 file(s)
  source: hxc_silesia_results.csv; rivals\rival3_results.csv;
  HX-Fast-vs-mzip-Speed-2026-11-03.txt; setsort hcm.csv/kanzi.csv; hcm-v36-windows; memory
  /areas/hcm-benchmarks.md
  Machines differ by row (marked); rank speeds only among rows of the same machine.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Complete, every route (router	*	3,547,196	2.39×	186.6	0.045	n/r	n/r
16-bit image route (JPEG XL lossless inside HX Complete); compress = sum of all route times; decode n/r							
JPEG-LS tuned (CharLS) (cloud)	m	3,574,443	2.37×	0.089	95.22	0.100	84.74
JPEG-LS (CharLS) (cloud)		3,591,153	2.36×	0.115	73.69	0.120	70.62
JPEG XL -d 0 -e 7 (cloud)		3,631,359	2.33×	2.03	4.17	0.296	28.63
HX Fast (cloud 2 thr)	*	3,651,990	2.32×	5.28	1.60	0.061	138.9
HX Fast (desktop 12 thr, in memor	*	3,651,990	2.32×	2.84	2.98	0.013	642.0
HX Complete (cloud)	*	3,651,990	2.32×	8.56	0.990	0.062	136.7
HCM v3.7 max (cloud)	r	3,654,730	2.32×	48.29	0.175	12.34	0.687
mzip (desktop)		~3,657,092	2.32×	33.68	0.252	0.671	12.63
bzip3 (cloud)		3,657,095	2.32×	1.46	5.82	1.38	6.15
HCM v3.6 fast single core (deskto	*	3,666,372	2.31×	10.60	0.799	3.70	2.29
JPEG XL -d 0 -e 9 (cloud)		3,669,155	2.31×	n/r	n/r	0.345	24.56
ZPAQ -m5 (cloud) from Highlights-2026-11-04.txt		3,669,704	2.31×	n/r	n/r	38.00	0.223
Sets + mzip 12 jobs 4 MB (desktop r		~3,695,760	2.29×	11.78	0.719	0.283	29.97
lpaq1 (cloud) from Highlights-2026-11-04.txt		3,712,145	2.28×	n/r	n/r	19.00	0.446
kanzi -l 9 (cloud)		3,728,578	2.27×	6.72	1.26	6.70	1.26
BSC (cloud)		3,756,574	2.26×	1.12	7.55	1.05	8.07
7z-hi (Delta 2)	m	~4,197,350	2.02×	n/r	n/r	n/r	n/r

7-Zip -mx9		~4,479,873	1.89×	n/r	n/r	n/r	n/r
xz -9e (desktop)		~4,491,279	1.89×	2.01	4.22	0.193	43.86
zstd -19 + record delta 2 (mutant r		4,508,494	1.88×	n/r	n/r	0.020	419.5
8 paired runs, decode median; encode ~4.7 -> ~6.1 s							
zstd -19 (cloud, in-process)		5,129,823	1.65×	n/r	n/r	0.025	344.5
cmix v20 (published)		3,503,868	2.42×	n/r	n/r	n/r	n/r
Compared via existing metrics, not actually tested - not ranked							

CORPUS: Silesia mr

Silesia mr (9,970,564 B, MRI DICOM) - every recorded tool

machine: mixed: see notes

original: 9,970,564 bytes in 1 file(s)

source: same sources as x-ray table

Machines differ by row (marked); rank speeds only among rows of the same machine.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
JPEG XL -d 0 -e 9 + xz header (cl		1,777,016	5.61×	3.47	2.87	0.252	39.57
HX Complete, every route (router	*	1,777,401	5.61×	158.5	0.063	n/r	n/r
16-bit image route (JPEG XL lossless inside HX Complete); compress = sum of all route times; decode n/r							
JPEG XL -d 0 -e 7 + xz header (cl		1,839,741	5.42×	0.953	10.46	0.221	45.12
JPEG-LS (CharLS) + xz header (clo		2,021,475	4.93×	0.127	78.51	0.077	129.5
HX Fast (cloud 2 thr)	*	2,069,305	4.82×	5.13	1.94	0.066	151.1
HX Fast (desktop 12 thr, in memor	*	2,069,305	4.82×	3.30	3.02	0.014	722.5
mzip (desktop)		~2,117,678	4.71×	38.87	0.257	0.714	13.96
bzip3 (cloud)		2,120,002	4.70×	1.36	7.35	1.58	6.32
HCM v3.7 max (cloud)	r	2,123,303	4.70×	37.80	0.264	9.67	1.03
HCM v3.6 fast single core (deskto	*	2,137,406	4.66×	10.20	0.978	3.50	2.85
kanzi -l 9 (cloud)		2,148,534	4.64×	4.41	2.26	4.33	2.30
ZPAQ -m5 (cloud)		2,181,313	4.57×	n/r	n/r	62.00	0.161
Highlights-2026-11-04.txt							
lpaq1 (cloud)		2,203,847	4.52×	n/r	n/r	30.00	0.332
Highlights-2026-11-04.txt							
BSC (cloud)		2,206,002	4.52×	0.751	13.28	0.632	15.78
7z-hi (Delta 2)	m	~2,592,097	3.85×	n/r	n/r	n/r	n/r
7-Zip -mx9		~2,748,256	3.63×	n/r	n/r	n/r	n/r
xz -9e (desktop)		~2,751,906	3.62×	3.94	2.53	0.111	89.99
cmix v20 (published)		1,827,572	5.46×	n/r	n/r	n/r	n/r
Compared via existing metrics, not actually tested - not ranked							

CORPUS: Calgary

Calgary corpus, 18 files - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 3,251,493 bytes in 18 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		731,676	4.44×	9.58	0.340	7.51	0.433
HX Complete	*	796,770	4.08×	9.06	0.359	0.315	10.32
bzip3		803,007	4.05×	0.480	6.77	0.438	7.42
BSC		805,634	4.04×	0.390	8.34	0.337	9.65
7-Zip PPMd (-mx9 -m0=PPMd)		810,275	4.01×	0.560	5.81	0.636	5.11
Brotli 11		856,816	3.79×	6.68	0.487	0.079	41.16
7-Zip -mx9 (LZMA2)		887,055	3.67×	0.700	4.64	0.186	17.48
HX Fast	*	913,802	3.56×	8.48	0.383	0.080	40.64
gzip -9		1,059,335	3.07×	0.410	7.93	0.093	34.96
LZ4 (default)		1,683,270	1.93×	0.090	36.13	0.084	38.71

Calgary corpus (18 files, each compressed on its own) - Matt Mahoney test sets race (2 Nov 2026)

machine: desktop (run in the PC test folder HCM-tests), wall clock

original: 3,251,493 bytes in 18 file(s)

source: HCM-tests\mahoney-sets-2026-11-02\results_summary.txt (GitHub tools\mahoney_race_2026-11-02)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
hcmx max	*	707,392	4.60×	11.70	0.278	2.70	1.20
lpaq1-h 9	r	709,793	4.58×	61.20	0.053	9.80	0.332
HCM max	*	710,258	4.58×	6.40	0.508	2.50	1.30
HCM Sets max	*	710,798	4.57×	6.30	0.516	2.60	1.25
zpaq-hi -m5	m	716,354	4.54×	12.50	0.260	5.70	0.570

HCM fast	*	717,557	4.53×	4.80	0.677	1.90	1.71
HCM Sets fast	*	718,097	4.53×	5.20	0.625	2.00	1.63
ZPAQ 7.15 -m5		718,255	4.53×	5.50	0.591	5.50	0.591
lpaq1 9		730,373	4.45×	11.60	0.280	10.40	0.313
ZPAQ 7.15 -m4		767,905	4.23×	1.50	2.17	1.60	2.03

CORPUS: Canterbury

Canterbury corpus, 11 files - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 2,810,784 bytes in 11 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		412,514	6.81×	5.09	0.553	5.05	0.556
HX Complete	*	444,353	6.33×	7.02	0.400	0.175	16.06
BSC		457,510	6.14×	0.240	11.71	0.220	12.78
bzip3		459,469	6.12×	0.370	7.60	0.310	9.07
Brotli 11		490,750	5.73×	5.64	0.498	0.051	55.11
7-Zip -mx9 (LZMA2)		494,853	5.68×	0.610	4.61	0.118	23.82
HX Fast	*	516,471	5.44×	6.47	0.434	0.049	57.36
7-Zip PPMd (-mx9 -m0=PPMd)		575,177	4.89×	0.390	7.21	0.419	6.71
gzip -9		730,625	3.85×	0.810	3.47	0.062	45.34
LZ4 (default)		1,223,226	2.30×	0.050	56.22	0.055	51.11

Canterbury - 14 tools, one fresh same-machine run (29 Oct 2026)
machine: cloud, wall clock per file, summed
original: 2,810,784 bytes in 11 file(s)
source: benchmark-charts-2026-10-29\results.json (GitHub benchmarks\charts-2026-10-29)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.4 max	*	350,326	8.02×	9.90	0.284	3.42	0.822
ZPAQ -m5		350,542	8.02×	11.57	0.243	11.52	0.244
hcmx fast	*	350,749	8.01×	14.79	0.190	2.98	0.945
HCM v3.4 fast	*	353,387	7.95×	7.82	0.359	2.29	1.23
lpaq1-h	r	359,786	7.81×	22.93	0.123	3.98	0.706
lpaq1 -6		392,132	7.17×	4.41	0.638	3.77	0.746
xz-hi	m	462,832	6.07×	154.5	0.018	0.067	41.83
xz -9e		493,080	5.70×	2.03	1.38	0.074	37.83
zstd -19		516,401	5.44×	1.17	2.41	0.044	63.74
bzip2-hi	m	531,899	5.28×	69.66	0.040	0.110	25.48
bzip2 -9		542,710	5.18×	0.186	15.13	0.103	27.37
zstd -3		634,602	4.43×	0.103	27.40	0.056	50.55
gzip -9		730,732	3.85×	0.662	4.24	0.053	53.23
lz4 -1		1,223,226	2.30×	0.047	59.42	0.066	42.78

Canterbury, 11 files - HCM line and standard rivals as first recorded (25-29 Oct 2026)
machine: cloud; sizes recorded relative to the original (bytes derived); decode where recorded
original: 2,810,784 bytes in 11 file(s)
source: memory /areas/combary.md, /areas/hcm-benchmarks.md; PC HCM_versions_and_results.md, HCM_Wins_Under_Standard.txt
Decode for xz/bzip2/LZMA rows = summed in-process decode of all 11 files. HCM v3 decode ~6.3-6.5x faster than ZPAQ -m5; HCM v3.3 decode median 1.85-1.87 s (v3 1.88 s).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
paq8px v217 -8 (full)		~302,019	9.31×	n/r	n/r	n/r	n/r
paq8px lab run of 25 Oct (kennedy 414 s -> 181 s); see section 8							
paq8px lite model set (lab)	r	~302,913	9.28×	n/r	n/r	n/r	n/r
paq8px lab run of 25 Oct (kennedy 414 s -> 181 s); see section 8							
hcm-trit v2 max + cmix dictionary	r	~332,468	8.45×	n/r	n/r	n/r	n/r
dictionary is GPL-3 and English-only (research)							
HCM v3.3 fast + cmix English dict	r	~337,322	8.33×	n/r	n/r	n/r	n/r
dictionary is GPL-3 and English-only (research)							
hcmx max best-of	*	~347,697	8.08×	n/r	n/r	n/r	n/r
hcm-trit v2 max	r	~348,135	8.07×	n/r	n/r	n/r	n/r
HCM v3.3 max	*	~350,434	8.02×	n/r	n/r	n/r	n/r
hcmx fast best-of	*	~350,873	8.01×	n/r	n/r	n/r	n/r
hcm-trit v2 (pointer) 6-order	r	~351,179	8.00×	n/r	n/r	n/r	n/r
HCM v3.3 fast	*	~353,493	7.95×	n/r	n/r	n/r	n/r
HCM v3.2 max	*	~356,284	7.89×	n/r	n/r	n/r	n/r
HCM v3.1 max	*	~357,467	7.86×	n/r	n/r	n/r	n/r
HCM-Bin v1 (overall)	r	~357,602	7.86×	n/r	n/r	n/r	n/r
HCM v3.2 fast	*	~358,884	7.83×	n/r	n/r	n/r	n/r
zpaq-hi (one file per archive)	m	~359,426	7.82×	n/r	n/r	n/r	n/r

HCM v3.1 fast	*	~361,093	7.78×	n/r	n/r	n/r	n/r
hcm-trit v1.1	r	~361,711	7.77×	n/r	n/r	n/r	n/r
ZPAQ -m5		~361,793	7.77×	n/r	n/r	n/r	n/r
HCM v3 max	*	~412,328	6.82×	n/r	n/r	n/r	n/r
HCM v3 fast	*	~414,602	6.78×	n/r	n/r	n/r	n/r
hcm-trit v1	r	~414,903	6.77×	n/r	n/r	n/r	n/r
HCM v2	*	~416,994	6.74×	n/r	n/r	n/r	n/r
xz-hi	m	~462,832	6.07×	n/r	n/r	0.025	113.3
xz-hi (king race)	m	~462,832	6.07×	n/r	n/r	0.029	97.56
7z-hi	m	~464,423	6.05×	n/r	n/r	n/r	n/r
lzma-hi	m	~486,294	5.78×	n/r	n/r	0.029	98.45
xz -9e		~493,079	5.70×	n/r	n/r	0.032	87.56
LZMA -9 (7-Zip SDK, king)		~493,422	5.70×	n/r	n/r	0.029	98.11
7-Zip Ultra -mx9		~494,900	5.68×	n/r	n/r	n/r	n/r
bzip2 rebuild + candidate escape	r	~503,338	5.58×	n/r	n/r	n/r	n/r
bzip2-style rebuild + arithmetic	r	~522,162	5.38×	n/r	n/r	n/r	n/r
bzip2-hi	m	~531,899	5.28×	n/r	n/r	0.055	50.74
bzip2 -9		~542,709	5.18×	n/r	n/r	0.060	47.08

Canterbury - zstd-hi / Brotli-hi encoder search, stock decoders (26 Oct 2026)
machine: cloud; sizes only
original: 2,810,784 bytes in 11 file(s)
source: memory /areas/hcm-benchmarks.md (zstd-hi / Brotli-hi)
Decode with stock decoders; speeds not recorded per level (no-pain checks passed).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
Brotli-hi 11	m	~485,917	5.78×	n/r	n/r	n/r	n/r
Brotli 11		~490,662	5.73×	n/r	n/r	n/r	n/r
zstd-hi -22	m	~515,627	5.45×	n/r	n/r	n/r	n/r
zstd-hi -19	m	~516,113	5.45×	n/r	n/r	n/r	n/r
zstd -22		~516,237	5.44×	n/r	n/r	n/r	n/r
zstd -19		~516,358	5.44×	n/r	n/r	n/r	n/r
Brotli-hi 8	m	~545,095	5.16×	n/r	n/r	n/r	n/r
Brotli 8		~545,660	5.15×	n/r	n/r	n/r	n/r
Brotli-hi 5	m	~564,495	4.98×	n/r	n/r	n/r	n/r
Brotli 5		~565,499	4.97×	n/r	n/r	n/r	n/r
zstd-hi -9	m	~591,597	4.75×	n/r	n/r	n/r	n/r
zstd -9		~596,707	4.71×	n/r	n/r	n/r	n/r
zstd-hi -3	m	~630,189	4.46×	n/r	n/r	n/r	n/r
zstd -3		~634,557	4.43×	n/r	n/r	n/r	n/r
zstd-hi -1	m	~678,886	4.14×	n/r	n/r	n/r	n/r
zstd -1		~686,781	4.09×	n/r	n/r	n/r	n/r

Canterbury, 11 files - zpaq-hi vs stock ZPAQ 7.15 -m5, one file per archive (29 Oct 2026)
machine: cloud, wall clock
original: 2,810,784 bytes in 11 file(s)
source: zpaq-hi\canterbury_results.txt
Per-file times rounded to 0.1 s in the source; totals summed from them.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zpaq-hi -m5	m	359,424	7.82×	20.90	0.134	14.10	0.199
ZPAQ 7.15 -m5		361,792	7.77×	11.50	0.244	11.80	0.238

CORPUS: kennedy.xls

Canterbury kennedy.xls (1,029,744 B, spreadsheet, record length 13) - recorded tools
machine: cloud; in-process decode in microseconds where marked
original: 1,029,744 bytes in 1 file(s)
source: hxc_calgary_canterbury_large_synthetic.csv; results.json;
zpaq-hi\canterbury_results.txt; memory /areas/combary.md; HCM_Wins_Under_Standard.txt
kennedy.xls is an extreme outlier (record data); totals elsewhere are given with it included.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zpaq-hi -m5	m	16,002	64.35×	n/r	n/r	5.50	0.187
HCM-Bin v1	r	16,763	61.43×	n/r	n/r	n/r	n/r
HCM-Bin v1 max (hcmx)	*	16,797	61.31×	n/r	n/r	n/r	n/r
ZPAQ -m5		16,822	61.21×	n/r	n/r	4.30	0.239
HCM v3.3 fast	*	18,680	55.13×	n/r	n/r	n/r	n/r
lpaq1-h	r	21,644	47.58×	n/r	n/r	n/r	n/r
xz-hi (delta 13)	m	23,656	43.53×	n/r	n/r	n/r	n/r
zstd -22 + record delta 13	r	26,398	39.01×	n/r	n/r	0.001	1,980
gzip -9 + change map	r	35,558	28.96×	n/r	n/r	n/r	n/r
lpaq1 -6		37,943	27.14×	n/r	n/r	n/r	n/r
gzip -9 + record delta 13	r	42,972	23.96×	n/r	n/r	0.001	977.0
lz4 -12 + record delta 13	r	43,654	23.59×	n/r	n/r	0.001	1,420

xz -9e		51,868	19.85×	n/r	n/r	n/r	n/r
HCM v3 fast	*	53,899	19.11×	n/r	n/r	n/r	n/r
LZFSE + record delta 13	r	~54,844	18.78×	n/r	n/r	0.001	1,343
zstd -22		64,815	15.89×	n/r	n/r	0.001	757.7
HX Fast	*	64,838	15.88×	1.92	0.537	0.006	171.6
BSC		74,010	13.91×	0.054	19.07	0.045	22.88
bzip3		76,918	13.39×	0.129	7.98	0.098	10.51
bzip2-hi (100k blocks)	m	120,482	8.55×	n/r	n/r	n/r	n/r
Snappy + record delta 13	r	~129,763	7.94×	n/r	n/r	0.001	1,436
bzip2 -9		130,280	7.90×	n/r	n/r	n/r	n/r
LZFSE		~136,282	7.56×	n/r	n/r	0.001	967.8
gzip -9		209,721	4.91×	n/r	n/r	0.002	472.6
lz4 -12		324,136	3.18×	n/r	n/r	0.001	1,337
Snappy		~422,640	2.44×	n/r	n/r	0.001	1,066

CORPUS: Canterbury large

Canterbury large corpus, 3 files - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 11,159,482 bytes in 3 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		2,162,555	5.16×	6.08	1.84	5.15	2.17
HX Complete	*	2,249,710	4.96×	20.57	0.543	1.40	7.95
bzip3		2,255,366	4.95×	1.47	7.59	1.61	6.95
BSC		2,258,712	4.94×	0.860	12.98	0.656	17.01
7-Zip PPMd (-mx9 -m0=PPMd)		2,320,547	4.81×	3.07	3.64	2.44	4.57
Brotli 11		2,502,134	4.46×	33.07	0.337	0.076	146.8
HX Fast	*	2,547,279	4.38×	17.86	0.625	0.034	328.2
7-Zip -mx9 (LZMA2)		2,558,272	4.36×	5.45	2.05	0.182	61.32
gzip -9		3,197,094	3.49×	6.71	1.66	0.107	104.3
LZ4 (default)		5,753,483	1.94×	0.530	21.06	0.048	232.5

Canterbury large corpus (E.coli, bible.txt, world192.txt) - Matt Mahoney test sets race (2 Nov 2026)
machine: desktop (run in the PC test folder HCM-tests), wall clock
original: 11,159,482 bytes in 3 file(s)
source: HCM-tests\mahoney-sets-2026-11-02\results_summary.txt (GitHub tools\mahoney_race_2026-11-02)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zpaq-hi -m5	m	2,151,206	5.19×	50.00	0.223	17.00	0.656
ZPAQ 7.15 -m5		2,153,108	5.18×	16.00	0.697	15.90	0.702
lpaq1-h 9	r	2,168,206	5.15×	51.40	0.217	9.60	1.16
lpaq1 9		2,169,454	5.14×	10.20	1.09	9.80	1.14
hcmx max	*	2,180,478	5.12×	27.70	0.403	5.90	1.89
HCM max	*	2,188,255	5.10×	15.00	0.744	5.30	2.11
HCM Sets max	*	2,188,345	5.10×	14.70	0.759	5.30	2.11
HCM fast	*	2,208,053	5.05×	13.40	0.833	4.80	2.32
HCM Sets fast	*	2,208,143	5.05×	12.90	0.865	4.80	2.32
ZPAQ 7.15 -m4		2,242,719	4.98×	3.40	3.28	3.40	3.28

CORPUS: enwik8

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enwik8 (100,000,000 B) - race of 27 Oct 2026
machine: cloud, 2 cores, decode seconds
original: 100,000,000 bytes in 1 file(s)
source: memory /areas/hcm-benchmarks.md (enwik8 race, 27 Oct)
Compression times not recorded. Also recorded elsewhere: xman (encode.su) ran HCM v3.4 max on his Xeon: 21,477,536 B (4.66×); HCM v3.6 max 21,692,518 B (4.61×), 1.8x faster compress than v3.4 max (no absolute times recorded). lpaq1 -9 whole file in the pre-training run ~19,755,900 B (5.06×, no times).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
ZPAQ -m5		~19,625,000	5.10×	n/r	n/r	282.0	0.355
ZPAQ -m4		~20,740,500	4.82×	n/r	n/r	102.0	0.980
BSC -b64e2		~21,480,500	4.66×	n/r	n/r	6.00	16.67
HCM v3.3 max	*	~21,701,000	4.61×	n/r	n/r	115.0	0.870
HCM v3.3 fast	*	~22,187,200	4.51×	n/r	n/r	99.00	1.01
bzip3		~22,677,500	4.41×	n/r	n/r	13.90	7.19

CORPUS: enwik9

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enwik9 (1,000,000,000 bytes) - Matt Mahoney test sets race (2 Nov 2026)
machine: desktop (run in the PC test folder HCM-tests), wall clock
original: 1,000,000,000 bytes in 1 file(s)
source: HCM-tests\mahoney-sets-2026-11-02\results_summary.txt (GitHub tools\mahoney_race_2026-11-02)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
lpaq1 9		164,508,919	6.08×	643.4	1.55	675.6	1.48
zpaq-hi -m5	m	168,215,145	5.94×	868.2	1.15	468.8	2.13
ZPAQ 7.15 -m5		168,590,741	5.93×	412.7	2.42	422.5	2.37
ZPAQ 7.15 -m4		179,455,249	5.57×	135.9	7.36	148.2	6.75
HCM max	*	187,028,826	5.35×	1,403	0.713	501.2	2.00
HCM Sets max	*	192,181,500	5.20×	289.1	3.46	100.3	9.97
HCM Sets fast	*	196,048,761	5.10×	250.6	3.99	92.00	10.87

CORPUS: GCB (uiq2)

Random Turing machine output (uiq2, Generic Compression Benchmark data), 3 seeds - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 19,552,881 bytes in 3 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
lpaq1 8		9,162,095	2.13×	29.81	0.656	33.84	0.578
HX Complete BSC	*	9,185,811	2.13×	29.62	0.660	2.71	7.22
7-Zip PPMd (-mx9 -m0=PPMd)		9,385,990	2.08×	8.64	2.26	8.60	2.27
kanzi -l 9		9,417,958	2.08×	19.05	1.03	17.71	1.10
Brotli 11		10,077,742	1.94×	50.54	0.387	0.214	91.37
HX Fast repeat line	*	10,180,161	1.92×	21.67	0.902	0.082	238.4
7-Zip -mx9 (LZMA2)		10,181,320	1.92×	4.79	4.08	0.640	30.55
bzip2 -9		11,047,198	1.77×	1.82	10.74	1.63	12.00
gzip -9		11,824,901	1.65×	2.69	7.27	0.206	94.92
LZ4 (default)		16,138,858	1.21×	0.100	195.5	0.067	291.8

Generic Compression Benchmark (uiq2 seeds 0, 1, 2) - Matt Mahoney test sets race (2 Nov 2026)
machine: desktop (run in the PC test folder HCM-tests), wall clock
original: 19,552,881 bytes in 3 file(s)
source: HCM-tests\mahoney-sets-2026-11-02\results_summary.txt (GitHub tools\mahoney_race_2026-11-02)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
lpaq1-h 9	r	9,121,594	2.14×	107.5	0.182	19.90	0.983
zpaq-hi -m5	m	9,137,324	2.14×	69.80	0.280	36.00	0.543
ZPAQ 7.15 -m5		9,139,174	2.14×	34.10	0.573	35.30	0.554
lpaq1 9		9,152,676	2.14×	18.10	1.08	19.30	1.01
hcmx max	*	9,307,087	2.10×	107.9	0.181	20.40	0.958
HCM max	*	9,368,500	2.09×	33.70	0.580	12.80	1.53
ZPAQ 7.15 -m4		9,404,729	2.08×	9.20	2.13	9.80	2.00
HCM Sets max	*	9,411,934	2.08×	32.00	0.611	11.30	1.73
HCM fast	*	9,539,490	2.05×	29.20	0.670	10.70	1.83
HCM Sets fast	*	9,576,420	2.04×	26.00	0.752	9.30	2.10

CORPUS: Synthetic (ours)

Synthetic test files, 5 files - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 152,544,864 bytes in 5 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Complete	*	91,662,977	1.66×	209.5	0.728	6.87	22.21
7-Zip -mx9 (LZMA2)		92,571,973	1.65×	22.02	6.93	3.42	44.66
HX Fast	*	92,960,242	1.64×	115.5	1.32	0.462	330.2
kanzi -l 9		94,967,953	1.61×	141.4	1.08	108.5	1.41
Brotli 11		96,816,928	1.58×	551.2	0.277	1.19	128.6

BSC		100,680,826	1.52×	24.96	6.11	21.33	7.15
bzip3		101,606,743	1.50×	27.18	5.61	28.16	5.42
7-Zip PPMd (-mx9 -m0=PPMd)		114,662,957	1.33×	81.81	1.86	84.94	1.80
gzip -9		125,366,209	1.22×	14.89	10.24	1.66	92.17
LZ4 (default)		137,086,744	1.11×	0.460	331.6	0.283	539.0
ZPAQ -m5		68,301,824	1.94×	617.1	0.215	630.0	0.210
partial: 4 of 5 files (132,544,864 B) - not ranked							
HCM max	*	68,674,323	1.93×	1,494	0.089	317.9	0.417
partial: 4 of 5 files (132,544,864 B) - not ranked							
lpaq1 -9		72,574,570	1.83×	304.4	0.435	303.2	0.437
partial: 4 of 5 files (132,544,864 B) - not ranked							
xz -9e		72,593,648	1.83×	77.03	1.72	3.69	35.93
partial: 4 of 5 files (132,544,864 B) - not ranked							
zstd -19 --long		80,834,596	1.64×	66.69	1.99	0.325	407.8
partial: 4 of 5 files (132,544,864 B) - not ranked							
zstd -19		80,909,994	1.64×	54.15	2.45	0.311	426.2
partial: 4 of 5 files (132,544,864 B) - not ranked							
bzip2 -9		101,105,036	1.31×	24.31	5.45	20.43	6.49
partial: 4 of 5 files (132,544,864 B) - not ranked							

CORPUS: Goodwin synthetic

Synthetic (Goodwin) - 14 tools, one fresh same-machine run (29 Oct 2026)
machine: cloud, wall clock per file, summed
original: 17,999,934 bytes in 18 file(s)
source: benchmark-charts-2026-10-29\results.json (GitHub benchmarks\charts-2026-10-29)

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
lpaq1-h	r	8,113,745	2.22×	149.4	0.120	26.71	0.674
HCM v3.4 max	*	8,127,874	2.21×	99.90	0.180	23.70	0.760
hcmx fast	*	8,168,432	2.20×	191.6	0.094	21.64	0.832
HCM v3.4 fast	*	8,190,545	2.20×	79.03	0.228	18.75	0.960
ZPAQ -m5		8,201,855	2.19×	82.27	0.219	83.12	0.217
lpaq1 -6		8,656,077	2.08×	24.51	0.734	25.51	0.706
xz-hi	m	9,162,824	1.96×	762.9	0.024	0.569	31.62
xz -9e		9,342,648	1.93×	9.04	1.99	0.540	33.30
bzip2-hi	m	9,628,268	1.87×	826.6	0.022	1.29	14.00
bzip2 -9		9,717,583	1.85×	1.45	12.38	1.28	14.03
zstd -19		9,852,387	1.83×	6.15	2.93	0.107	168.9
zstd -3		10,795,883	1.67×	0.149	120.8	0.085	212.8
gzip -9		11,075,558	1.63×	10.29	1.75	0.216	83.18
lz4 -1		15,060,740	1.20×	0.183	98.52	0.120	150.0

CORPUS: Biased test files, 8 MB each

Biased test files, 8 MB each - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 67,108,864 bytes in 8 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
lpaq1 -9		33,236,487	2.02×	173.5	0.387	180.3	0.372
HX Fast	*	33,239,986	2.02×	120.1	0.559	0.440	152.5
HX Complete	*	33,239,986	2.02×	176.9	0.379	0.453	148.1
ZPAQ -m5		33,295,222	2.02×	377.1	0.178	370.0	0.181
HCM max	*	33,365,082	2.01×	498.5	0.135	143.2	0.469
HCM Sets max + cascade v3	*	33,365,322	2.01×	519.2	0.129	133.0	0.505
bzip3		33,583,203	2.00×	12.90	5.20	14.90	4.51
BSC		34,299,188	1.96×	13.28	5.05	13.00	5.16
Brotli 11		34,371,767	1.95×	177.5	0.378	0.660	101.7
HX lines + Sets	*	34,466,859	1.95×	25.99	2.58	0.462	145.3
HX lines	*	34,780,240	1.93×	11.65	5.76	0.420	159.8
xz -9e		35,288,552	1.90×	71.64	0.937	2.88	23.29
zstd -19		35,969,040	1.87×	57.50	1.17	0.184	364.7
hs_zstd 1.3 Sets -19, 16 MB block	*	35,975,525	1.87×	92.65	0.724	0.482	139.2
cascade v1.1 -19, 16 MB blocks	*	35,975,533	1.87×	180.9	0.371	0.562	119.4
7-Zip -mx9 (LZMA2)		36,072,298	1.86×	23.82	2.82	2.14	31.37
7-Zip PPMd (-mx9 -m0=PPMd)		36,277,949	1.85×	28.40	2.36	30.40	2.21
hs_zstd 1.3 Sets -19	*	37,279,984	1.80×	41.06	1.63	0.444	151.1
cascade v1.1 -19	*	37,280,048	1.80×	79.58	0.843	0.463	144.9
adaptive arithmetic (ac0)		37,476,874	1.79×	4.48	14.98	3.99	16.83

FSE (tANS) 128 KB		37,936,832	1.77×	0.280	239.7	0.155	433.0
rANS 1 MB		38,033,652	1.76×	0.740	90.69	0.444	151.1
FSE (tANS) 1 MB		38,206,480	1.76×	0.320	209.7	0.145	462.8
huff0 (Huffman) 128 KB		38,823,051	1.73×	0.260	258.1	0.090	745.7
hs_zstd 1.3 Sets -3	*	39,592,763	1.69×	1.38	48.63	0.451	148.8
cascade v1.1 -3	*	39,592,827	1.69×	1.82	36.87	0.483	138.9
gzip -9		40,285,818	1.67×	40.88	1.64	0.676	99.27
LZ4 (default)		55,622,364	1.21×	0.210	319.6	0.189	355.1
kanzi -l 9		21,016,437	2.00×	34.57	1.21	28.38	1.48
partial: 5 of 8 files - not ranked [rivals\rival3_results.csv]							
mzip		32,833,668	1.79×	451.0	0.130	7.25	8.09
partial: 7 of 8 files (58,720,256 B) - not ranked							
Sets + mzip 16 MB blocks	r	32,834,046	1.79×	448.1	0.131	7.41	7.93
partial: 7 of 8 files (58,720,256 B) - not ranked							
Sets + mzip 4 MB blocks	r	33,222,105	1.77×	200.2	0.293	4.11	14.28
partial: 7 of 8 files (58,720,256 B) - not ranked							

CORPUS: NIST SP 800-90B real noise samples

NIST SP 800-90B real noise samples - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 7,000,000 bytes in 7 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Complete	*	2,810,459	2.49×	20.29	0.345	0.129	54.26
HX lines	*	2,811,481	2.49×	1.11	6.31	0.029	241.4
HX lines + Sets	*	2,811,488	2.49×	1.72	4.07	0.023	304.3
HX Fast	*	2,811,537	2.49×	17.23	0.406	0.074	94.59
lpaq1 -9		2,813,763	2.49×	17.53	0.399	19.44	0.360
HCM max	*	2,818,867	2.48×	31.76	0.220	7.62	0.918
HCM Sets max + cascade v3	*	2,819,100	2.48×	25.24	0.277	7.14	0.980
ZPAQ -m5		2,821,719	2.48×	30.47	0.230	29.26	0.239
bzip3		2,824,395	2.48×	1.47	4.76	1.44	4.85
BSC		2,826,982	2.48×	1.55	4.52	1.10	6.36
rANS 1 MB		2,854,935	2.45×	0.100	70.00	0.042	166.7
adaptive arithmetic (ac0)		2,855,331	2.45×	0.730	9.59	0.530	13.21
FSE (tANS) 1 MB		2,859,086	2.45×	0.050	140.0	0.012	583.3
FSE (tANS) 128 KB		2,860,984	2.45×	0.030	233.3	0.014	500.0
Brotli 11		2,869,839	2.44×	13.81	0.507	0.060	116.7
zstd -19		2,877,895	2.43×	5.49	1.28	0.014	500.0
cascade v1.1 -19	*	2,878,206	2.43×	12.15	0.576	0.088	79.55
cascade v1.1 -19, 16 MB blocks	*	2,878,206	2.43×	13.53	0.517	0.108	64.81
hs_zstd 1.3 Sets -19	*	2,878,217	2.43×	7.10	0.986	0.087	80.46
hs_zstd 1.3 Sets -19, 16 MB block	*	2,878,217	2.43×	6.57	1.07	0.086	81.40
xz -9e		2,916,908	2.40×	8.48	0.825	0.192	36.46
7-Zip -mx9 (LZMA2)		2,933,402	2.39×	1.83	3.83	0.200	35.00
7-Zip PPMd (-mx9 -m0=PPMd)		2,971,671	2.36×	1.53	4.58	1.60	4.38
huff0 (Huffman) 128 KB		3,003,497	2.33×	0.030	233.3	0.006	1,167
gzip -9		3,015,159	2.32×	9.38	0.746	0.080	87.50
cascade v1.1 -3	*	3,070,894	2.28×	0.280	25.00	0.119	58.82
hs_zstd 1.3 Sets -3	*	3,070,905	2.28×	0.190	36.84	0.122	57.38
LZ4 (default)		4,461,510	1.57×	0.050	140.0	0.051	137.3
mzip		78,179	12.79×	29.02	0.034	0.172	5.81
partial: 1 of 7 files (1,000,000 B) - not ranked							
Sets + mzip 16 MB blocks	r	78,233	12.78×	28.58	0.035	0.199	5.03
partial: 1 of 7 files (1,000,000 B) - not ranked							
Sets + mzip 4 MB blocks	r	78,233	12.78×	31.13	0.032	0.196	5.10
partial: 1 of 7 files (1,000,000 B) - not ranked							

CORPUS: Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR)

Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR) -
same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 11,255,022 bytes in 38 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG
/ MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		1,124,004	10.01×	18.25	0.617	16.14	0.697
7-Zip PPMd (-mx9 -m0=PPMd)		1,295,714	8.69×	1.41	7.98	1.47	7.67

HX Complete + audio/colour lines	*	1,306,230	8.62×	46.73	0.241	0.584	19.27
BSC		1,311,678	8.58×	0.690	16.31	0.593	18.98
bzip3		1,318,058	8.54×	1.11	10.14	1.06	10.59
Brotli 11		1,418,253	7.94×	26.57	0.424	0.184	61.17
xz -9e		1,430,096	7.87×	6.72	1.67	0.253	44.49
7-Zip -mx9 (LZMA2)		1,461,711	7.70×	2.56	4.40	0.413	27.25
zstd -19		1,508,880	7.46×	6.44	1.75	0.158	71.23
HX Fast + audio/colour lines repeat line	*	1,508,898	7.46×	44.73	0.252	0.106	106.2
gzip -9		1,907,293	5.90×	1.18	9.54	0.223	50.47
LZ4 (default)		3,132,684	3.59×	0.200	56.28	0.187	60.19

CORPUS: Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc++, cc1plus, git)

Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc++, cc1plus, git) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 51,401,568 bytes in 9 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		12,993,502	3.96×	45.93	1.12	39.49	1.30
7-Zip -mx9 (LZMA2)		14,703,863	3.50×	12.27	4.19	1.25	41.29
7-Zip PPMd (-mx9 -m0=PPMd)		15,683,430	3.28×	11.60	4.43	11.35	4.53
HX Fast + audio/colour lines	*	16,893,333	3.04×	138.8	0.370	0.255	201.6
xz -9e		16,895,880	3.04×	32.04	1.60	1.30	39.51
HX Complete + audio/colour lines	*	17,063,456	3.01×	147.1	0.349	5.98	8.60
Brotli 11		17,246,650	2.98×	165.7	0.310	0.417	123.3
zstd -19		18,141,912	2.83×	24.79	2.07	0.228	225.4
bzip3		18,718,447	2.75×	7.52	6.84	7.51	6.84
BSC		18,922,226	2.72×	5.83	8.82	5.28	9.73
gzip -9		21,577,700	2.38×	8.75	5.87	0.576	89.24
LZ4 (default)		30,904,188	1.66×	0.210	244.8	0.148	347.3

CORPUS: Speech, WAVs, 8 kHz mono (FSDD)

Speech, WAVs, 8 kHz mono (FSDD) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 20,997,112 bytes in 6 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
OptimFROG --preset max		10,122,381	2.07×	42.32	0.496	8.05	2.61
OptimFROG --preset 2		10,282,663	2.04×	1.68	12.47	1.06	19.88
WavPack (ffmpeg, level 8) samples checked; +44 header bytes		10,611,462	1.98×	310.8	0.068	0.900	23.33
WavPack -hhx6 (official, max)		10,629,948	1.98×	21.55	0.974	0.332	63.24
WavPack -hh (official)		10,839,610	1.94×	0.730	28.76	0.413	50.84
HX Complete + audio/colour lines	*	10,962,708	1.92×	72.12	0.291	0.577	36.39
HX Complete	*	10,963,787	1.92×	28.31	0.742	0.475	44.20
HX Fast + audio/colour lines Sets + Exchange lines	*	10,983,665	1.91×	63.59	0.330	0.206	101.9
HX Fast Sets + Exchange lines	*	10,984,744	1.91×	21.68	0.969	0.194	108.2
flac -8 -e -p (max)		11,102,873	1.89×	8.07	2.60	0.180	116.7
flac -8		11,162,777	1.88×	0.330	63.63	0.178	118.0
Monkey's Audio -c4000		11,729,340	1.79×	3.83	5.48	0.546	38.43
Monkey's Audio -c5000		11,962,372	1.76×	1.27	16.49	1.26	16.69
bzip3		12,022,016	1.75×	3.61	5.82	3.17	6.61
TTA (ffmpeg) samples checked; +44 header bytes		12,240,480	1.72×	1.18	17.79	0.627	33.49
BSC		12,243,536	1.71×	3.04	6.91	3.01	6.98
Brotli 11		12,520,021	1.68×	54.70	0.384	0.261	80.45
7-Zip PPMd (-mx9 -m0=PPMd)		12,522,629	1.68×	6.25	3.36	6.92	3.03
ALAC (ffmpeg) samples checked; +44 header bytes		12,546,016	1.67×	1.53	13.72	0.806	26.05
7-Zip -mx9 (LZMA2)		12,653,377	1.66×	3.65	5.75	0.961	21.85
xz -9e		13,408,280	1.57×	10.12	2.07	0.937	22.41
zstd -19		14,663,529	1.43×	7.87	2.67	0.107	196.2

gzip -9		15,372,052	1.37×	3.46	6.07	0.243	86.41
LZ4 (default)		19,152,362	1.10×	0.090	233.3	0.068	308.8

CORPUS: Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data)

Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 124,700,736 bytes in 17 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
Monkey's Audio -c5000		43,718,100	2.85×	6.80	18.33	6.59	18.93
Monkey's Audio -c4000		43,780,848	2.85×	16.20	7.70	2.95	42.28
WavPack -hhx6 (official, max)		47,642,738	2.62×	126.5	0.986	1.97	63.17
flac -8 -e -p (max)		47,797,169	2.61×	71.90	1.73	0.974	128.0
flac -8		47,965,326	2.60×	2.92	42.71	0.986	126.5
HX Complete + audio/colour lines	*	48,155,341	2.59×	336.8	0.370	1.72	72.46
HX Fast + audio/colour lines	*	48,177,563	2.59×	281.1	0.444	1.60	78.08
Sets + Exchange lines							
WavPack -hh (official)		49,503,738	2.52×	4.03	30.94	2.12	58.68
TTA (ffmpeg)		49,636,215	2.51×	4.95	25.19	2.33	53.52
samples checked; +44 header bytes							
ALAC (ffmpeg)		53,544,126	2.33×	6.36	19.61	3.04	40.98
samples checked; +44 header bytes							
HX Complete	*	56,124,398	2.22×	162.4	0.768	1.17	106.5
HX Fast	*	56,148,900	2.22×	105.6	1.18	1.05	118.2
Sets + Exchange lines							
7-Zip -mx9 (LZMA2)		77,027,067	1.62×	21.50	5.80	5.59	22.30
7-Zip PPMd (-mx9 -m0=PPMd)		78,132,858	1.60×	49.71	2.51	55.82	2.23
Brotli 11		91,705,360	1.36×	537.6	0.232	1.64	76.18
bzip3		92,975,719	1.34×	24.92	5.00	24.71	5.05
xz -9e		94,138,428	1.32×	54.79	2.28	6.70	18.61
BSC		94,738,072	1.32×	23.80	5.24	23.92	5.21
zstd -19		102,900,417	1.21×	42.80	2.91	0.471	264.8
gzip -9		104,572,320	1.19×	14.53	8.58	1.29	96.82
LZ4 (default)		118,618,519	1.05×	0.300	415.7	0.279	447.0
OptimFROG --preset max		36,822,808	3.39×	221.2	0.564	41.29	3.02
1 of 17 files not exact - not ranked [rivals\rival3_results.csv]							
OptimFROG --preset 2		40,128,175	3.11×	7.57	16.47	5.61	22.21
1 of 17 files not exact - not ranked [rivals\rival3_results.csv]							
WavPack (ffmpeg, level 8)		11,298,377	2.76×	508.9	0.061	1.29	24.12
partial: 7 of 17 files (31,181,178 B) - not ranked							

CORPUS: Environmental recordings, WAVs, 44.1 kHz mono (ESC-50)

Environmental recordings, WAVs, 44.1 kHz mono (ESC-50) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 44,104,400 bytes in 100 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Fast + audio/colour lines	*	18,742,900	2.35×	185.6	0.238	1.15	38.52
Sets + Exchange lines							
HX Complete + audio/colour lines	*	18,742,900	2.35×	204.4	0.216	1.16	37.96
Sets + Exchange lines							
WavPack -hhx6 (official, max)		18,849,418	2.34×	48.71	0.905	1.02	43.07
WavPack -hh (official)		19,220,070	2.29×	1.94	22.73	1.10	40.09
TTA (ffmpeg)		19,864,325	2.22×	7.64	5.77	5.90	7.47
samples checked; +44 header bytes							
flac -8 -e -p (max)		19,888,780	2.22×	14.62	3.02	0.564	78.20
flac -8		19,933,673	2.21×	0.910	48.47	0.557	79.18
HX Fast	*	20,572,733	2.14×	79.38	0.556	0.827	53.33
Sets + Exchange lines							
HX Complete	*	20,572,733	2.14×	97.60	0.452	0.835	52.82
Sets + Exchange lines							
ALAC (ffmpeg)		20,611,953	2.14×	8.13	5.42	6.24	7.07
samples checked; +44 header bytes							
7-Zip -mx9 (LZMA2)		24,466,865	1.80×	5.36	8.23	2.56	17.24
7-Zip PPMd (-mx9 -m0=PPMd)		24,554,154	1.80×	9.19	4.80	9.77	4.52

bzip3	25,023,455	1.76×	6.74	6.54	5.52	7.99
kanzi -l 9	25,124,613	1.76×	72.01	0.612	66.47	0.664
BSC	25,822,812	1.71×	6.29	7.01	6.75	6.54
Brotli 11	26,341,840	1.67×	129.1	0.342	0.770	57.28
xz -9e	28,076,996	1.57×	18.43	2.39	2.39	18.43
zstd -19	30,262,061	1.46×	9.69	4.55	0.474	93.05
gzip -9	31,553,455	1.40×	4.26	10.35	0.756	58.34
LZ4 (default)	37,571,021	1.17×	0.430	102.6	0.434	101.6

CORPUS: Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC

Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 22,814,423 bytes in 20 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
precomp -cn + HX Complete	r	22,175,488	1.03×	34.99	0.652	3.70	6.16
precomp (lzma)		22,255,696	1.03×	9.03	2.53	1.69	13.50
HX Complete	*	22,524,890	1.01×	34.38	0.664	2.88	7.93
bzip3		22,542,480	1.01×	4.23	5.39	4.31	5.29
BSC		22,546,944	1.01×	3.98	5.73	4.61	4.94
Brotli 11		22,569,901	1.01×	40.44	0.564	0.217	105.1
zstd -19		22,603,216	1.01×	3.94	5.79	0.109	209.3
HX Fast	*	22,603,238	1.01×	23.95	0.953	0.086	265.3
gzip -9		22,609,366	1.01×	1.30	17.55	0.238	95.86
xz -9e		22,642,076	1.01×	8.18	2.79	0.482	47.33
7-Zip -mx9 (LZMA2)		22,643,436	1.01×	1.73	13.19	0.548	41.63
LZ4 (default)		22,759,796	1.00×	0.120	190.1	0.118	193.3
7-Zip PPMd (-mx9 -m0=PPMd)		22,943,081	0.99×	7.67	2.97	8.50	2.68
packMP3		2,866,250	1.15×	1.33	2.49	1.35	2.45
partial: 4 of 20 files (3,310,411 B) - not ranked							

CORPUS: Colour photos, raw RGB (Kodak 24, PPM)

Colour photos, raw RGB (Kodak 24, PPM) - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 28,311,912 bytes in 24 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Complete + specialists	*	9,996,882	2.83×	409.5	0.069	2.08	13.64
JPEG XL lossless (e9)		10,008,374	2.83×	53.63	0.528	2.05	13.82
JPEG XL lossless (e7)		10,131,884	2.79×	17.41	1.63	2.06	13.76
WebP lossless (z9)		11,204,040	2.53×	218.3	0.130	0.360	78.64
HX Complete	*	12,207,358	2.32×	63.12	0.449	3.77	7.51
bzip3		12,210,516	2.32×	4.26	6.65	3.69	7.66
BSC		12,806,370	2.21×	3.39	8.35	3.46	8.19
7-Zip PPMd (-mx9 -m0=PPMd)		13,612,113	2.08×	5.47	5.18	5.69	4.97
7-Zip -mx9 (LZMA2)		14,464,352	1.96×	4.17	6.79	1.08	26.14
xz -9e		14,475,300	1.96×	12.22	2.32	1.00	28.20
HX Fast	*	14,493,007	1.95×	52.22	0.542	0.340	83.27
Brotli 11		15,132,197	1.87×	71.56	0.396	0.334	84.77
zstd -19		15,190,569	1.86×	8.23	3.44	0.185	153.0
JPEG-LS (ffmpeg)		15,376,142	1.84×	2.80	10.11	4.07	6.95
PNG (level 9)		15,813,970	1.79×	5.57	5.08	0.531	53.32
gzip -9		17,683,055	1.60×	2.21	12.81	0.381	74.31
LZ4 (default)		24,001,278	1.18×	0.180	157.3	0.147	192.6
HX Complete + audio/colour lines	*	10,633,879	2.55×	131.8	0.206	1.20	22.59
partial: 23 of 24 files (27,132,249 B) - not ranked							
HX Fast + audio/colour lines	*	8,412,542	2.52×	92.13	0.230	0.398	53.35
partial: 18 of 24 files (21,233,934 B) - not ranked							

CORPUS: Rendered document pages / screenshots, raw (PGM/PPM)

Rendered document pages / screenshots, raw (PGM/PPM) - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 22,061,574 bytes in 9 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
7-Zip PPMd (-mx9 -m0=PPMd)		609,027	36.22×	0.650	33.94	0.740	29.81
kanzi -l 9		624,463	35.33×	6.25	3.53	5.53	3.99
HX Complete + specialists	*	633,346	34.83×	77.99	0.283	0.322	68.51
HX Complete	*	634,200	34.79×	23.57	0.936	0.365	60.44
HX Complete + audio/colour lines	*	634,200	34.79×	55.93	0.394	0.363	60.78
BSC		640,236	34.46×	0.320	68.94	0.294	75.04
bzip3		661,544	33.35×	0.580	38.04	0.472	46.74
xz -9e		665,864	33.13×	7.47	2.95	0.129	171.0
Brotli 11		673,056	32.78×	16.93	1.30	0.091	242.4
7-Zip -mx9 (LZMA2)		687,523	32.09×	1.45	15.21	0.143	154.3
WebP lossless (z9)		694,106	31.78×	44.76	0.493	0.139	158.7
HX Fast	*	723,697	30.48×	22.02	1.00	0.048	459.6
repeat line							
HX Fast + audio/colour lines	*	723,697	30.48×	54.78	0.403	0.046	479.6
repeat line							
zstd -19		728,775	30.27×	2.72	8.11	0.058	380.4
gzip -9		1,392,180	15.85×	2.04	10.81	0.157	140.5
JPEG XL lossless (e9)		1,629,387	13.54×	10.00	2.21	0.577	38.23
JPEG XL lossless (e7)		1,734,006	12.72×	2.33	9.47	0.574	38.43
LZ4 (default)		1,786,059	12.35×	0.070	315.2	0.072	306.4
JPEG-LS (ffmpeg)		2,852,972	7.73×	0.740	29.81	0.811	27.20
PNG (level 9)		1,176,639	18.69×	2.85	7.72	0.313	70.26
partial: 8 of 9 files (21,991,332 B) - not ranked							

CORPUS: Medical DICOM files (imageio test images)

Medical DICOM files (imageio test images) - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 2,482,413 bytes in 6 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HX Fast	*	787,635	3.15×	3.90	0.637	0.035	70.93
HX Complete	*	787,635	3.15×	4.66	0.533	0.035	70.93
bzip3		800,677	3.10×	0.330	7.52	0.329	7.55
BSC		826,864	3.00×	0.260	9.55	0.254	9.77
7-Zip PPMd (-mx9 -m0=PPMd)		852,564	2.91×	0.380	6.53	0.423	5.87
7-Zip -mx9 (LZMA2)		940,442	2.64×	0.380	6.53	0.103	24.10
xz -9e		945,264	2.63×	1.36	1.83	0.100	24.82
Brotli 11		970,775	2.56×	5.89	0.421	0.037	67.09
zstd -19		1,069,918	2.32×	1.07	2.32	0.031	80.08
gzip -9		1,197,361	2.07×	0.500	4.96	0.042	59.11
LZ4 (default)		1,740,433	1.43×	0.030	82.75	0.025	99.30

CORPUS: PNG files (Kodak 24)

PNG files (Kodak 24) - same-machine campaign 3-4 Nov 2026
machine: cloud, 2 cores (busy), wall clock
original: 15,394,305 bytes in 24 file(s)
source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
H-Deflate + HX Complete + image r	*	10,003,027	1.54×	271.8	0.057	9.69	1.59
precomp -cn + HX Complete	r	12,549,818	1.23×	71.40	0.216	10.91	1.41
bzip3							
precomp deflate-only + HX Complet	r	12,549,818	1.23×	151.7	0.101	11.61	1.33
bzip3							
precomp (lzma)		14,243,824	1.08×	24.96	0.617	7.91	1.95
HX Complete	*	15,260,803	1.01×	30.35	0.507	1.66	9.25
bzip3		15,274,416	1.01×	2.75	5.60	2.54	6.05
BSC		15,299,515	1.01×	2.73	5.64	2.27	6.80
7-Zip -mx9 (LZMA2)		15,369,356	1.00×	1.38	11.16	0.464	33.18
xz -9e		15,370,572	1.00×	5.90	2.61	0.376	40.94
HX Fast	*	15,388,229	1.00×	21.97	0.701	0.103	149.5
gzip -9		15,389,086	1.00×	1.10	13.99	0.189	81.45
zstd -19		15,392,862	1.00×	2.58	5.97	0.099	155.5
Brotli 11		15,394,617	1.00×	8.55	1.80	0.102	150.9
LZ4 (default)		15,394,761	1.00×	0.110	139.9	0.117	131.6
7-Zip PPMd (-mx9 -m0=PPMd)		15,576,773	0.99×	6.12	2.52	5.97	2.58

CORPUS: JPEG photos

JPEG photos - same-machine campaign 3-4 Nov 2026
  machine: cloud, 2 cores (busy), wall clock
  original: 43,888,544 bytes in 26 file(s)
  source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
packJPG		34,160,911	1.28×	20.42	2.15	18.99	2.31
HX Complete + specialists	*	34,161,292	1.28×	282.0	0.156	19.32	2.27
format specialist							
Lepton		34,394,594	1.28×	18.92	2.32	5.83	7.53
precomp -cn + HX Complete	r	34,508,754	1.27×	64.00	0.686	5.97	7.35
precomp (lzma)		34,514,096	1.27×	18.54	2.37	6.07	7.23
HX Complete	*	41,766,818	1.05×	69.03	0.636	8.37	5.24
bzip3		41,779,658	1.05×	8.57	5.12	8.95	4.91
BSC		41,965,245	1.05×	7.98	5.50	10.20	4.30
7-Zip PPMd (-mx9 -m0=PPMd)		42,474,873	1.03×	14.19	3.09	15.79	2.78
HX Fast	*	43,475,294	1.01×	46.88	0.936	0.287	152.9
zstd -19		43,557,789	1.01×	7.94	5.53	0.190	231.0
gzip -9		43,608,331	1.01×	2.13	20.60	0.460	95.41
7-Zip -mx9 (LZMA2)		43,640,450	1.01×	3.27	13.42	1.41	31.15
Brotli 11		43,657,923	1.01×	43.24	1.01	0.264	166.2
xz -9e		43,663,544	1.01×	15.17	2.89	1.23	35.83
LZ4 (default)		43,867,703	1.00×	0.180	243.8	0.175	250.8

CORPUS: GIF and TIFF files

GIF and TIFF files - same-machine campaign 3-4 Nov 2026
  machine: cloud, 2 cores (busy), wall clock
  original: 2,943,033 bytes in 6 file(s)
  source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
precomp (lzma)		2,120,932	1.39×	1.48	1.99	0.204	14.43
precomp -cn + HX Complete	r	2,134,770	1.38×	6.20	0.475	0.575	5.12
HX Complete	*	2,265,234	1.30×	4.57	0.644	0.485	6.07
bzip3		2,265,270	1.30×	0.620	4.75	0.519	5.67
Brotli 11		2,285,286	1.29×	6.42	0.458	0.049	60.06
xz -9e		2,305,136	1.28×	1.25	2.35	0.178	16.53
7-Zip -mx9 (LZMA2)		2,307,804	1.28×	0.280	10.51	0.197	14.94
BSC		2,311,698	1.27×	0.510	5.77	0.613	4.80
7-Zip PPMd (-mx9 -m0=PPMd)		2,344,644	1.26×	0.730	4.03	0.827	3.56
HX Fast	*	2,381,457	1.24×	3.20	0.920	0.039	75.46
zstd -19		2,477,984	1.19×	0.600	4.91	0.033	89.18
gzip -9		2,520,202	1.17×	0.180	16.35	0.047	62.62
LZ4 (default)		2,819,353	1.04×	0.030	98.10	0.026	113.2

CORPUS: Raw video (Y4M; frames decoded from H.264 sources + one raw test clip)

Raw video (Y4M; frames decoded from H.264 sources + one raw test clip) - same-machine campaign 3-4 Nov 2026
  machine: cloud, 2 cores (busy), wall clock
  original: 128,594,781 bytes in 4 file(s)
  source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
x264 lossless (veryslow)		7,628,505	16.86×	9.21	13.96	1.16	110.7
frames checked; decode time includes wri							
HX Complete + specialists + x264	r	7,629,062	16.86×	206.4	0.623	1.47	87.30
format specialist							
HX Complete + specialists	*	11,101,193	11.58×	197.9	0.650	1.94	66.39
FFV1 (level 3)		11,104,477	11.58×	1.61	79.87	1.61	79.97
frames checked; decode time includes wri							
HX Fast	*	12,282,763	10.47×	59.86	2.15	1.00	128.2
Sets + Exchange lines							
HX Complete	*	12,282,763	10.47×	76.53	1.68	0.998	128.9
Sets + Exchange lines							
bzip3		15,860,095	8.11×	9.37	13.72	9.55	13.47
BSC		16,015,970	8.03×	5.69	22.60	5.05	25.44

xz -9e	16,194,660	7.94×	127.0	1.01	1.55	83.07
7-Zip PPMd (-mx9 -m0=PPMd)	16,589,146	7.75×	15.60	8.24	16.10	7.99
7-Zip -mx9 (LZMA2)	16,661,105	7.72×	39.74	3.24	1.39	92.65
Brotli 11	17,452,940	7.37×	286.5	0.449	0.523	245.9
zstd -19	18,826,551	6.83×	70.87	1.81	0.329	390.9
MagicYUV -pred median (FFmpeg)	22,804,095	5.64×	0.639	201.2	0.667	192.8
MagicYUV -pred gradient (FFmpeg)	22,894,267	5.62×	0.714	180.0	0.439	293.1
gzip -9	25,095,840	5.12×	21.21	6.06	0.819	157.0
Ut Video	28,716,960	4.48×	0.490	262.4	0.456	282.0
frames checked; decode time includes wri						
LZ4 (default)	39,957,839	3.22×	0.280	459.3	0.186	691.4
x265 lossless (slow)	11,149,138	11.53×	23.00	5.59	1.44	89.09
partial: 3 of 4 files (128,563,974 B) - not ranked						

CORPUS: Compressed video files (H.264, HEVC, VP9)

Compressed video files (H.264, HEVC, VP9) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 1,223,197 bytes in 4 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
Brotli 11		1,171,889	1.04×	3.32	0.368	0.022	55.60
kanzi -l 9		1,174,018	1.04×	2.87	0.426	2.04	0.600
zstd -19		1,174,177	1.04×	0.200	6.12	0.023	53.18
HX Fast	*	1,174,182	1.04×	1.97	0.621	0.010	122.3
HX Complete	*	1,174,182	1.04×	2.55	0.480	0.010	122.3
gzip -9		1,175,885	1.04×	0.060	20.39	0.022	55.60
BSC		1,178,032	1.04×	0.210	5.82	0.242	5.05
bzip3		1,180,468	1.04×	0.220	5.56	0.206	5.94
xz -9e		1,182,280	1.03×	0.660	1.85	0.068	17.99
7-Zip -mx9 (LZMA2)		1,182,634	1.03×	0.130	9.41	0.077	15.89
LZ4 (default)		1,184,108	1.03×	0.020	61.16	0.017	71.95
7-Zip PPMd (-mx9 -m0=PPMd)		1,207,923	1.01×	0.400	3.06	0.447	2.74

CORPUS: PDF files from the pdf.js test set

PDF files from the pdf.js test set - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 24,453,200 bytes in 40 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
H-Deflate + HX Complete + image r	*	14,502,416	1.69×	252.1	0.097	3.89	6.29
precomp deflate-only + HX Complet	r	14,582,469	1.68×	226.6	0.108	4.45	5.50
precomp -cn + HX Complete	r	14,700,932	1.66×	77.19	0.317	3.81	6.41
precomp (lzma)		14,751,756	1.66×	21.87	1.12	2.95	8.28
HX Complete	*	18,071,735	1.35×	46.89	0.522	1.20	20.43
bzip3		18,205,173	1.34×	4.17	5.86	4.04	6.05
Brotli 11		18,243,414	1.34×	58.65	0.417	0.327	74.78
BSC		18,263,948	1.34×	3.90	6.27	3.91	6.25
7-Zip -mx9 (LZMA2)		18,290,235	1.34×	2.79	8.76	0.904	27.05
xz -9e		18,290,684	1.34×	10.77	2.27	0.828	29.53
HX Fast	*	18,380,607	1.33×	36.81	0.664	0.138	177.2
zstd -19		18,381,696	1.33×	5.51	4.44	0.189	129.4
7-Zip PPMd (-mx9 -m0=PPMd)		18,562,596	1.32×	6.99	3.50	7.52	3.25
gzip -9		19,072,184	1.28×	2.05	11.93	0.312	78.38
LZ4 (default)		20,100,865	1.22×	0.200	122.3	0.189	129.4

CORPUS: Packet captures, public pcapng files

Packet captures, public pcapng files - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 25,037,740 bytes in 73 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		17,642,423	1.42×	51.08	0.490	47.65	0.525
Brotli 11		17,914,569	1.40×	74.75	0.335	0.432	57.96

xz -9e		18,073,208	1.39×	13.15	1.90	1.59	15.72
7-Zip -mx9 (LZMA2)		18,082,068	1.38×	2.88	8.69	1.65	15.18
HX Complete	*	18,180,352	1.38×	40.04	0.625	2.62	9.57
HX Fast	*	18,323,393	1.37×	27.99	0.895	0.242	103.5
zstd -19		18,329,902	1.37×	8.25	3.03	0.341	73.42
bzip3		18,351,424	1.36×	5.46	4.59	5.32	4.70
BSC		18,388,276	1.36×	4.91	5.10	5.02	4.98
7-Zip PPMd (-mx9 -m0=PPMd)		18,679,977	1.34×	9.00	2.78	9.96	2.51
gzip -9		19,426,801	1.29×	1.87	13.39	0.427	58.64
LZ4 (default)		20,766,840	1.21×	0.290	86.34	0.277	90.39

CORPUS: Packet capture, one 113 MB merged pcap

Packet capture, one 113 MB merged pcap - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 112,917,841 bytes in 1 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
7-Zip -mx9 (LZMA2)		81,457,377	1.39×	14.00	8.07	5.82	19.40
HX Fast	*	81,974,307	1.38×	93.38	1.21	0.270	418.2
repeat line							
HX Complete	*	81,974,307	1.38×	157.1	0.719	0.243	464.7
repeat line							
BSC		86,709,224	1.30×	24.89	4.54	20.89	5.41
bzip3		88,928,019	1.27×	23.95	4.71	24.87	4.54
Brotli 11		89,534,946	1.26×	429.1	0.263	1.14	98.88
xz -9e		89,584,976	1.26×	91.29	1.24	6.16	18.34
zstd -19		90,900,415	1.24×	42.59	2.65	0.315	358.5
7-Zip PPMd (-mx9 -m0=PPMd)		92,313,955	1.22×	63.03	1.79	67.68	1.67
gzip -9		93,935,040	1.20×	4.98	22.67	0.968	116.7
LZ4 (default)		98,330,093	1.15×	0.130	868.6	0.132	855.4

CORPUS: Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data)

Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 23,473,678 bytes in 177 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
H-Deflate + HX Complete + image r	*	11,405,352	2.06×	335.2	0.070	5.22	4.50
precomp -cn + HX Complete	r	12,091,505	1.94×	104.7	0.224	5.49	4.27
precomp (lzma)		12,217,429	1.92×	30.23	0.777	3.66	6.42
precomp deflate-only + HX Complet	r	12,237,507	1.92×	277.0	0.085	5.73	4.10
HX Complete	*	15,314,128	1.53×	45.60	0.515	1.64	14.32
Brotli 11		15,328,058	1.53×	60.28	0.389	0.726	32.33
xz -9e		15,366,712	1.53×	13.57	1.73	1.22	19.18
7-Zip -mx9 (LZMA2)		15,372,299	1.53×	3.68	6.38	1.62	14.46
HX Fast	*	15,445,428	1.52×	36.01	0.652	0.417	56.29
zstd -19		15,511,705	1.51×	5.50	4.27	0.634	37.02
bzip3		15,522,734	1.51×	3.93	5.97	3.92	5.98
BSC		15,741,533	1.49×	4.01	5.85	4.58	5.13
7-Zip PPMd (-mx9 -m0=PPMd)		16,177,029	1.45×	6.97	3.37	7.40	3.17
gzip -9		16,443,522	1.43×	2.38	9.86	0.726	32.33
LZ4 (default)		17,322,346	1.36×	0.720	32.60	0.665	35.30

CORPUS: SQLite database (Chinook)

SQLite database (Chinook) - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 1,007,616 bytes in 1 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt; kanzi / Lepton / Monkey's Audio / OptimFROG / MagicYUV rows: rivals\rival3_results.csv

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
kanzi -l 9		190,841	5.28×	1.33	0.760	0.973	1.04
7-Zip -mx9 (LZMA2)		216,977	4.64×	0.100	10.08	0.025	40.30
xz -9e		221,336	4.55×	0.430	2.34	0.030	33.59
precomp (lzma)		221,881	4.54×	0.380	2.65	0.033	30.53

Brotli 11		231,468	4.35×	4.33	0.233	0.012	83.97
bzip3		273,889	3.68×	0.150	6.72	0.121	8.33
HX Complete	*	273,894	3.68×	2.12	0.475	0.122	8.26
bzip3							
precomp -cn + HX Complete	r	273,914	3.68×	2.11	0.478	0.111	9.08
bzip3							
BSC		289,356	3.48×	0.120	8.40	0.086	11.72
zstd -19		299,154	3.37×	0.280	3.60	0.006	167.9
HX Fast	*	299,324	3.37×	1.69	0.596	0.005	201.5
repeat line							
7-Zip PPMd (-mx9 -m0=PPMd)		318,497	3.16×	0.150	6.72	0.148	6.81
gzip -9		400,247	2.52×	0.460	2.19	0.015	67.17
LZ4 (default)		543,288	1.85×	0.010	100.8	0.006	167.9

CORPUS: Archives: ZIP, TAR, TAR.GZ

Archives: ZIP, TAR, TAR.GZ - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 36,095,732 bytes in 7 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
precomp -cn + HX Complete	r	21,937,054	1.65×	148.9	0.242	21.88	1.65
bzip3							
precomp deflate-only + HX Complet	r	21,939,313	1.65×	217.7	0.166	21.76	1.66
bzip3							
H-Deflate + HX Complete + image	r *	21,947,789	1.64×	357.6	0.101	15.30	2.36
precomp (lzma)		23,768,901	1.52×	58.16	0.621	13.78	2.62
HX Complete	*	28,100,846	1.28×	49.53	0.729	3.47	10.41
BSC		28,114,459	1.28×	5.79	6.23	3.95	9.15
Brotli 11		28,128,248	1.28×	57.91	0.623	0.167	216.1
bzip3		28,149,251	1.28×	7.12	5.07	7.38	4.89
xz -9e		28,167,808	1.28×	15.22	2.37	0.358	100.8
7-Zip -mx9 (LZMA2)		28,176,976	1.28×	4.22	8.55	0.419	86.15
HX Fast	*	28,182,146	1.28×	34.85	1.04	0.097	372.1
zstd -19		28,182,262	1.28×	10.88	3.32	0.114	316.6
7-Zip PPMd (-mx9 -m0=PPMd)		28,386,082	1.27×	12.65	2.85	14.37	2.51
gzip -9		28,835,961	1.25×	1.94	18.61	0.275	131.3
LZ4 (default)		29,817,059	1.21×	0.120	300.8	0.097	372.1

CORPUS: Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.coli), FITS, NumPy

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Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.coli), FITS, NumPy - same-machine campaign 3-4 Nov 2026

machine: cloud, 2 cores (busy), wall clock

original: 192,360,420 bytes in 23 file(s)

source: plans\All-Units-vs-All-Units-2026-11-04.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
precomp -cn + HX Complete	r	70,654,674	2.72×	451.6	0.426	17.52	10.98
HX Complete + audio/colour lines	*	70,655,868	2.72×	452.0	0.426	17.32	11.11
bzip3		71,805,265	2.68×	22.21	8.66	22.80	8.44
BSC		72,637,134	2.65×	18.34	10.49	19.17	10.03
7-Zip -mx9 (LZMA2)		72,897,638	2.64×	28.77	6.69	5.07	37.95
xz -9e		73,422,812	2.62×	104.1	1.85	4.80	40.04
precomp (lzma)		73,462,177	2.62×	80.00	2.40	4.86	39.60
Brotli 11		73,674,604	2.61×	463.6	0.415	1.42	135.6
7-Zip PPMd (-mx9 -m0=PPMd)		74,521,490	2.58×	47.78	4.03	52.17	3.69
HX Fast + audio/colour lines	*	75,145,212	2.56×	396.6	0.485	0.769	250.1
zstd -19		77,282,914	2.49×	58.62	3.28	0.545	353.0
gzip -9		81,571,333	2.36×	28.69	6.70	1.54	125.0
LZ4 (default)		100,134,791	1.92×	0.500	384.7	0.423	454.8

CORPUS: Silesia 4 GB

Silesia repeated to 4 GB (4,026,833,020 B) - hs_zstd race (1 Nov 2026)

machine: desktop 12 threads, decode median of 5

original: 4,026,833,020 bytes in 1 file(s)

source: hs-zstd-race-4gb-2026-11-01\results-silesia-4gb.txt

Compress times as recorded by the race script; zstd --long decode swung widely between rounds.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zstd -3 --long=27 -T12		1,253,900,488	3.21×	4.76	846.0	3.79	1,062
zstd -3, one thread		1,264,836,194	3.18×	14.71	273.7	3.73	1,080
zstd -3 -T12		1,264,836,194	3.18×	2.93	1,374	4.25	947.5
hs_zstd plain-blocks control 16 M	*	1,265,088,590	3.18×	3.24	1,243	2.39	1,685
hs_zstd 1.2 rule, 16 MB	*	1,265,608,808	3.18×	3.13	1,287	2.39	1,685
hs_zstd 1.3 (Sets, raw off, safe)	*	1,265,694,589	3.18×	3.50	1,151	2.51	1,604
pzstd -3 -p 12		1,268,506,060	3.17×	4.09	984.6	2.85	1,413
hs_zstd plain-blocks control 4 MB	*	1,269,379,994	3.17×	3.07	1,312	2.20	1,830
hs_zstd 1.3 (Sets, raw off, safe)	*	1,270,027,824	3.17×	2.94	1,370	2.27	1,774

CORPUS: Mixed 3.2 GB tar

Mixed public tar 3.2 GB (3,179,306,496 B: Linux, Blender, pdf.js, NYT CSV) - hs_zstd race

machine: desktop 12 threads, decode median of 5

original: 3,179,306,496 bytes in 1 file(s)

source: hs-zstd-race-4gb-2026-11-01\results-mixed-3.2gb.txt

Compress times as recorded by the race script; zstd --long decode swung widely between rounds.

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zstd -3 --long=27 -T12		741,136,596	4.29×	3.26	975.2	7.00	454.2
hs_zstd 1.3 (Sets, raw off, safe)	*	776,824,933	4.09×	2.16	1,472	1.75	1,817
zstd -3, one thread		778,820,145	4.08×	7.90	402.4	2.23	1,426
zstd -3 -T12		778,820,145	4.08×	2.17	1,465	2.19	1,452
hs_zstd plain-blocks control 16 M	*	779,945,863	4.08×	2.14	1,486	1.67	1,904
hs_zstd 1.2 rule, 16 MB	*	782,627,226	4.06×	2.21	1,439	1.72	1,848
pzstd -3 -p 12		784,902,640	4.05×	2.83	1,123	1.92	1,656
hs_zstd 1.3 (Sets, raw off, safe)	*	785,946,871	4.05×	1.90	1,673	1.75	1,817
hs_zstd plain-blocks control 4 MB	*	787,198,376	4.04×	1.97	1,614	1.78	1,786

CORPUS: Plant data

Industrial plant-data exports, tar (163,870,720 B, 7 public UCI/NAB data sets) - 1 Nov 2026

machine: cloud, 2 cores

original: 163,870,720 bytes in 7 file(s)

source: plant-data-2026-11-01\results-plant.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6.1 Harrington's Sets 16 M	*	17,675,237	9.27×	240.7	0.681	62.11	2.64
HCM v3.6.1 fast (one core)	*	17,987,330	9.11×	638.9	0.256	120.7	1.36
xz -9		21,919,100	7.48×	161.4	1.02	2.21	74.15
zstd -19		22,989,339	7.13×	135.4	1.21	0.340	482.0
zstd -19 -T2		22,989,339	7.13×	79.60	2.06	0.220	744.9
hs_zstd -19 plain blocks 16 MB	*	23,010,511	7.12×	67.60	2.42	0.290	565.1
hs_zstd -19 Harrington's Sets 16	*	23,052,992	7.11×	65.40	2.51	0.280	585.3
hs_zstd -19 plain blocks 4 MB	*	23,372,624	7.01×	51.20	3.20	0.270	606.9
hs_zstd -19 Harrington's Sets 4 M	*	23,375,395	7.01×	52.40	3.13	0.240	682.8
gzip -9		30,563,416	5.36×	94.50	1.73	2.05	79.94

CORPUS: Seismic

Seismic miniSEED, 8 EarthScope station-days (132,250,680 B as int32 samples) - 1 Nov 2026

machine: cloud, 2 cores

original: 132,250,680 bytes in 8 file(s)

source: bank-seismic-2026-11-01\results-seismic.txt

Steim2 compress time not recorded (archive files as published).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6.1 on raw samples	*	28,594,207	4.63×	569.9	0.232	145.0	0.912
difference+shuffle, xz -9		31,139,192	4.25×	86.70	1.53	3.03	43.65
difference+shuffle, HCM v3.6.1	*	33,443,709	3.95×	379.3	0.349	101.5	1.30
difference+shuffle, zstd -19		34,782,226	3.80×	66.50	1.99	0.750	176.3
xz -9 on raw samples		38,160,844	3.47×	120.6	1.10	3.93	33.65
Steim2 in miniSEED (archive stand		42,413,056	3.12×	n/r	n/r	0.390	339.1
zstd -19 on raw samples		43,266,256	3.06×	82.90	1.60	0.580	228.0
record delta + xz -9	r	49,853,596	2.65×	106.3	1.24	4.24	31.19
record delta + zstd -19	r	58,013,095	2.28×	79.70	1.66	1.06	124.8

CORPUS: Bank data

Bank data: ACH files tar (64,143,360 B, synthetic NACHA) - 1 Nov 2026
  machine: cloud, 2 cores
  original: 64,143,360 bytes in 1 file(s)
  source: bank-seismic-2026-11-01\results-bank.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6.1 fast	*	10,051,902	6.38×	229.3	0.280	59.34	1.08
xz -9		12,875,868	4.98×	91.60	0.700	1.43	44.86
zstd -19		13,577,145	4.72×	70.00	0.916	0.160	400.9
record delta + xz -9	r	16,466,516	3.90×	85.50	0.750	1.83	35.05
record delta + zstd -19	r	17,707,264	3.62×	68.10	0.942	0.400	160.4
gzip -9		18,378,293	3.49×	16.60	3.86	0.770	83.30

Bank data: ATM electronic journals tar (55,552,000 B, synthetic) - 1 Nov 2026
  machine: cloud, 2 cores
  original: 55,552,000 bytes in 1 file(s)
  source: bank-seismic-2026-11-01\results-bank.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6.1 fast	*	1,945,687	28.55×	152.9	0.363	41.01	1.35
zstd -19		3,634,839	15.28×	65.90	0.843	0.090	617.2
xz -9		3,723,224	14.92×	35.40	1.57	0.470	118.2
gzip -9		4,910,341	11.31×	2.00	27.78	0.320	173.6
record delta + xz -9	r	4,921,624	11.29×	32.50	1.71	0.700	79.36
record delta + zstd -19	r	5,697,147	9.75×	52.20	1.06	0.270	205.7

Bank data: Mixed bank backup tar (119,685,120 B, synthetic) - 1 Nov 2026
  machine: cloud, 2 cores
  original: 119,685,120 bytes in 1 file(s)
  source: bank-seismic-2026-11-01\results-bank.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
HCM v3.6.1 fast	*	12,126,393	9.87×	385.9	0.310	102.2	1.17
xz -9		16,950,308	7.06×	126.3	0.948	1.87	64.00
zstd -19		17,413,034	6.87×	140.1	0.854	0.210	569.9
record delta + xz -9	r	22,761,988	5.26×	115.2	1.04	2.66	44.99
gzip -9		23,367,914	5.12×	18.90	6.33	0.840	142.5
record delta + zstd -19	r	25,334,678	4.72×	111.6	1.07	0.710	168.6

CORPUS: Silesia 10 GB

Silesia repeated to 10 GB (10,000,000,000 B) - gzip -6 alone vs hs_wrap 1.4 (31 Oct 2026)
  machine: desktop 12 jobs, 16 MB blocks, average of 2
  original: 10,000,000,000 bytes in 1 file(s)
  source: HCM-tests\gzip-direct-2026-10-31\results.txt

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
gzip -6 (zlib 1.3.1 minigzip)		3,222,055,858	3.10×	283.3	35.30	35.00	285.7
hs_wrap 1.4 + gzip -6	*	3,223,099,680	3.10×	61.60	162.3	10.80	925.9

CORPUS: Silesia 0.5 GB

Silesia repeated to 0.5 GB (500,000,000 B) - Tier-1 check (31 Oct 2026)
  machine: desktop 12 jobs, decode median of 3
  original: 500,000,000 bytes in 1 file(s)
  source: HCM-tests\lz4-tier-2026-10-31\results.txt
  The zstd row is from the second run in the same file (its own LZ4 median also 0.42 s).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zstd 1.5.6 default		159,257,968	3.14×	1.70	294.1	0.480	1,042
hs_wrap 1.4 + gzip -6	*	163,598,643	3.06×	3.50	142.9	0.610	819.7
LZ4 1.10.0 default		240,984,906	2.07×	0.400	1,250	0.420	1,190

CORPUS: Silesia 2 GB

Silesia repeated to 2 GB - hs_zstd vs pzstd (31 Oct 2026)
  machine: desktop 12 threads, decode median
  original: 2,000,000,000 bytes in 1 file(s)
  source: HCM-tests\pzstd-2026-10-31\results.txt + memory /areas/combary.md
  Compression times not carried into the summary; 10 GB run of the same race: pzstd 6.87 s vs
  hs_zstd 4 MB 4.00 s decode (1.72x).

tool	g	compressed B	ratio	comp s	comp MB/s	dec s	dec MB/s
zstd -3 (1 thread)		~626,668,000	3.19×	n/r	n/r	1.67	1,198
hs_zstd 1.1, 16 MB blocks	*	~627,304,000	3.19×	n/r	n/r	0.690	2,899
pzstd -3 -p 12		~628,524,000	3.18×	n/r	n/r	0.940	2,128
hs_zstd 1.1, 4 MB blocks	*	~629,122,000	3.18×	n/r	n/r	0.630	3,175

3. HX - ALL HX FAST AND HX COMPLETE RESULTS

What they are. HX splits a file into independent segments of about 1 MB, each decoded on its own core; for every segment it tries the Harrington Exchange lines (byte lanes, 12-bit samples, context lines, audio predictors), Harrington's Sets and the cascade's repeat line, and keeps the smallest. HX Fast is the speed-first mode. HX Complete adds whole-file and split routes with BSC, bzip3, LZMA (BCJ2 for x86), PPMd and Brotli, the x86 filter, and a learned router. The per-slice build (routing slice by slice) is under construction and has no results yet. Source-file build labels: hxf 8 / 9 / 11 = HX Fast builds, hxc / 11 / 12 / 13b = HX Complete builds, hx15 = HX Complete with the learned router, hx / hxs = the HX lines bench (entbench).

3.1 SILESIA, PER FILE (cloud, 2 threads, wall clock; decode best of 3; all exact)

file	orig B	HX Fast	ratio	dec s	HX Compl.	ratio	dec s	chose
dickens	10,192,446	2,849,388	3.58×	0.034	2,233,951	4.56×	1.473	bzip3
mozilla	51,220,480	14,973,527	3.42×	0.153	14,973,527	3.42×	0.155	repeat line
mr	9,970,564	2,069,305	4.82×	0.066	2,069,305	4.82×	0.065	Sets + Exch
nci	33,553,445	1,617,072	20.75×	0.057	1,210,339	27.72×	0.654	BSC
ooffice	6,152,192	2,240,864	2.75×	0.028	2,123,508	2.90×	0.863	x86 filter + b
osdb	10,085,684	3,099,946	3.25×	0.047	2,224,783	4.53×	0.737	BSC
reymont	6,627,202	1,348,534	4.91×	0.021	980,476	6.76×	0.970	bzip3
samba	21,606,400	3,878,607	5.57×	0.050	3,878,607	5.57×	0.049	repeat line
sao	7,251,944	5,000,585	1.45×	0.035	4,673,324	1.55×	1.346	bzip3
webster	41,458,703	8,592,737	4.82×	0.120	6,448,535	6.43×	2.111	BSC
xml	5,345,280	452,811	11.80×	0.014	393,183	13.59×	0.139	BSC
x-ray	8,474,240	3,651,990	2.32×	0.061	3,651,990	2.32×	0.062	Sets + Exch
TOTAL	211,938,580	49,775,366	4.26×	0.69	44,861,528	4.72×	8.62	

Compress totals: HX Fast 235.5 s, HX Complete 279.2 s. ooffice rows use the x86-filter build (HX Fast 2,240,864 B, HX Complete 2,123,508 B; without the filter 2,594,986 / 2,526,899 B). Set totals in the same run: mzip 4.92× (dec 24.76 s, comp 2,209.4 s), bzip3 46,426,674 B = 4.57× (28.03 s), BSC 4.55× (13.66 s). Source: HX-Complete-and-HX-Fast-9-Silesia-2026-11-03.txt, hxc_*.csv.

3.2 x86 FILTER ON ooffice (6,152,192 B; cloud) source: hxc_ooffice_x86_filter.csv

HX Fast + x86 filter	2,240,864 B	2.75×	comp	9.48 s	dec	0.028 s
HX Complete (x86 filter + bzip3)	2,123,508 B	2.90×	comp	9.85 s	dec	0.863 s
BSC	2,559,098 B	2.40×	comp	0.67 s	dec	0.665 s
bzip3	2,526,894 B	2.43×	comp	0.94 s	dec	0.921 s

Without the filter: HX Fast 2,594,986 B (2.37×), HX Complete 2,526,899 B (2.43×).

3.3 CALGARY, CANTERBURY, CANTERBURY LARGE, SYNTHETIC - PER FILE (cloud, 2 threads)

source: HU-pretrain-results\hx-complete\hxc_calgary_canterbury_large_synthetic.csv

Calgary

file	orig B	HX Fast	ratio	dec s	HX Compl.	ratio	dec s	chose
bib	111,261	31,948	3.48×	0.004	26,005	4.28×	0.012	BSC
book1	768,771	264,778	2.90×	0.006	211,428	3.64×	0.091	bzip3
book2	610,856	173,862	3.51×	0.005	145,337	4.20×	0.042	BSC
geo	102,400	55,669	1.84×	0.005	51,930	1.97×	0.017	bzip3
news	377,109	122,711	3.07×	0.005	109,441	3.45×	0.048	bzip3
obj1	21,504	9,842	2.18×	0.004	9,842	2.18×	0.004	repeat line
obj2	246,814	70,232	3.51×	0.005	70,232	3.51×	0.005	repeat line
paper1	53,161	17,630	3.02×	0.004	15,775	3.37×	0.009	BSC
paper2	82,199	27,570	2.98×	0.004	23,737	3.46×	0.011	BSC
paper3	46,526	17,196	2.71×	0.004	15,127	3.08×	0.009	BSC
paper4	13,286	5,382	2.47×	0.004	5,051	2.63×	0.008	BSC
paper5	11,954	4,878	2.45×	0.004	4,705	2.54×	0.007	BSC
paper6	38,105	12,741	2.99×	0.004	11,743	3.24×	0.009	BSC
pic	513,216	43,627	11.76×	0.005	43,627	11.76×	0.005	repeat line
progc	39,611	12,803	3.09×	0.004	11,995	3.30×	0.009	BSC
progl	71,646	15,164	4.72×	0.005	14,685	4.88×	0.011	BSC
progp	49,379	10,463	4.72×	0.004	9,893	4.99×	0.008	BSC
trans	93,695	17,306	5.41×	0.004	16,217	5.78×	0.010	BSC
TOTAL	3,251,493	913,802	3.56×	0.08	796,770	4.08×	0.32	

compress totals: HX Fast 8.5 s, HX Complete 9.1 s

Canterbury

file	orig B	HX Fast	ratio	dec s	HX Compl.	ratio	dec s	chose

alice29.txt	152,089	49,224	3.09×	0.004	40,243	3.78×	0.015	BSC
asyoulik.txt	125,179	45,150	2.77×	0.004	37,239	3.36×	0.016	BSC
cp.html	24,603	7,726	3.18×	0.004	7,331	3.36×	0.007	BSC
fields.c	11,150	3,028	3.68×	0.004	2,961	3.77×	0.007	BSC
grammar.lsp	3,721	1,224	3.04×	0.004	1,224	3.04×	0.004	repeat line
kennedy.xls	1,029,744	64,838	15.88×	0.006	64,838	15.88×	0.007	repeat line
lcet10.txt	426,754	121,349	3.52×	0.005	99,409	4.29×	0.049	bzip3
plrabn12.txt	481,861	167,451	2.88×	0.005	134,631	3.58×	0.055	bzip3
ptt5	513,216	43,627	11.76×	0.005	43,627	11.76×	0.005	repeat line
sum	38,240	11,117	3.44×	0.004	11,117	3.44×	0.004	repeat line
xargs.1	4,227	1,737	2.43×	0.004	1,733	2.44×	0.006	BSC
TOTAL	2,810,784	516,471	5.44×	0.05	444,353	6.33×	0.18	

compress totals: HX Fast 6.5 s, HX Complete 7.0 s

Canterbury large

file	orig B	HX Fast	ratio	dec s	HX Compl.	ratio	dec s	chose
E.coli	4,638,690	1,151,097	4.03×	0.012	1,125,311	4.12×	0.709	bzip3
bible.txt	4,047,392	892,268	4.54×	0.013	726,176	5.57×	0.573	bzip3
world192.txt	2,473,400	503,914	4.91×	0.009	398,223	6.21×	0.121	BSC
TOTAL	11,159,482	2,547,279	4.38×	0.03	2,249,710	4.96×	1.40	

compress totals: HX Fast 17.9 s, HX Complete 20.6 s

Synthetic (ours)

file	orig B	HX Fast	ratio	dec s	HX Compl.	ratio	dec s	chose
dup_random_chunk	50,331,648	20,746,659	2.43×	0.092	19,564,686	2.57×	5.816	Exch + BSC
floats_counters.	34,000,000	27,988,484	1.21×	0.169	27,988,484	1.21×	0.152	Sets + Exch
hi16_mix.bin	23,047,392	12,267,317	1.88×	0.078	12,152,025	1.90×	0.784	Exch + BSC
random20.bin	20,000,000	20,000,013	1.00×	0.039	20,000,013	1.00×	0.042	stored
switch_1mb.bin	25,165,824	11,957,769	2.10×	0.084	11,957,769	2.10×	0.074	Exch + repe
TOTAL	152,544,864	92,960,242	1.64×	0.46	91,662,977	1.66×	6.87	

compress totals: HX Fast 115.5 s, HX Complete 209.5 s

3.4 HX ON EVERY DATA SET OF THE 3-4 NOV CAMPAIGN (cloud; best HX Fast and best HX Complete row per set)

source: plans\All-Units-vs-All-Units-2026-11-04.txt (full tables in section 2)

set	HX Fast	dec MB/s	HX Compl.	dec MB/s	best other	ratio
Biased test files, 8 MB each	2.02×	152.5	2.02×	148.1	lpaq1 -9	2.02×
NIST SP 800-90B real noise s	2.49×	94.59	2.49×	54.26	lpaq1 -9	2.49×
Random Turing machine output	1.92×	238.4	2.13×	7.22	lpaq1 8	2.13×
Text: server logs (16 system	7.46×	106.2	8.62×	19.27	kanzi -l 9	10.01×
Programs: Windows x64 EXE (z	3.04×	201.6	3.01×	8.60	kanzi -l 9	3.96×
Silesia, 12 files	4.26×	308.9	4.72×	24.58	kanzi -l 9	5.20×
Calgary corpus, 18 files	3.56×	40.64	4.08×	10.32	kanzi -l 9	4.44×
Canterbury corpus, 11 files	5.44×	57.36	6.33×	16.06	kanzi -l 9	6.81×
Canterbury large corpus, 3 f	4.38×	328.2	4.96×	7.95	kanzi -l 9	5.16×
Synthetic test files, 5 file	1.64×	330.2	1.66×	22.21	7-Zip -mx9 (LZM	1.65×
Speech, WAVs, 8 kHz mono (FS	1.91×	101.9	1.92×	36.39	OptimFROG --pre	2.07×
Music and sounds, WAVs, 44.1	2.59×	78.08	2.59×	72.46	Monkey's Audio	2.85×
Environmental recordings, WA	2.35×	38.52	2.35×	37.96	WavPack -hhx6 (	2.34×
Already-compressed audio fil	1.01×	265.3	1.01×	7.93	precomp (lzma)	1.03×
Colour photos, raw RGB (Koda	1.95×	83.27	2.83×	13.64	JPEG XL lossles	2.83×
Rendered document pages / sc	30.48×	459.6	34.83×	68.51	7-Zip PPMd (-mx	36.22×
Medical DICOM files (imageio	3.15×	70.93	3.15×	70.93	bzip3	3.10×
PNG files (Kodak 24)	1.00×	149.5	1.54×	1.59	precomp (lzma)	1.08×
JPEG photos	1.01×	152.9	1.28×	2.27	packJPG	1.28×
GIF and TIFF files	1.24×	75.46	1.30×	6.07	precomp (lzma)	1.39×
Raw video (Y4M; frames decod	10.47×	128.2	16.86×	87.30	x264 lossless (	16.86×
Compressed video files (H.26	1.04×	122.3	1.04×	122.3	Brotli 11	1.04×
PDF files from the pdf.js te	1.33×	177.2	1.69×	6.29	precomp (lzma)	1.66×
Packet captures, public pcap	1.37×	103.5	1.38×	9.57	kanzi -l 9	1.42×
Packet capture, one 113 MB m	1.38×	418.2	1.38×	464.7	7-Zip -mx9 (LZM	1.39×
Office files: DOCX/XLSX/PPTX	1.52×	56.29	2.06×	4.50	precomp (lzma)	1.92×
SQLite database (Chinook)	3.37×	201.5	3.68×	8.26	kanzi -l 9	5.28×
Archives: ZIP, TAR, TAR.GZ	1.28×	372.1	1.64×	2.36	precomp (lzma)	1.52×
Scientific: climate NetCDF/G	2.56×	250.1	2.72×	11.11	bzip3	2.68×

3.5 HX FAST ON THE DESKTOP - 1 vs 12 THREADS, IN-MEMORY DECODE (3 Nov 2026, best of 3, all exact)

source: HU-pretrain-results\speed-desktop\HX-Fast-vs-mzip-Speed-2026-11-03.txt (sizes recorded as ratios)

file	orig B	ratio	dec 1 thr	dec 12 thr	MB/s 12	mzip ratio	mzip dec
x-ray	8,474,240	2.32×	0.0675s	0.0132s	642	2.32×	0.6707s
mr	9,970,564	4.82×	0.0639s	0.0138s	723	4.71×	0.7142s
E_16bit_hi_biased.bin	8,388,608	1.60×	0.0092s	0.0051s	1,645	1.60×	0.0486s
F_12bit_gauss_packed.bin	8,388,608	1.88×	0.0129s	0.0062s	1,353	1.87×	0.8242s

G_markov_bits.bin	8,388,608	2.75×	0.0514s	0.0121s	693		2.71×	0.6685s
D_bias_per_window.bin	8,388,608	1.57×	0.0326s	0.0158s	531		1.53×	0.2505s
ringOsc-nist.bin	1,000,000	12.56×	0.0040s	0.0044s	250		12.79×	0.0883s
H_zipf_with_repeats.bin	8,388,608	1.76×	0.0064s	0.0064s	1,311		1.76×	0.0215s
A_bits_p10.bin	8,388,608	2.13×	0.0486s	0.0130s	645		2.11×	0.6763s
sao	7,251,944	1.45×	0.0087s	0.0087s	834		1.64×	0.2005s
osdb	10,085,684	3.25×	0.0078s	0.0081s	1,293		4.48×	0.7804s
xml	5,345,280	11.80×	0.0031s	0.0031s	1,724		15.04×	0.2631s
dickens	10,192,446	3.58×	0.0101s	0.0095s	1,073		4.54×	0.6964s

HX Fast compress (desktop): x-ray 2.84 s, mr 3.30 s, biased files 2.0-3.9 s, sao 2.33 s, osdb 2.98 s, xml 1.70 s, dickens 4.28 s. mzip compress 12-75 s. HX Fast decodes 3x to 133x faster than mzip on every file here.

3.6 HX LINES vs THE ENTROPY CODERS - BIASED DATA WITH A KNOWN FLOOR (desktop, 128 MB files, in memory)

source: HU-pretrain-results\biased-desktop\Biased-Data-vs-Entropy-Coders-2026-11-03.txt

source	floor	best coder	ratio	MB/s	HX lines	MB/s 1 / 12 thr
E 16-bit, high byte biased	1.60×	Huff0	1.35×	1,915	1.60×	2,048 / 9,804
F 12-bit noise, packed	1.88×	FSE	1.32×	865	1.88×	880 / 6,328
G bits with memory (Markov)	2.76×	rANS	2.63×	223	2.75×	172 / 1,571
D bias changes every 64 KB	1.63×	adaptive arith	1.60×	23	1.62×	336 / 1,800

Floor = the exact entropy of the source, shown as the best possible ratio. D uses HX lines + Sets (its speeds are recorded as ~336 MB/s on 1 thread and ~1,800 MB/s on 12). Plain biased bits and Zipf bytes (A, B, C) tie the best coder on the floor. NIST ringOsc (1 MB, cloud): HX lines 12.56×, decode 4.4 ms, vs best coder FSE 8.00×; BSC 12.74× (71 ms) and bzip3 12.72× (102 ms).

3.7 HX FAST ON SPEECH vs FLAC (cloud, 6 FSDD WAVs, 20,997,112 B; decode best of 5)

source: HU-pretrain-results\audio-speed\FLAC-vs-HX-Fast-Speed-2026-11-03.txt (sizes recorded as ratios)

tuned FLAC (flac-hi, modified)	1.93×	comp	n/r s	dec 0.171 s (123 MB/s)
HX Fast + audio lines	1.89×	comp	79.54 s	dec 0.271 s (77 MB/s)
flac -8	1.88×	comp	0.380 s	dec 0.169 s (124 MB/s)

HX Fast build comparison (cloud, 22 files, 144,816,920 B): earlier build 2.23× comp 374 s dec 0.83 s; current build 2.24× comp 201 s dec 0.84 s (1.9x faster compress). Source: HX-Fast-8-vs-7-2026-11-03.txt.

3.8 THE LEARNED ROUTER (HX Complete, whole-file routing) - 83 training files + 10 unseen files

source: hx15\hx15_routes.csv replayed by hx15\hx15_eval.py (held-out by set; no compression re-run); hx15_live.csv

Try every route on all 83 files: 128,913,421 B (2.39×), route time 6,792 s. [ ] = files in pass 1 / 2 / 3.

router, plain	130,386,842 B	2.36×	4,529 s	(33.3% saved) size +1.143% passes 57 / 13 / 13
router + specialist rule	129,793,900 B	2.37×	4,369 s	(35.7% saved) size +0.683% passes 60 / 15 / 8
router + specialist + novelty rule	129,008,241 B	2.39×	5,029 s	(26.0% saved) size +0.074% passes 41 / 9 / 33

The same 0.074% / 0.683% figures and pass counts are recorded in the provisional draft (section 9.7).

Live run on files the router had never seen (learning on; each decoded and compared byte for byte):

set	file	orig B	HX B	ratio	comp s	dec s
bin	linux_python3.11	6,644,312	1,642,175	4.05×	67.68	8.201
text	csv_womens-world-cup-predictions_	3,142	872	3.60×	0.08	0.012
pcap_small	synscan.pcapng	185,072	22,023	8.40×	7.04	0.193
speech	yweweler.wav	2,833,384	1,312,492	2.16×	69.83	0.088
calgary	paper5	11,954	4,088	2.92×	0.48	0.004
canterbury	asyoulik.txt	125,179	34,150	3.67×	3.10	0.170
biased	ringOsc-nist.bin	1,000,000	78,477	12.74×	9.68	0.434
dicom	dicom_file90.dcm	68,404	66,791	1.02×	1.77	0.005
jpg	018.jpg	1,876,687	1,463,385	1.28×	34.95	0.850
coded	admiralbob77___Choice___Drum_bass	369,614	369,627	1.00×	4.58	0.006
TOTAL (10 of 30 listed files run)		13,117,748	4,994,080	2.63×	199.2	9.96

3.9 HX COMPLETE TRYING EVERY ROUTE (learned-router build, training run; cloud, 2 threads)

source: hx15\hx15_routes.csv (smallest route per file; time = sum of all route times; decode not recorded here)

set	file	orig B	HX Complete B	ratio	winning route
sqlite	Chinook_Sqlite.sqlite	1,007,616	158,875	6.34×	HCM v3.7 route
	set total	1,007,616	158,875	6.34×	route time 25 s
bin	linux_git	4,066,232	1,328,915	3.06×	HCM v3.7 route
bin	linux_libstdc++.so.6	2,592,224	608,767	4.26×	HCM v3.7 route

bin	win_bgzip.exe	1,014,272	378,732	2.68×	HCM v3.7 route
bin	win_pzstd.exe	1,128,960	364,537	3.10×	HCM v3.7 route
	set total	8,801,688	2,680,951	3.28×	route time 224 s
text	chinese_poetry_1.json	2,955	665	4.44×	HCM route
text	csv_fandango_fandango_score_	15,144	4,344	3.49×	HCM v3.7 route
text	csv_womens-world-cup-predict	2,315	506	4.58×	HCM v3.7 route
text	log_HealthApp_2k.log	187,456	8,837	21.21×	HCM v3.7 route
	set total	207,870	14,352	14.48×	route time 3 s
screen_raw	pg_issue11045-1.ppm	3,394,066	5,809	584.28×	media specialist
screen_raw	pg_issue11131_reduced-1.ppm	3,394,066	1,584	2142.72×	Brotli
screen_raw	tm_gray-02.pgm	2,103,767	68,994	30.49×	HCM v3.7 route
screen_raw	tm_gray-04.pgm	2,103,767	59,019	35.65×	HCM route
	set total	10,995,666	135,406	81.21×	route time 144 s
pcap_small	3e80211_wepauth.pcapng	1,544	899	1.72×	PPMd
pcap_small	dns_query_response.pcapng	280	167	1.68×	PPMd
pcap_small	icmpv6_neighbor_solicitation	1,048	276	3.80×	PPMd
pcap_small	passiveosfingerprinting.pcap	252	131	1.92×	PPMd
	set total	3,124	1,473	2.12×	route time 0 s
vcoded	cockatoo.mp4	728,751	680,447	1.07×	Brotli
vcoded	cockatoo_hevc.mp4	178,136	176,347	1.01×	Brotli
vcoded	cockatoo_vp9.webm	219,488	219,494	1.00×	Brotli
vcoded	realshort.mp4	96,822	95,631	1.01×	Brotli
	set total	1,223,197	1,171,919	1.04×	route time 35 s
speech	george.wav	3,533,784	2,290,195	1.54×	media specialist
speech	jackson.wav	4,131,724	2,603,285	1.59×	media specialist
speech	lucas.wav	4,593,732	2,464,053	1.86×	media specialist
speech	nicolas.wav	2,793,546	440,841	6.34×	HCM v3.7 route
	set total	15,052,786	7,798,374	1.93×	route time 375 s
music	147793__setuniman__sweet_wal	8,678,924	2,652,107	3.27×	media specialist
music	456440__inspectorj__bird_whi	476,080	185,006	2.57×	HX lines
music	Kevin_MacLeod___P_I_Tchaikov	21,146,156	7,894,574	2.68×	HX lines
music	drese_tempo.wav	2,330,084	904,221	2.58×	HX lines
	set total	32,631,244	11,635,908	2.80×	route time 823 s
gcb	uiq2_seed0.bin	6,515,209	3,061,282	2.13×	BSC
gcb	uiq2_seed1.bin	6,517,929	3,061,586	2.13×	BSC
gcb	uiq2_seed2.bin	6,519,743	3,062,904	2.13×	BSC
	set total	19,552,881	9,185,772	2.13×	route time 441 s
calgary	bib	111,261	22,627	4.92×	HCM v3.7 route
calgary	news	377,109	97,743	3.86×	HCM v3.7 route
calgary	paper2	82,199	21,231	3.87×	HCM route
calgary	paper6	38,105	10,271	3.71×	HCM v3.7 route
	set total	608,674	151,872	4.01×	route time 13 s
canterbury	alice29.txt	152,089	36,267	4.19×	HCM route
canterbury	cp.html	24,603	6,387	3.85×	HCM route
canterbury	grammar.lsp	3,721	1,038	3.58×	HCM route
canterbury	lcet10.txt	426,754	88,378	4.83×	HCM route
	set total	607,167	132,070	4.60×	route time 12 s
large	E.coli	4,638,690	1,122,061	4.13×	HCM route
large	bible.txt	4,047,392	696,783	5.81×	HCM v3.7 route
large	world192.txt	2,473,400	364,927	6.78×	HCM v3.7 route
	set total	11,159,482	2,183,771	5.11×	route time 212 s
silesia	dickens	10,192,446	2,179,294	4.68×	HCM v3.7 route
silesia	mr	9,970,564	1,777,401	5.61×	JPEG XL 16-bit image
silesia	ooffice	6,152,192	1,857,185	3.31×	HCM v3.7 route
silesia	osdb	10,085,684	2,224,770	4.53×	BSC
silesia	reymont	6,627,202	980,463	6.76×	bzip3
silesia	samba	21,606,400	3,339,453	6.47×	HCM v3.7 route
silesia	sao	7,251,944	3,890,324	1.86×	HCM v3.7 route
silesia	x-ray	8,474,240	3,547,196	2.39×	JPEG XL 16-bit image
silesia	xml	5,345,280	348,364	15.34×	HCM v3.7 route
	set total	85,705,952	20,144,450	4.25×	route time 1,559 s
esc	1-100032-A-0.wav	441,044	14,953	29.50×	media specialist
esc	1-13613-A-37.wav	441,044	245,265	1.80×	HX lines
esc	1-18655-A-31.wav	441,044	31,827	13.86×	media specialist
esc	1-27403-A-28.wav	441,044	235,258	1.87×	HX lines
	set total	1,764,176	527,303	3.35×	route time 35 s
synthetic	hi16_mix.bin	23,047,392	11,929,223	1.93×	HCM v3.7 route
synthetic	random20.bin	20,000,000	20,000,013	1.00×	HCM route
synthetic	switch_1mb.bin	25,165,824	11,744,815	2.14×	HCM route
	set total	68,213,216	43,674,051	1.56×	route time 1,717 s
biased	A_bits_p10.bin	8,388,608	3,940,241	2.13×	HX lines
biased	D_bias_per_window.bin	8,388,608	4,934,512	1.70×	HX lines
biased	G_markov_bits.bin	8,388,608	3,048,408	2.75×	HX lines
biased	biased-random-bytes.bin	1,000,000	290,359	3.44×	HX lines

	set total	26,165,824	12,213,520	2.14×	route time 495 s
dicom	dicom_file01.dcm	549,626	116,474	4.72×	JPEG XL 16-bit image
dicom	dicom_file02.dcm	1,591,962	526,957	3.02×	JPEG XL 16-bit image
dicom	dicom_file03.dcm	263,014	2,123	123.89×	JPEG XL 16-bit image
dicom	dicom_file04.dcm	4,637	3,910	1.19×	HCM route
	set total	2,409,239	649,464	3.71×	route time 50 s
photo_raw	01.ppm	1,179,663	454,779	2.59×	media specialist
photo_raw	07.ppm	1,179,663	357,122	3.30×	media specialist
photo_raw	13.ppm	1,179,663	548,928	2.15×	media specialist
photo_raw	19.ppm	1,179,663	437,879	2.69×	media specialist
	set total	4,718,652	1,798,708	2.62×	route time 191 s
jpg	001.jpg	758,785	562,910	1.35×	media specialist
jpg	007.jpg	2,253,385	1,781,516	1.26×	media specialist
jpg	013.jpg	2,368,480	1,883,750	1.26×	media specialist
jpg	019.jpg	2,098,093	1,649,515	1.27×	media specialist
	set total	7,478,743	5,877,691	1.27×	route time 208 s
coded	Hungarian_Dance_number_5___A	3,505,191	3,492,223	1.00×	HCM v3.7 route
coded	Kevin_MacLeod___P_I_Tchaikov	1,553,340	1,515,377	1.03×	HCM v3.7 route
coded	admiralbob77___Choice___Drum	1,847,212	1,836,754	1.01×	HCM v3.7 route
coded	sorohanro___solo_trumpet_06.	233,834	221,151	1.06×	Brotli
	set total	7,139,577	7,065,505	1.01×	route time 161 s
gif_tif	fei-sem-rbc.tif	1,701,683	1,257,762	1.35×	HCM v3.7 route
gif_tif	geotiff.tif	650,466	452,822	1.44×	HCM v3.7 route
gif_tif	imageio_issue245.gif	1,292	902	1.43×	HCM v3.7 route
gif_tif	multipage.tif	940	500	1.88×	LZMA
	set total	2,354,381	1,711,986	1.38×	route time 69 s

Standouts: Silesia mr 5.61× and x-ray 2.39× (both smaller than JPEG-LS and stock JPEG XL), ooffice 3.31×, samba 6.47×, xml 15.34×. The HCM v3.7 route is a private research build.

4. MODIFIED TOOLS (-hi tools) - stock decoders read every output

Full corpus rows for these tools are in section 2 (Silesia LZMA/7-Zip/RAR race, Canterbury tables, Silesia 1 MB sample, Goodwin synthetic, Mahoney race, x-ray, mr, kennedy.xls). Results recorded only outside those tables:

xz-hi: Canterbury 5.70× -> 6.07×; decode 32.1 -> 24.8 ms; Goodwin G_audio16 439,256 -> 382,084 B (decode 23.1 -> 18.1 ms); modern data 12 files 4,053,776 -> 4,040,884 B, decode 389 -> 360 ms.

bzip2-hi: Canterbury 5.18× -> 5.28×, decode 59.7 -> 55.4 ms; compress much slower (plrabn12 0.07 -> 22.9 s).

lzma-hi: Canterbury 5.70× -> 5.78×, 28.65 -> 28.55 ms; nci 19.27× -> 23.24× and 1.22x faster decode.

7z-hi: Canterbury 5.68× -> 6.05×; ooffice 2.54× -> 3.04× (1.25x faster, BCJ2); nci 19.27× -> 23.23× (1.20x faster); mr 3.63× -> 3.85×; x-ray 1.89× -> 2.02×.

zstd-hi / Brotli-hi: small no-pain gains only (Canterbury table, section 2); zstd-hi counties export compress 72.9 -> 36.3 s.

jxl-hi (stock cjxl effort 7, other settings searched): x-ray: +0.3% size with 5.3x faster decode, or 2.3% smaller with 1.35x faster; RG1 +2.9% / 7.5x; MR2 +3.6% / 4.8x; RG3 0.5% smaller / 1.30x; OBXXXX1A +4.8% / 1.37x. Lost on 693, MR-SIEMENS, ALOKA, eCT.

JPEG-LS tuned (CharLS): x-ray 3,591,153 -> 3,574,443 B (2.36× -> 2.37×), decode 103.6 -> 87.6 ms (cloud rival3 run: 0.120 -> 0.100 s).

flac-hi (tuned FLAC): 6 speech WAVs: flac -8 1.88× -> 1.92×, decode 136 -> 123 ms; later run 1.93×, decode 0.171 s.

RAR no-pain search: Silesia 3.99× -> 4.06×, decode 1,059 -> 1,019 ms; sao 1.31× -> 1.47× with -mcd-, decode 68.0 -> 61.5 ms.

zpaq-hi: Canterbury 361,792 -> 359,424 B (one file per archive), compress 11.5 -> 20.9 s, decode 11.8 -> 14.1 s (size win only); Calgary 716,354 B, Canterbury large 2,151,206 B (smallest of all there), GCB 9,137,324 B, enwik9 168,215,145 B (section 2, Mahoney race). Same-block multi-core: 0.1-1.1% smaller than stock, decodes 1-37% slower.

PDF stream rewrite: 139 pdf.js PDFs: originals 1.00×, qpdf known optimisation 1.17×, ours 1.19×; render 11.22 / 11.35 / 11.60 s; strict no-pain pick 1.18× at 11.20 s render.

WebP / PNG searches: WebP lossless -z 9 on 24 Kodak images: 2.53× -> 2.53× (no real gain); PNG per-image filter + zlib strategy 1.79× -> 1.82×, decode 225 -> 202 ms (known optipng work, not

counted).

source for this section: memory /areas/hcm-benchmarks.md and /areas/combary.md; PC folders xz-hi, bzip2-hi,
zpaq-hi, tools*, verification 2026-10-28, HCM_Wins_Under_Standard.txt.

5. MUTANTS AND RESEARCH BUILDS

lpaq1-h (lpaq1 + HCM record detector and contexts): Canterbury 392,132 -> 359,786 B (7.17× -> 7.81×), decode 4.21 -> 4.18 s; Silesia 1 MB sample 3,643,900 -> 3,559,695 B; Goodwin synthetic 8,113,745 B = smallest of all 14 tools (compress 146.8 s, decode 25.6 s); GCB 9,121,594 B (2.14×, smallest in the Mahoney race). Parallel 2 MB blocks: reymont decode 6.7 -> 4.1 s, x-ray 12.5 -> 6.5 s, dickens 12.4 -> 7.0 s at 1-6% size cost.

HCM v3.7 (research, private): HCM 3.6.1 + two kanzi-style contexts + kanzi EXE filter: ooffice 2,251,050 -> 1,857,184 B (3.31×); samba 3,355,530 -> 3,296,662 B; full Silesia max 41.74 MB (section 2 table).

Sets reorder (setsort, private): Silesia xml HCM 3.7 max 348,363 -> 341,969 B (15.63×), decode 4.91 -> 4.98 s, beats kanzi -l 9 342,833; 16 KB windows 338,588 B (15.79×). pcap 112,917,841 B: HCM 3.7 max 88,498,954 -> 80,616,598 B (1.40×), decode 117.4 -> 102.5 s; kanzi 88,755,437 -> 80,692,125; Brotli 11 89,598,288 -> 81,206,385; zstd -9 91,398,395 -> 82,701,553; zstd -3 92,254,442 -> 83,401,428; LZ4 HC -12 95,392,369 -> 87,638,425; bzip2 -9 92,613,792 -> 87,080,870; gzip -9 94,023,309 -> 92,702,776; BSC 86,687,460 -> 82,881,374.

cascade (cascading lines; zstd build v1.1 and HCM build v3, private): desktop: hi16_mix 1.14× -> 1.89×, dup_random_chunks 1.37× -> 2.32× (beats zstd --long 2.32×), random stored raw 1.00×; 3.2 GB tar 4.09× vs 4.09× (v1.1 now keeps the smaller). HCM cascade v3 (desktop, Sets fast off -> on): hi16 1.17× -> 1.93× (decode 1.76 -> 3.20 s), switch 1.76× -> 2.14× (4.65 -> 4.80 s), dup 1.39× -> 2.57× (5.04 -> 8.01 s).

record delta front ends (Mutated): kennedy.xls zstd -22 64,815 -> 26,398 B with 2.6x faster decode; x-ray zstd -19 5,129,823 -> 4,508,494 B, decode 24.6 -> 20.2 ms; zstd -1 x-ray 1.25× -> 1.69×; Snappy kennedy 2.44× -> 7.94× (966 -> 717 us); LZFSE kennedy 7.56× -> 18.78× (1,064 -> 767 us). Silesia totals: section 2.

change map (formerly activity map): kennedy gzip -9 209,721 -> 35,558 B, decode 2,148 -> ~1,313-1,476 us; slower on lz4 (39,997 B, 739 -> ~1,000 us); Snappy x-ray 1.00× -> 1.64× but 11.4 vs 1.1 ms decode.

Harrington Exchange (swap + juggle) front ends: synthetic packed 12-bit noise 324,507 B (1.62×; best known method 1.42×), decode 5.1 ms (zstd 0.89, xz 19.2 ms); NIST biased bits 19,722 B vs xz 21,772 B, decode 1.49 ms (xz 1.13 ms). On real images renaming hurts (x-ray 12-bit split-only 342,496 B is best).

Parquet / ORC with record columns: DuckDB Parquet zstd: kennedy 6.31× -> 14.98×; x-ray 1.04× -> 1.60× (read 137 -> 102 ms); ORC zlib: x-ray 1.37× -> 1.80× (46.1 -> 25.6 ms); kennedy 3.88× -> 19.87× (2.8 -> 1.8 ms).

hcm-trit, HCM-Bin, hcmx: Canterbury table in section 2; kennedy HCM-Bin v1 16,763 B (ZPAQ 16,822) decode 0.79 s vs ZPAQ 6.35 s; ptt5 27,623 B (ZPAQ 28,563) 0.26 s vs 3.29 s. Calgary hcmx max 707,392 B (4.60×), smallest in the Mahoney race.

HCM parallel and Sets modes (v3.5 / v3.6): Silesia samba v3.4 3,400,668 B (62.9 / 16.8 s) -> v3.5 blocks 3,443,819 (59.2 / 13.9 s) -> v3.6 Sets 3,406,084 (42.5 / 9.7 s); mozilla v3.5 13,412,925 (130.2 / 32.7 s). Decoder memory, Silesia desktop: single stream 86.8 s / 107 MB, blocks 16 MB 58.8 s / 230 MB, Sets 16 MB 47.3 s / 271 MB, Sets 4 MB 26.6 s / 549 MB.

IBM-style splitting controls (US8782019, US9792350) vs Sets, HCM fast, cloud: samba Sets 3,406,080 B (40.5 / 9.2 s) vs equal blocks 3,426,219 (62.3 / 9.4 s), IBM-A 3,536,707 (38.7 / 10.5 s), IBM-B 3,475,209 (57.7 / 12.6 s); mozilla Sets 13,419,755 (103.2 / 28.8 s) vs IBM-A 13,489,282 (94.4 / 24.9 s). Source ibm-comparison-2026-10-29\results.txt.

paq8px / cmix labs: paq8px lite model set ~2x faster (kennedy 414 -> 181 s) for 9.28× vs 9.31× on Canterbury; cmix run in 2.5-4 GB instead of 32 GB with no size change on fields.c (2,087 B).

mzip research builds (AGPL, research only): Silesia Sets + mzip 16 MB 4.84× decode 16.8 s vs mzip 4.92× 24.8 s (cloud); HX Complete + mzip 4.81× on 9 of 12 files. Section 2.

sources: memory /areas/hcm-benchmarks.md, /areas/combary.md, /areas/harringtons-sets.md; PC sets-cascade-2026-11-03,
setsort-reorder-2026-11-04, lpaq1-h, hcm37, hcm-v35/36, HCM-tests\cascade-2026-11-03,

tools-archive 2026-10-28.

6. TASK-LEVEL TESTS (Harrington's Sets applied to software that is not a compressor)

Speed-up = plain single run time / our run time on the same machine. "Identical" = output compared with the plain run.

test	machine	plain	ours	speed	identical / note
Game: autosave (gzip -6 of 131 MB save)	cloud 2c	0.84 s	0.45 s	1.86x	yes (+0.031% size)
Game: patch creation (xdelta3, 600 MB)	cloud 2c	31.4 s	19.1 s	1.64x	yes (+721 B)
Game: patch apply (xdelta3)	cloud 2c	1.31 s	2.63 s	0.50x	yes (loss)
Game: save-game load (131 MB XML)	cloud 2c	0.701 s	0.527 s	1.33x	yes (771,200 obj.)
Game: sprite/texture load (3,020 PNG)	cloud 2c	7.61 s	3.91 s	1.95x	yes (queue 4.03 s)
Game: LOD generation (30.6M tris)	cloud 2c	31.03 s	14.96 s	2.07x	yes (= plain queue)
Game: navmesh bake (Recast, 11,022 tiles)	cloud 2c	18.4 s	9.2 s	2.0x	yes (queue 9.0 s)
Game: asset bundle load (zstd, 600 MB)	cloud 2c	1.242 s	0.415 s	2.99x	chunks 3.65x faster
Zeek 8.0.4, 903 MB pcap, 12 shards	desktop	6.54 s	4.35 s	1.50x	yes, all 35,920 rows
Zeek 8.0.4, 7.2 GB pcap, 12 shards	desktop	50.34 s	36.32 s	1.39x	yes, 284,416 rows
Zeek 8.0.4, 903 MB pcap, 2 shards	cloud 2c	10.44 s	8.12 s	1.28x	2 logs differed
Snort 3.12.4, 4 shards	cloud 2c	8.91 s	8.58 s	1.04x	content alerts yes
ClamAV 1.5.1, 811 MB image, 4 pieces	cloud 2c	162.1 s	149.4 s	1.08x	yes (detections)
binwalk 3.1.0, 2 pieces	cloud 2c	2.72 s	5.60 s	0.49x	NO (loss)
YARA (hs_scan), 150 MB, 400 rules	cloud 2c	10.8 s	5.8 s	1.86x	yes (offsets)
Wireshark tshark 4.6.9 (hs_shark direct)	desktop	16.0 s	5.6 s	2.9x	yes, 964,896 lines
Tesseract OCR, 180 pages (hs_ocr)	desktop	236.9 s	35.4 s	6.6x	yes (text bytes)
whisper.cpp, 25.6 min audio (hs_whisper)	cloud 2c	383.8 s	331.4 s	1.16x	WER 22.1 -> 17.1%
Apache Tika + OCR, clean_mixed.pdf	desktop	95.8 s	16.2 s	5.9x	yes (body text)
jq 1.7, 800 MB JSONL, summarize	cloud 2c	30.8 s	16.3 s	1.88x	yes
jq 1.7 via hs_lines, 52 MB	cloud 2c	3.65 s	2.61 s	1.40x	yes
river anomaly detector via hs_lines	cloud 2c	27.4 s	27.2 s	1.01x	AUC .518 -> .652
GNU grep -o regex, 800 MB (hs_grep)	cloud 2c	20.6 s	10.7 s	1.93x	yes (bytes)
Ghostscript render, 409 pages (hs_gs)	desktop	56.5 s	11.1 s	5.1x	yes (page images)
Build: heavy C++ project (hs_build)	desktop	78.7 s	74.7 s	1.05x	yes (objects)
3D slicing, PrusaSlicer (first failure)	cloud 2c	n/a	n/a	none	no gain found

Sources by row (PC Documents\data compression\...):

Game: autosave (gzip -6 of 131 MB save)	hs-games-2026-10-31\autosave
Game: patch creation (xdelta3, 600 MB)	hs-games-2026-10-31\patch
Game: patch apply (xdelta3)	hs-games-2026-10-31\patch
Game: save-game load (131 MB XML)	hs-games\savegame-load
Game: sprite/texture load (3,020 PNG)	hs-games\sprites
Game: LOD generation (30.6M tris)	hs-games\lod
Game: navmesh bake (Recast, 11,022 tiles)	hs-navmesh-2026-10-31
Game: asset bundle load (zstd, 600 MB)	hs-games\asset-bundle
Zeek 8.0.4, 903 MB pcap, 12 shards	security-tools\ZEEK-EXACT
Zeek 8.0.4, 7.2 GB pcap, 12 shards	security-tools\ZEEK-EXACT
Zeek 8.0.4, 903 MB pcap, 2 shards	security-tools\RESULTS
Snort 3.12.4, 4 shards	security-tools\RESULTS
ClamAV 1.5.1, 811 MB image, 4 pieces	security-tools\RESULTS
binwalk 3.1.0, 2 pieces	security-tools\RESULTS
YARA (hs_scan), 150 MB, 400 rules	hs-scan-2026-10-30
Wireshark tshark 4.6.9 (hs_shark direct)	hs-shark-2026-10-30
Tesseract OCR, 180 pages (hs_ocr)	hs-ocr-2026-10-31
whisper.cpp, 25.6 min audio (hs_whisper)	hs-whisper-2026-10-31
Apache Tika + OCR, clean_mixed.pdf	hs-tika-2026-10-31
jq 1.7, 800 MB JSONL, summarize	hs-jq-2026-10-31
jq 1.7 via hs_lines, 52 MB	hs-lines-2026-10-30
river anomaly detector via hs_lines	hs-lines-2026-10-30
GNU grep -o regex, 800 MB (hs_grep)	hs-grep-2026-10-31
Ghostscript render, 409 pages (hs_gs)	hs-gs-2026-10-30
Build: heavy C++ project (hs_build)	hs-build-2026-10-31
3D slicing, PrusaSlicer (first failure)	slicer-test-2026-10-31

Notes: OpenSSL build 1.02-1.09x (cloud), none on desktop; ClamAV pieces cost 2.3x more total work; the game LOD, sprite and navmesh gains equal an ordinary shared work queue; whisper output is not byte-identical by design (word error rate shown). Compression-side Sets tests (hs_wrap, hs_zstd, HCM Sets) are in section 2 and 7.

7. OUR PROGRAMS - OTHER RECORDED RESULTS (desktop speed records and early versions)

hs_zstd 1.0 (20 GB Silesia, desktop, 3 rounds): decode 10.71 / 19.23 / 10.50 s vs zstd 19.46 / 28.72 / 28.51 s (safe claim 1.5-1.9x); compress 15.6 s vs zstd 68.2 s (1 core) / 18.8 s (-T0);

3.180× vs 3.183×.

hs_zstd 1.1 vs pzstd, 10 GB (desktop): decode pzstd 6.87 s vs hs_zstd 4 MB 4.00 s (1.72x).

hs_zstd core scaling (4 GB Silesia, zstd -3, desktop): decode at 1/2/4/6/8/12 jobs: hs_zstd 4 MB 7.52 / 4.87 / 3.24 / 2.71 / 2.55 / 2.43 s; pzstd 4.77 / 4.04 / 3.60 / 3.78 / 3.49 / 3.53 s. Compress hs_zstd 19.59 -> 2.89 s (6.8x), pzstd 15.58 -> 3.78 s (4.1x).

hs_zstd 1.3 ablation (16 MB blocks): 4 types switching every 1 MB: 1.2 1.57×, plain blocks 1.71×, 1.3 1.88×, zstd -3 --long 1.71×; dup random chunks: 1.00× / 1.31× / 1.37× / 2.31×; Silesia joined 3.20× / 3.20× / 3.20× / 3.23×.

hs_wrap 1.5 + gzip, fair gzip baselines (4 GB Silesia, level 6, desktop): decode gzip 14.16 s, pigz 9.55-9.77 s, rapidgzip 2.88 s, bgzip -@12 3.36 s (2.99×), hs_wrap 1.5 Sets 4.02 s (3.11×), plain blocks 4.03 s. rapidgzip beats hs_wrap + gzip on decode.

hs_wrap 1.4 memory (2 GB, desktop): 16 MB blocks + hold 14: decode 1.9 s at 276 MB; 8 MB + hold 14: 2.1 s at 165 MB; gzip alone 6.0 s at 9 MB.

HCM v3 / v2 speed: v3 fast 3.96x compress / 4.10x decompress vs v2; memory fast 42 MB, max 68 MB, v2 121 MB.

HCM v3.6 Windows, samba (desktop): single 6.35× (33.4 / 8.1 s); blocks 6.27× (24.0 / 6.3 s); Sets 16 MB 6.34× (15.2 / 4.0 s); Sets 4 MB 6.19× (7.5 / 2.1 s).

Early Python HCM (200 KB Python docs text): context-balance coder 6.73× (29,725 B), ~27 s each way; restore point BWT route 5.65× (35,378 B, 0.9 s); ZPAQ -m5 7.12× (28,101 B, 1.0 s), bzip2 5.92×, xz 5.40×.

H-Deflate v1 (4 deflate-heavy sets, cloud): PNG 24 files 1.54× (dec 10.0 s) vs precomp route 1.23×; PDF 40 files 1.69× (3.89 s) vs 1.66×; Office 177 files 2.06× (5.22 s) vs 1.94×; archives 1.645× (15.3 s) vs 1.645× (21.9 s). Full tables in section 2.

sources: memory /areas/combary.md and /areas/hcm-benchmarks.md; PC HCM-tests\zstd20-2026-10-31, pzstd-2026-10-31, hs-zstd-scaling-2026-11-01, hs-zstd-ablation-1.3, HCM-tests\gzfair-2026-11-01, gzip-mem-2026-10-31, hcm-v36-windows, h-deflate-2026-11-04\H-Deflate-Results-2026-11-04.txt.

8. cmix AND paq8px - Compared via existing metrics, not actually tested

Silesia (211,938,580 B): published cmix v20 28,302,847 B = 7.49×. Our best measured: HCM v3.6 fast on the desktop 42,661,041 B = 4.97×; HCM v3.7 max (research) 4.06× file order; HX Complete 4.72× at 8.6 s decode. Compared via existing metrics, not actually tested.

Silesia images: cmix mr 1,827,572 B (5.46×) vs HX 2,069,305 B (4.82×); cmix x-ray 3,503,868 B (2.42×) vs HX 3,651,990 B (2.32×). Compared via existing metrics, not actually tested.

Random Turing machine data (19,552,881 B): predicted paq8px (2009) ~8.74 MB (2.24×) vs HX Complete 9,185,811 B (2.13×) and lpaq1 9,162,095 B (2.13×). Compared via existing metrics, not actually tested.

Canterbury: cmix published 9.61× and paq8px v217 -8 9.31×; our best hcm-trit v2 max + dictionary 8.45×, hcmx max 8.08×. Compared via existing metrics, not actually tested.

Biased files with a known floor: HX is within 0.06-0.31% of the floor (0.74% on the repeats file), so cmix or paq8px can be at most that much smaller there. Compared via existing metrics, not actually tested.

Recorded exception: paq8px -1 / -6 was run once inside the hs_wrap speed matrix on 3 MB heads of 10 Silesia files (section 2, "Silesia 3 MB heads"), and the paq8px / cmix lab runs of 25 Oct (section 5). No full paq8px or cmix benchmark was run.

sources: plans\All-Units-vs-All-Units-2026-11-04.txt section 12; HU-pretrain-results\published\; memory /areas/combary.md (paq8px / cmix lab, cmix dictionary).

9. FINAL RANKINGS - every table, by ratio and separately by decode speed

Only tools with an exact result on every file of the table are ranked; rows without a recorded decode time appear in the ratio list only. Marks: * ours, m modified, r mutant/research, blank original. Speed = decode MB/s (original bytes per second). Compress MB/s is in the CSV.

[Silesia] Silesia, 12 files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	5.20×	LZ4 (default)	482.8
2	mzip	4.92×	*HX Fast	308.9
3	rSets + mzip 16 MB blocks	4.84×	Brotli 11	176.2
4	rSets + mzip 4 MB blocks	4.73×	gzip -9	135.4
5	*HX Complete	4.72×	7-Zip -mx9 (LZMA2)	62.19
6	bzip3	4.57×	*HX Complete	24.58
7	BSC	4.55×	BSC	15.52
8	7-Zip PPMd (-mx9 -m0=PPMd)	4.44×	rSets + mzip 16 MB blocks	12.62
9	7-Zip -mx9 (LZMA2)	4.35×	rSets + mzip 4 MB blocks	11.07
10	Brotli 11	4.28×	mzip	8.56
11	*HX Fast	4.26×	bzip3	7.56
12	gzip -9	3.13×	7-Zip PPMd (-mx9 -m0=PPMd)	4.26
13	LZ4 (default)	2.11×	kanzi -l 9	2.19

[Silesia] Silesia, 12 files - HX run of 3 Nov 2026 (HX Fast, HX Complete, BSC, bzip3, mzip)

#	by ratio	ratio	by decode speed	dec MB/s
1	mzip	4.92×	*HX Fast	308.9
2	*HX Complete	4.72×	*HX Complete	24.58
3	bzip3	4.57×	BSC	15.49
4	BSC	4.55×	mzip	8.56
5	*HX Fast	4.26×	bzip3	7.54

[Silesia] Silesia, 12 files - HCM v3.6 fast on the 12-thread desktop (30 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6 fast, single core	4.97×	*HCM v3.6 fast, Harrington's Sets 4	7.82
2	*HCM v3.6 fast, parallel 16 MB bloc	4.94×	*HCM v3.6 fast, Harrington's Sets 1	4.56
3	*HCM v3.6 fast, Harrington's Sets 1	4.93×	*HCM v3.6 fast, parallel 16 MB bloc	3.69
4	*HCM v3.6 fast, Harrington's Sets 4	4.86×	*HCM v3.6 fast, single core	2.54

[Silesia] Silesia, 12 files - hs_zstd 1.1 corpus run on the desktop (31 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	zstd default (-3)	3.18×	*hs_zstd 1.1, 4 MB blocks	440.6
2	*hs_zstd 1.1, 16 MB blocks	3.18×	*hs_zstd 1.1, 16 MB blocks	368.6
3	*hs_zstd 1.1, 4 MB blocks	3.18×	zstd default (-3)	185.4
4	LZ4 default	2.10×	LZ4 default	181.9

[Silesia] Silesia, 12 files - HCM v3.7 max (research build) vs kanzi -l 9, file order and Sets reorder (4 Nov)

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	5.20×	rkanzi -l 9 + Sets reorder	2.33
2	rkanzi -l 9 + Sets reorder	5.13×	kanzi -l 9	2.28
3	rHCM v3.7 max, reorder kept only if	5.08×	rHCM v3.7 max, file order	1.07
4	rHCM v3.7 max, file order	5.08×	rHCM v3.7 max + Sets reorder (all f	1.03
5	rHCM v3.7 max + Sets reorder (all f	5.03×		

[Silesia] Silesia, 12 files - Tier-4 race (27 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	ZPAQ -m5	5.42×	BSC	25.53
2	*HCM v3.3 max + x86 E8/E9 filter	5.05×	BSC -b64e2	16.96
3	*hcmx-s fast (x86 inside format)	5.04×	bzip3	9.21
4	*HCM v3.3 max	5.02×	ZPAQ -m4	1.32
5	*hcmx fast best-of incl. HCM-Bin	5.01×	*HCM v3.3 fast	1.14
6	*hcmx fast (trit2)	4.98×	*HCM v3.3 max	0.977
7	*hcmx-s trit2 lite	4.97×	*hcmx fast (trit2)	0.879
8	ZPAQ -m4	4.97×	*hcmx fast best-of incl. HCM-Bin	0.768
9	*HCM v3.3 fast + x86 E8/E9 filter	4.97×	ZPAQ -m5	0.383
10	*HCM v3.3 fast	4.94×		
11	BSC -b64e2	4.55×		
12	bzip3	4.52×		
13	BSC	4.47×		

[Silesia] Silesia, 12 files - LZMA / 7-Zip / RAR king races and record delta (26-27 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	m7z-hi	4.48×	mRAR no-pain search (rar-hi)	208.0
2	mbest of lzma-hi / xz-hi per file	4.46×	RAR 7.30b2 -m5 -ma5	200.1
3	rxz -9e + record delta	4.41×	mlzma-hi	73.36
4	mlzma-hi	4.40×	mbest of lzma-hi / xz-hi per file	72.48
5	xz -9e	4.37×	LZMA -9 (7-Zip SDK, king)	72.11
6	LZMA -9 (7-Zip SDK, king)	4.35×	m7z-hi	63.95
7	7-Zip Ultra -mx9	4.35×	7-Zip Ultra -mx9	60.85
8	mRAR no-pain search (rar-hi)	4.06×		

9	rzstd -19 + record delta	4.03×	
10	zstd -19	4.01×	
11	RAR 7.30b2 -m5 -ma5	3.99×	
12	rxz -0 + record delta	3.42×	
13	xz -0	3.39×	
14	rgzip -9 + record delta	3.19×	
15	gzip -9	3.13×	
16	rzstd -1 + record delta	2.96×	
17	zstd -1	2.89×	
18	rgzip -1 + record delta	2.76×	
19	gzip -1	2.73×	

[Silesia (silesia.tar)] silesia.tar (211,957,760 B) - TurboBench 2026.11.01 and our tools (1 Nov 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	zpaq 5	5.24×	zstd 3	874.7
2	kanzi 9	5.06×	zstd 22	730.8
3	*HCM v3.6.1 max	4.93×	*hs_zstd -19 plain blocks 16 MB, 2	642.3
4	*HCM v3.6.1 Harrington's Sets 16 MB	4.91×	*hs_zstd -19 Harrington's Sets 16 M	623.4
5	*HCM v3.6.1 fast	4.84×	zstd 19	562.5
6	lzma 9	4.35×	zlib 9	196.1
7	xz 9	4.35×	brotli 11	190.6
8	bsc 0	4.30×	xz 9	71.92
9	brotli 11	4.28×	lzma 9	65.82
10	zstd 22	4.05×	bsc 0	16.11
11	zstd 19	4.01×	bzip3 0	4.60
12	*hs_zstd -19 Harrington's Sets 16 M	3.96×	*HCM v3.6.1 Harrington's Sets 16 MB	1.97
13	*hs_zstd -19 plain blocks 16 MB, 2	3.96×	kanzi 9	1.65
14	bzip3 0	3.57×	*HCM v3.6.1 fast	1.08
15	zstd 3	3.19×	*HCM v3.6.1 max	0.936
16	zlib 9	3.13×	zpaq 5	0.320

[Silesia 1 MB sample] Silesia 1 MB sample - 14 tools, one fresh same-machine run (29 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	ZPAQ -m5	3.68×	zstd -3	194.2
2	*HCM v3.4 max	3.55×	zstd -19	182.9
3	*hcmx fast	3.54×	lz4 -1	174.3
4	rlpaq1-h	3.53×	gzip -9	97.92
5	*HCM v3.4 fast	3.53×	mxz-hi	39.37
6	lpaq1 -6	3.45×	xz -9e	38.30
7	mxz-hi	3.14×	mbzip2-hi	18.85
8	mbzip2-hi	3.08×	bzip2 -9	18.07
9	bzip2 -9	3.07×	*HCM v3.4 fast	0.953
10	xz -9e	3.03×	lpaq1 -6	0.799
11	zstd -19	2.83×	*HCM v3.4 max	0.776
12	gzip -9	2.51×	rlpaq1-h	0.774
13	zstd -3	2.49×	*hcmx fast	0.708
14	lz4 -1	1.77×	ZPAQ -m5	0.230

[Silesia 3 MB heads] First 3 MB of 10 Silesia files (31,457,280 B) - paq8px through hs_wrap (30 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	*hs_wrap+paq8px -6 1 MB 6 jobs	4.56×	*hs_wrap+paq8px -1 1 MB 12 jobs	0.017
2	paq8px -1 single stream	4.50×	*hs_wrap+paq8px -1 1 MB 3 jobs	0.017
3	*hs_wrap+paq8px -1 16 MB Sets 12 jo	4.49×	*hs_wrap+paq8px -1 1 MB 6 jobs	0.016
4	*hs_wrap+paq8px -1 2 MB 12 jobs	4.43×	*hs_wrap+paq8px -6 1 MB 6 jobs	0.015
5	*hs_wrap+paq8px -1 1 MB 12 jobs	4.39×	*hs_wrap+paq8px -1 2 MB 12 jobs	0.011
6	*hs_wrap+paq8px -1 1 MB 6 jobs	4.39×	*hs_wrap+paq8px -1 16 MB Sets 12 jo	0.009
7	*hs_wrap+paq8px -1 1 MB 3 jobs	4.39×	paq8px -1 single stream	0.008
8	*hs_wrap+paq8px -1 1 MB 1 job	4.39×	*hs_wrap+paq8px -1 1 MB 1 job	0.008

[Silesia x-ray] Silesia x-ray (8,474,240 B, 12-bit radiograph) - every recorded tool

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Complete, every route (router b	2.39×	*HX Fast (desktop 12 thr, in memory	642.0
2	mJPEG-LS tuned (CharLS) (cloud)	2.37×	rzstd -19 + record delta 2 (mutant)	419.5
3	JPEG-LS (CharLS) (cloud)	2.36×	zstd -19 (cloud, in-process)	344.5
4	JPEG XL -d 0 -e 7 (cloud)	2.33×	*HX Fast (cloud 2 thr)	138.9
5	*HX Fast (cloud 2 thr)	2.32×	*HX Complete (cloud)	136.7
6	*HX Fast (desktop 12 thr, in memory	2.32×	mJPEG-LS tuned (CharLS) (cloud)	84.74
7	*HX Complete (cloud)	2.32×	JPEG-LS (CharLS) (cloud)	70.62
8	rHCM v3.7 max (cloud)	2.32×	xz -9e (desktop)	43.86
9	mzip (desktop)	2.32×	rSets + mzip 12 jobs 4 MB (desktop)	29.97
10	bzip3 (cloud)	2.32×	JPEG XL -d 0 -e 7 (cloud)	28.63
11	*HCM v3.6 fast single core (desktop	2.31×	JPEG XL -d 0 -e 9 (cloud)	24.56

12	JPEG XL -d 0 -e 9 (cloud)	2.31×	mzip (desktop)	12.63
13	ZPAQ -m5 (cloud)	2.31×	BSC (cloud)	8.07
14	rSets + mzip 12 jobs 4 MB (desktop)	2.29×	bzip3 (cloud)	6.15
15	lpaq1 (cloud)	2.28×	*HCM v3.6 fast single core (desktop	2.29
16	kanzi -l 9 (cloud)	2.27×	kanzi -l 9 (cloud)	1.26
17	BSC (cloud)	2.26×	rHCM v3.7 max (cloud)	0.687
18	m7z-hi (Delta 2)	2.02×	lpaq1 (cloud)	0.446
19	7-Zip -mx9	1.89×	ZPAQ -m5 (cloud)	0.223
20	xz -9e (desktop)	1.89×		
21	rzstd -19 + record delta 2 (mutant)	1.88×		
22	zstd -19 (cloud, in-process)	1.65×		

[Silesia mr] Silesia mr (9,970,564 B, MRI DICOM) - every recorded tool

#	by ratio	ratio	by decode speed	dec MB/s
1	JPEG XL -d 0 -e 9 + xz header (clo	5.61×	*HX Fast (desktop 12 thr, in memory	722.5
2	*HX Complete, every route (router b	5.61×	*HX Fast (cloud 2 thr)	151.1
3	JPEG XL -d 0 -e 7 + xz header (clo	5.42×	JPEG-LS (CharLS) + xz header (clou	129.5
4	JPEG-LS (CharLS) + xz header (clou	4.93×	xz -9e (desktop)	89.99
5	*HX Fast (cloud 2 thr)	4.82×	JPEG XL -d 0 -e 7 + xz header (clo	45.12
6	*HX Fast (desktop 12 thr, in memory	4.82×	JPEG XL -d 0 -e 9 + xz header (clo	39.57
7	mzip (desktop)	4.71×	BSC (cloud)	15.78
8	bzip3 (cloud)	4.70×	mzip (desktop)	13.96
9	rHCM v3.7 max (cloud)	4.70×	bzip3 (cloud)	6.32
10	*HCM v3.6 fast single core (desktop	4.66×	*HCM v3.6 fast single core (desktop	2.85
11	kanzi -l 9 (cloud)	4.64×	kanzi -l 9 (cloud)	2.30
12	ZPAQ -m5 (cloud)	4.57×	rHCM v3.7 max (cloud)	1.03
13	lpaq1 (cloud)	4.52×	lpaq1 (cloud)	0.332
14	BSC (cloud)	4.52×	ZPAQ -m5 (cloud)	0.161
15	m7z-hi (Delta 2)	3.85×		
16	7-Zip -mx9	3.63×		
17	xz -9e (desktop)	3.62×		

[Calgary] Calgary corpus, 18 files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	4.44×	Brotli 11	41.16
2	*HX Complete	4.08×	*HX Fast	40.64
3	bzip3	4.05×	LZ4 (default)	38.71
4	BSC	4.04×	gzip -9	34.96
5	7-Zip PPMd (-mx9 -m0=PPMd)	4.01×	7-Zip -mx9 (LZMA2)	17.48
6	Brotli 11	3.79×	*HX Complete	10.32
7	7-Zip -mx9 (LZMA2)	3.67×	BSC	9.65
8	*HX Fast	3.56×	bzip3	7.42
9	gzip -9	3.07×	7-Zip PPMd (-mx9 -m0=PPMd)	5.11
10	LZ4 (default)	1.93×	kanzi -l 9	0.433

[Calgary] Calgary corpus (18 files, each compressed on its own) - Matt Mahoney test sets race (2 Nov 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	*hcmx max	4.60×	ZPAQ 7.15 -m4	2.03
2	rlpaq1-h 9	4.58×	*HCM fast	1.71
3	*HCM max	4.58×	*HCM Sets fast	1.63
4	*HCM Sets max	4.57×	*HCM max	1.30
5	mzpaq-hi -m5	4.54×	*HCM Sets max	1.25
6	*HCM fast	4.53×	*hcmx max	1.20
7	*HCM Sets fast	4.53×	ZPAQ 7.15 -m5	0.591
8	ZPAQ 7.15 -m5	4.53×	mzpaq-hi -m5	0.570
9	lpaq1 9	4.45×	rlpaq1-h 9	0.332
10	ZPAQ 7.15 -m4	4.23×	lpaq1 9	0.313

[Canterbury] Canterbury corpus, 11 files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	6.81×	*HX Fast	57.36
2	*HX Complete	6.33×	Brotli 11	55.11
3	BSC	6.14×	LZ4 (default)	51.11
4	bzip3	6.12×	gzip -9	45.34
5	Brotli 11	5.73×	7-Zip -mx9 (LZMA2)	23.82
6	7-Zip -mx9 (LZMA2)	5.68×	*HX Complete	16.06
7	*HX Fast	5.44×	BSC	12.78
8	7-Zip PPMd (-mx9 -m0=PPMd)	4.89×	bzip3	9.07
9	gzip -9	3.85×	7-Zip PPMd (-mx9 -m0=PPMd)	6.71
10	LZ4 (default)	2.30×	kanzi -l 9	0.556

[Canterbury] Canterbury - 14 tools, one fresh same-machine run (29 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s

1	*HCM v3.4 max	8.02×		zstd -19	63.74
2	ZPAQ -m5	8.02×		gzip -9	53.23
3	*hcmx fast	8.01×		zstd -3	50.55
4	*HCM v3.4 fast	7.95×		lz4 -1	42.78
5	rlpaq1-h	7.81×	mxz-hi		41.83
6	lpaq1 -6	7.17×		xz -9e	37.83
7	mxz-hi	6.07×		bzip2 -9	27.37
8	xz -9e	5.70×	mbzip2-hi		25.48
9	zstd -19	5.44×	*HCM v3.4 fast		1.23
10	mbzip2-hi	5.28×	*hcmx fast		0.945
11	bzip2 -9	5.18×	*HCM v3.4 max		0.822
12	zstd -3	4.43×		lpaq1 -6	0.746
13	gzip -9	3.85×	rlpaq1-h		0.706
14	lz4 -1	2.30×		ZPAQ -m5	0.244

[Canterbury] Canterbury, 11 files - HCM line and standard rivals as first recorded (25-29 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	paq8px v217 -8 (full)	9.31×	mxz-hi	113.3
2	rpaq8px lite model set (lab)	9.28×	mlzma-hi	98.45
3	rhcm-trit v2 max + cmix dictionary	8.45×	LZMA -9 (7-Zip SDK, king)	98.11
4	rHCM v3.3 fast + cmix English dicti	8.33×	mxz-hi (king race)	97.56
5	*hcmx max best-of	8.08×	xz -9e	87.56
6	rhcm-trit v2 max	8.07×	mbzip2-hi	50.74
7	*HCM v3.3 max	8.02×	bzip2 -9	47.08
8	*hcmx fast best-of	8.01×		
9	rhcm-trit v2 (pointer) 6-order	8.00×		
10	*HCM v3.3 fast	7.95×		
11	*HCM v3.2 max	7.89×		
12	*HCM v3.1 max	7.86×		
13	rHCM-Bin v1 (overall)	7.86×		
14	*HCM v3.2 fast	7.83×		
15	mzpaq-hi (one file per archive)	7.82×		
16	*HCM v3.1 fast	7.78×		
17	rhcm-trit v1.1	7.77×		
18	ZPAQ -m5	7.77×		
19	*HCM v3 max	6.82×		
20	*HCM v3 fast	6.78×		
21	rhcm-trit v1	6.77×		
22	*HCM v2	6.74×		
23	mxz-hi	6.07×		
24	mxz-hi (king race)	6.07×		
25	m7z-hi	6.05×		
26	mlzma-hi	5.78×		
27	xz -9e	5.70×		
28	LZMA -9 (7-Zip SDK, king)	5.70×		
29	7-Zip Ultra -mx9	5.68×		
30	rbzip2 rebuild + candidate escape f	5.58×		
31	rbzip2-style rebuild + arithmetic c	5.38×		
32	mbzip2-hi	5.28×		
33	bzip2 -9	5.18×		

[Canterbury] Canterbury, 11 files - zpaq-hi vs stock ZPAQ 7.15 -m5, one file per archive (29 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	mzpaq-hi -m5	7.82×	ZPAQ 7.15 -m5	0.238
2	ZPAQ 7.15 -m5	7.77×	mzpaq-hi -m5	0.199

[kennedy.xls] Canterbury kennedy.xls (1,029,744 B, spreadsheet, record length 13) - recorded tools

#	by ratio	ratio	by decode speed	dec MB/s
1	mzpaq-hi -m5	64.35×	rzstd -22 + record delta 13	1,980
2	rHCM-Bin v1	61.43×	rSnappy + record delta 13	1,436
3	*HCM-Bin v1 max (hcmx)	61.31×	rlz4 -12 + record delta 13	1,420
4	ZPAQ -m5	61.21×	rLZFSE + record delta 13	1,343
5	*HCM v3.3 fast	55.13×	lz4 -12	1,337
6	rlpaq1-h	47.58×	Snappy	1,066
7	mxz-hi (delta 13)	43.53×	rgzip -9 + record delta 13	977.0
8	rzstd -22 + record delta 13	39.01×	LZFSE	967.8
9	rgzip -9 + change map	28.96×	zstd -22	757.7
10	lpaq1 -6	27.14×	gzip -9	472.6
11	rgzip -9 + record delta 13	23.96×	*HX Fast	171.6
12	rlz4 -12 + record delta 13	23.59×	BSC	22.88
13	xz -9e	19.85×	bzip3	10.51
14	*HCM v3 fast	19.11×	ZPAQ -m5	0.239

15	rLZFSE + record delta 13	18.78×	mzpaq-hi -m5	0.187
16	zstd -22	15.89×		
17	*HX Fast	15.88×		
18	BSC	13.91×		
19	bzip3	13.39×		
20	mbzip2-hi (100k blocks)	8.55×		
21	rSnappy + record delta 13	7.94×		
22	bzip2 -9	7.90×		
23	LZFSE	7.56×		
24	gzip -9	4.91×		
25	lz4 -12	3.18×		
26	Snappy	2.44×		

[Canterbury large] Canterbury large corpus, 3 files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	5.16×	*HX Fast	328.2
2	*HX Complete	4.96×	LZ4 (default)	232.5
3	bzip3	4.95×	Brotli 11	146.8
4	BSC	4.94×	gzip -9	104.3
5	7-Zip PPMd (-mx9 -m0=PPMd)	4.81×	7-Zip -mx9 (LZMA2)	61.32
6	Brotli 11	4.46×	BSC	17.01
7	*HX Fast	4.38×	*HX Complete	7.95
8	7-Zip -mx9 (LZMA2)	4.36×	bzip3	6.95
9	gzip -9	3.49×	7-Zip PPMd (-mx9 -m0=PPMd)	4.57
10	LZ4 (default)	1.94×	kanzi -l 9	2.17

[Canterbury large] Canterbury large corpus (E.coli, bible.txt, world192.txt) - Matt Mahoney test sets race (2 Nov 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	mzpaq-hi -m5	5.19×	ZPAQ 7.15 -m4	3.28
2	ZPAQ 7.15 -m5	5.18×	*HCM fast	2.32
3	rlpaq1-h 9	5.15×	*HCM Sets fast	2.32
4	lpaq1 9	5.14×	*HCM max	2.11
5	*hcmx max	5.12×	*HCM Sets max	2.11
6	*HCM max	5.10×	*hcmx max	1.89
7	*HCM Sets max	5.10×	rlpaq1-h 9	1.16
8	*HCM fast	5.05×	lpaq1 9	1.14
9	*HCM Sets fast	5.05×	ZPAQ 7.15 -m5	0.702
10	ZPAQ 7.15 -m4	4.98×	mzpaq-hi -m5	0.656

[enwik8] enwik8 (100,000,000 B) - race of 27 Oct 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	ZPAQ -m5	5.10×	BSC -b64e2	16.67
2	ZPAQ -m4	4.82×	bzip3	7.19
3	BSC -b64e2	4.66×	*HCM v3.3 fast	1.01
4	*HCM v3.3 max	4.61×	ZPAQ -m4	0.980
5	*HCM v3.3 fast	4.51×	*HCM v3.3 max	0.870
6	bzip3	4.41×	ZPAQ -m5	0.355

[enwik9] enwik9 (1,000,000,000 bytes) - Matt Mahoney test sets race (2 Nov 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	lpaq1 9	6.08×	*HCM Sets fast	10.87
2	mzpaq-hi -m5	5.94×	*HCM Sets max	9.97
3	ZPAQ 7.15 -m5	5.93×	ZPAQ 7.15 -m4	6.75
4	ZPAQ 7.15 -m4	5.57×	ZPAQ 7.15 -m5	2.37
5	*HCM max	5.35×	mzpaq-hi -m5	2.13
6	*HCM Sets max	5.20×	*HCM max	2.00
7	*HCM Sets fast	5.10×	lpaq1 9	1.48

[GCB (uiq2)] Random Turing machine output (uiq2, Generic Compression Benchmark data), 3 seeds - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	lpaq1 8	2.13×	LZ4 (default)	291.8
2	*HX Complete	2.13×	*HX Fast	238.4
3	7-Zip PPMd (-mx9 -m0=PPMd)	2.08×	gzip -9	94.92
4	kanzi -l 9	2.08×	Brotli 11	91.37
5	Brotli 11	1.94×	7-Zip -mx9 (LZMA2)	30.55
6	*HX Fast	1.92×	bzip2 -9	12.00
7	7-Zip -mx9 (LZMA2)	1.92×	*HX Complete	7.22
8	bzip2 -9	1.77×	7-Zip PPMd (-mx9 -m0=PPMd)	2.27
9	gzip -9	1.65×	kanzi -l 9	1.10
10	LZ4 (default)	1.21×	lpaq1 8	0.578

[GCB (uiq2)] Generic Compression Benchmark (uiq2 seeds 0, 1, 2) - Matt Mahoney test sets race (2

Nov 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	rlpaq1-h 9	2.14×	*HCM Sets fast	2.10
2	mzpaq-hi -m5	2.14×	ZPAQ 7.15 -m4	2.00
3	ZPAQ 7.15 -m5	2.14×	*HCM fast	1.83
4	lpaq1 9	2.14×	*HCM Sets max	1.73
5	*hcmx max	2.10×	*HCM max	1.53
6	*HCM max	2.09×	lpaq1 9	1.01
7	ZPAQ 7.15 -m4	2.08×	rlpaq1-h 9	0.983
8	*HCM Sets max	2.08×	*hcmx max	0.958
9	*HCM fast	2.05×	ZPAQ 7.15 -m5	0.554
10	*HCM Sets fast	2.04×	mzpaq-hi -m5	0.543

[Synthetic (ours)] Synthetic test files, 5 files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Complete	1.66×	LZ4 (default)	539.0
2	7-Zip -mx9 (LZMA2)	1.65×	*HX Fast	330.2
3	*HX Fast	1.64×	Brotli 11	128.6
4	kanzi -l 9	1.61×	gzip -9	92.17
5	Brotli 11	1.58×	7-Zip -mx9 (LZMA2)	44.66
6	BSC	1.52×	*HX Complete	22.21
7	bzip3	1.50×	BSC	7.15
8	7-Zip PPMd (-mx9 -m0=PPMd)	1.33×	bzip3	5.42
9	gzip -9	1.22×	7-Zip PPMd (-mx9 -m0=PPMd)	1.80
10	LZ4 (default)	1.11×	kanzi -l 9	1.41

[Goodwin synthetic] Synthetic (Goodwin) - 14 tools, one fresh same-machine run (29 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	rlpaq1-h	2.22×	zstd -3	212.8
2	*HCM v3.4 max	2.21×	zstd -19	168.9
3	*hcmx fast	2.20×	lz4 -1	150.0
4	*HCM v3.4 fast	2.20×	gzip -9	83.18
5	ZPAQ -m5	2.19×	xz -9e	33.30
6	lpaq1 -6	2.08×	mxz-hi	31.62
7	mxz-hi	1.96×	bzip2 -9	14.03
8	xz -9e	1.93×	mbzip2-hi	14.00
9	mbzip2-hi	1.87×	*HCM v3.4 fast	0.960
10	bzip2 -9	1.85×	*hcmx fast	0.832
11	zstd -19	1.83×	*HCM v3.4 max	0.760
12	zstd -3	1.67×	lpaq1 -6	0.706
13	gzip -9	1.63×	rlpaq1-h	0.674
14	lz4 -1	1.20×	ZPAQ -m5	0.217

[Biased test files, 8 MB each] Biased test files, 8 MB each - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	lpaq1 -9	2.02×	huff0 (Huffman) 128 KB	745.7
2	*HX Fast	2.02×	FSE (tANS) 1 MB	462.8
3	*HX Complete	2.02×	FSE (tANS) 128 KB	433.0
4	ZPAQ -m5	2.02×	zstd -19	364.7
5	*HCM max	2.01×	LZ4 (default)	355.1
6	*HCM Sets max + cascade v3	2.01×	*HX lines	159.8
7	bzip3	2.00×	*HX Fast	152.5
8	BSC	1.96×	*hs_zstd 1.3 Sets -19	151.1
9	Brotli 11	1.95×	rANS 1 MB	151.1
10	*HX lines + Sets	1.95×	*hs_zstd 1.3 Sets -3	148.8
11	*HX lines	1.93×	*HX Complete	148.1
12	xz -9e	1.90×	*HX lines + Sets	145.3
13	zstd -19	1.87×	*cascade v1.1 -19	144.9
14	*hs_zstd 1.3 Sets -19, 16 MB blocks	1.87×	*hs_zstd 1.3 Sets -19, 16 MB blocks	139.2
15	*cascade v1.1 -19, 16 MB blocks	1.87×	*cascade v1.1 -3	138.9
16	7-Zip -mx9 (LZMA2)	1.86×	*cascade v1.1 -19, 16 MB blocks	119.4
17	7-Zip PPMd (-mx9 -m0=PPMd)	1.85×	Brotli 11	101.7
18	*hs_zstd 1.3 Sets -19	1.80×	gzip -9	99.27
19	*cascade v1.1 -19	1.80×	7-Zip -mx9 (LZMA2)	31.37
20	adaptive arithmetic (ac0)	1.79×	xz -9e	23.29
21	FSE (tANS) 128 KB	1.77×	adaptive arithmetic (ac0)	16.83
22	rANS 1 MB	1.76×	BSC	5.16
23	FSE (tANS) 1 MB	1.76×	bzip3	4.51
24	huff0 (Huffman) 128 KB	1.73×	7-Zip PPMd (-mx9 -m0=PPMd)	2.21
25	*hs_zstd 1.3 Sets -3	1.69×	*HCM Sets max + cascade v3	0.505
26	*cascade v1.1 -3	1.69×	*HCM max	0.469
27	gzip -9	1.67×	lpaq1 -9	0.372
28	LZ4 (default)	1.21×	ZPAQ -m5	0.181

[NIST SP 800-90B real noise samples] NIST SP 800-90B real noise samples - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Complete	2.49×	huff0 (Huffman) 128 KB	1,167
2	*HX lines	2.49×	FSE (tANS) 1 MB	583.3
3	*HX lines + Sets	2.49×	FSE (tANS) 128 KB	500.0
4	*HX Fast	2.49×	zstd -19	500.0
5	lpaq1 -9	2.49×	*HX lines + Sets	304.3
6	*HCM max	2.48×	*HX lines	241.4
7	*HCM Sets max + cascade v3	2.48×	rANS 1 MB	166.7
8	ZPAQ -m5	2.48×	LZ4 (default)	137.3
9	bzip3	2.48×	Brotli 11	116.7
10	BSC	2.48×	*HX Fast	94.59
11	rANS 1 MB	2.45×	gzip -9	87.50
12	adaptive arithmetic (ac0)	2.45×	*hs_zstd 1.3 Sets -19, 16 MB blocks	81.40
13	FSE (tANS) 1 MB	2.45×	*hs_zstd 1.3 Sets -19	80.46
14	FSE (tANS) 128 KB	2.45×	*cascade v1.1 -19	79.55
15	Brotli 11	2.44×	*cascade v1.1 -19, 16 MB blocks	64.81
16	zstd -19	2.43×	*cascade v1.1 -3	58.82
17	*cascade v1.1 -19	2.43×	*hs_zstd 1.3 Sets -3	57.38
18	*cascade v1.1 -19, 16 MB blocks	2.43×	*HX Complete	54.26
19	*hs_zstd 1.3 Sets -19	2.43×	xz -9e	36.46
20	*hs_zstd 1.3 Sets -19, 16 MB blocks	2.43×	7-Zip -mx9 (LZMA2)	35.00
21	xz -9e	2.40×	adaptive arithmetic (ac0)	13.21
22	7-Zip -mx9 (LZMA2)	2.39×	BSC	6.36
23	7-Zip PPMd (-mx9 -m0=PPMd)	2.36×	bzip3	4.85
24	huff0 (Huffman) 128 KB	2.33×	7-Zip PPMd (-mx9 -m0=PPMd)	4.38
25	gzip -9	2.32×	*HCM Sets max + cascade v3	0.980
26	*cascade v1.1 -3	2.28×	*HCM max	0.918
27	*hs_zstd 1.3 Sets -3	2.28×	lpaq1 -9	0.360
28	LZ4 (default)	1.57×	ZPAQ -m5	0.239

[Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR)] Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	10.01×	*HX Fast + audio/colour lines	106.2
2	7-Zip PPMd (-mx9 -m0=PPMd)	8.69×	zstd -19	71.23
3	*HX Complete + audio/colour lines	8.62×	Brotli 11	61.17
4	BSC	8.58×	LZ4 (default)	60.19
5	bzip3	8.54×	gzip -9	50.47
6	Brotli 11	7.94×	xz -9e	44.49
7	xz -9e	7.87×	7-Zip -mx9 (LZMA2)	27.25
8	7-Zip -mx9 (LZMA2)	7.70×	*HX Complete + audio/colour lines	19.27
9	zstd -19	7.46×	BSC	18.98
10	*HX Fast + audio/colour lines	7.46×	bzip3	10.59
11	gzip -9	5.90×	7-Zip PPMd (-mx9 -m0=PPMd)	7.67
12	LZ4 (default)	3.59×	kanzi -l 9	0.697

[Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc++, cc1plus, git)] Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc++, cc1plus, git) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	3.96×	LZ4 (default)	347.3
2	7-Zip -mx9 (LZMA2)	3.50×	zstd -19	225.4
3	7-Zip PPMd (-mx9 -m0=PPMd)	3.28×	*HX Fast + audio/colour lines	201.6
4	*HX Fast + audio/colour lines	3.04×	Brotli 11	123.3
5	xz -9e	3.04×	gzip -9	89.24
6	*HX Complete + audio/colour lines	3.01×	7-Zip -mx9 (LZMA2)	41.29
7	Brotli 11	2.98×	xz -9e	39.51
8	zstd -19	2.83×	BSC	9.73
9	bzip3	2.75×	*HX Complete + audio/colour lines	8.60
10	BSC	2.72×	bzip3	6.84
11	gzip -9	2.38×	7-Zip PPMd (-mx9 -m0=PPMd)	4.53
12	LZ4 (default)	1.66×	kanzi -l 9	1.30

[Speech, WAVs, 8 kHz mono (FSDD)] Speech, WAVs, 8 kHz mono (FSDD) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	OptimFROG --preset max	2.07×	LZ4 (default)	308.8
2	OptimFROG --preset 2	2.04×	zstd -19	196.2
3	WavPack (ffmpeg, level 8)	1.98×	flac -8	118.0
4	WavPack -hhx6 (official, max)	1.98×	flac -8 -e -p (max)	116.7
5	WavPack -hh (official)	1.94×	*HX Fast	108.2

6	*HX Complete + audio/colour lines	1.92×	*HX Fast + audio/colour lines	101.9
7	*HX Complete	1.92×	gzip -9	86.41
8	*HX Fast + audio/colour lines	1.91×	Brotli 11	80.45
9	*HX Fast	1.91×	WavPack -hhx6 (official, max)	63.24
10	flac -8 -e -p (max)	1.89×	WavPack -hh (official)	50.84
11	flac -8	1.88×	*HX Complete	44.20
12	Monkey's Audio -c4000	1.79×	Monkey's Audio -c4000	38.43
13	Monkey's Audio -c5000	1.76×	*HX Complete + audio/colour lines	36.39
14	bzip3	1.75×	TTA (ffmpeg)	33.49
15	TTA (ffmpeg)	1.72×	ALAC (ffmpeg)	26.05
16	BSC	1.71×	WavPack (ffmpeg, level 8)	23.33
17	Brotli 11	1.68×	xz -9e	22.41
18	7-Zip PPMd (-mx9 -m0=PPMd)	1.68×	7-Zip -mx9 (LZMA2)	21.85
19	ALAC (ffmpeg)	1.67×	OptimFROG --preset 2	19.88
20	7-Zip -mx9 (LZMA2)	1.66×	Monkey's Audio -c5000	16.69
21	xz -9e	1.57×	BSC	6.98
22	zstd -19	1.43×	bzip3	6.61
23	gzip -9	1.37×	7-Zip PPMd (-mx9 -m0=PPMd)	3.03
24	LZ4 (default)	1.10×	OptimFROG --preset max	2.61

[Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data)] Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	Monkey's Audio -c5000	2.85×	LZ4 (default)	447.0
2	Monkey's Audio -c4000	2.85×	zstd -19	264.8
3	WavPack -hhx6 (official, max)	2.62×	flac -8 -e -p (max)	128.0
4	flac -8 -e -p (max)	2.61×	flac -8	126.5
5	flac -8	2.60×	*HX Fast	118.2
6	*HX Complete + audio/colour lines	2.59×	*HX Complete	106.5
7	*HX Fast + audio/colour lines	2.59×	gzip -9	96.82
8	WavPack -hh (official)	2.52×	*HX Fast + audio/colour lines	78.08
9	TTA (ffmpeg)	2.51×	Brotli 11	76.18
10	ALAC (ffmpeg)	2.33×	*HX Complete + audio/colour lines	72.46
11	*HX Complete	2.22×	WavPack -hhx6 (official, max)	63.17
12	*HX Fast	2.22×	WavPack -hh (official)	58.68
13	7-Zip -mx9 (LZMA2)	1.62×	TTA (ffmpeg)	53.52
14	7-Zip PPMd (-mx9 -m0=PPMd)	1.60×	Monkey's Audio -c4000	42.28
15	Brotli 11	1.36×	ALAC (ffmpeg)	40.98
16	bzip3	1.34×	7-Zip -mx9 (LZMA2)	22.30
17	xz -9e	1.32×	Monkey's Audio -c5000	18.93
18	BSC	1.32×	xz -9e	18.61
19	zstd -19	1.21×	BSC	5.21
20	gzip -9	1.19×	bzip3	5.05
21	LZ4 (default)	1.05×	7-Zip PPMd (-mx9 -m0=PPMd)	2.23

[Environmental recordings, WAVs, 44.1 kHz mono (ESC-50)] Environmental recordings, WAVs, 44.1 kHz mono (ESC-50) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Fast + audio/colour lines	2.35×	LZ4 (default)	101.6
2	*HX Complete + audio/colour lines	2.35×	zstd -19	93.05
3	WavPack -hhx6 (official, max)	2.34×	flac -8	79.18
4	WavPack -hh (official)	2.29×	flac -8 -e -p (max)	78.20
5	TTA (ffmpeg)	2.22×	gzip -9	58.34
6	flac -8 -e -p (max)	2.22×	Brotli 11	57.28
7	flac -8	2.21×	*HX Fast	53.33
8	*HX Fast	2.14×	*HX Complete	52.82
9	*HX Complete	2.14×	WavPack -hhx6 (official, max)	43.07
10	ALAC (ffmpeg)	2.14×	WavPack -hh (official)	40.09
11	7-Zip -mx9 (LZMA2)	1.80×	*HX Fast + audio/colour lines	38.52
12	7-Zip PPMd (-mx9 -m0=PPMd)	1.80×	*HX Complete + audio/colour lines	37.96
13	bzip3	1.76×	xz -9e	18.43
14	kanzi -l 9	1.76×	7-Zip -mx9 (LZMA2)	17.24
15	BSC	1.71×	bzip3	7.99
16	Brotli 11	1.67×	TTA (ffmpeg)	7.47
17	xz -9e	1.57×	ALAC (ffmpeg)	7.07
18	zstd -19	1.46×	BSC	6.54
19	gzip -9	1.40×	7-Zip PPMd (-mx9 -m0=PPMd)	4.52
20	LZ4 (default)	1.17×	kanzi -l 9	0.664

[Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC] Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	rprecomp -cn + HX Complete	1.03×	*HX Fast	265.3

2	precomp (lzma)	1.03×	zstd -19	209.3
3	*HX Complete	1.01×	LZ4 (default)	193.3
4	bzip3	1.01×	Brotli 11	105.1
5	BSC	1.01×	gzip -9	95.86
6	Brotli 11	1.01×	xz -9e	47.33
7	zstd -19	1.01×	7-Zip -mx9 (LZMA2)	41.63
8	*HX Fast	1.01×	precomp (lzma)	13.50
9	gzip -9	1.01×	*HX Complete	7.93
10	xz -9e	1.01×	rprecomp -cn + HX Complete	6.16
11	7-Zip -mx9 (LZMA2)	1.01×	bzip3	5.29
12	LZ4 (default)	1.00×	BSC	4.94
13	7-Zip PPMd (-mx9 -m0=PPMd)	0.99×	7-Zip PPMd (-mx9 -m0=PPMd)	2.68

[Colour photos, raw RGB (Kodak 24, PPM)] Colour photos, raw RGB (Kodak 24, PPM) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Complete + specialists	2.83×	LZ4 (default)	192.6
2	JPEG XL lossless (e9)	2.83×	zstd -19	153.0
3	JPEG XL lossless (e7)	2.79×	Brotli 11	84.77
4	WebP lossless (z9)	2.53×	*HX Fast	83.27
5	*HX Complete	2.32×	WebP lossless (z9)	78.64
6	bzip3	2.32×	gzip -9	74.31
7	BSC	2.21×	PNG (level 9)	53.32
8	7-Zip PPMd (-mx9 -m0=PPMd)	2.08×	xz -9e	28.20
9	7-Zip -mx9 (LZMA2)	1.96×	7-Zip -mx9 (LZMA2)	26.14
10	xz -9e	1.96×	JPEG XL lossless (e9)	13.82
11	*HX Fast	1.95×	JPEG XL lossless (e7)	13.76
12	Brotli 11	1.87×	*HX Complete + specialists	13.64
13	zstd -19	1.86×	BSC	8.19
14	JPEG-LS (ffmpeg)	1.84×	bzip3	7.66
15	PNG (level 9)	1.79×	*HX Complete	7.51
16	gzip -9	1.60×	JPEG-LS (ffmpeg)	6.95
17	LZ4 (default)	1.18×	7-Zip PPMd (-mx9 -m0=PPMd)	4.97

[Rendered document pages / screenshots, raw (PGM/PPM)] Rendered document pages / screenshots, raw (PGM/PPM) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	7-Zip PPMd (-mx9 -m0=PPMd)	36.22×	*HX Fast + audio/colour lines	479.6
2	kanzi -l 9	35.33×	*HX Fast	459.6
3	*HX Complete + specialists	34.83×	zstd -19	380.4
4	*HX Complete	34.79×	LZ4 (default)	306.4
5	*HX Complete + audio/colour lines	34.79×	Brotli 11	242.4
6	BSC	34.46×	xz -9e	171.0
7	bzip3	33.35×	WebP lossless (z9)	158.7
8	xz -9e	33.13×	7-Zip -mx9 (LZMA2)	154.3
9	Brotli 11	32.78×	gzip -9	140.5
10	7-Zip -mx9 (LZMA2)	32.09×	BSC	75.04
11	WebP lossless (z9)	31.78×	*HX Complete + specialists	68.51
12	*HX Fast	30.48×	*HX Complete + audio/colour lines	60.78
13	*HX Fast + audio/colour lines	30.48×	*HX Complete	60.44
14	zstd -19	30.27×	bzip3	46.74
15	gzip -9	15.85×	JPEG XL lossless (e7)	38.43
16	JPEG XL lossless (e9)	13.54×	JPEG XL lossless (e9)	38.23
17	JPEG XL lossless (e7)	12.72×	7-Zip PPMd (-mx9 -m0=PPMd)	29.81
18	LZ4 (default)	12.35×	JPEG-LS (ffmpeg)	27.20
19	JPEG-LS (ffmpeg)	7.73×	kanzi -l 9	3.99

[Medical DICOM files (imageio test images)] Medical DICOM files (imageio test images) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HX Fast	3.15×	LZ4 (default)	99.30
2	*HX Complete	3.15×	zstd -19	80.08
3	bzip3	3.10×	*HX Fast	70.93
4	BSC	3.00×	*HX Complete	70.93
5	7-Zip PPMd (-mx9 -m0=PPMd)	2.91×	Brotli 11	67.09
6	7-Zip -mx9 (LZMA2)	2.64×	gzip -9	59.11
7	xz -9e	2.63×	xz -9e	24.82
8	Brotli 11	2.56×	7-Zip -mx9 (LZMA2)	24.10
9	zstd -19	2.32×	BSC	9.77
10	gzip -9	2.07×	bzip3	7.55
11	LZ4 (default)	1.43×	7-Zip PPMd (-mx9 -m0=PPMd)	5.87

[PNG files (Kodak 24)] PNG files (Kodak 24) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s

1	*H-Deflate + HX Complete + image ro	1.54×	1	zstd -19	155.5
2	rprecomp -cn + HX Complete	1.23×	2	Brotli 11	150.9
3	rprecomp deflate-only + HX Complete	1.23×	3	*HX Fast	149.5
4	precomp (lzma)	1.08×	4	LZ4 (default)	131.6
5	*HX Complete	1.01×	5	gzip -9	81.45
6	bzip3	1.01×	6	xz -9e	40.94
7	BSC	1.01×	7	7-Zip -mx9 (LZMA2)	33.18
8	7-Zip -mx9 (LZMA2)	1.00×	8	*HX Complete	9.25
9	xz -9e	1.00×	9	BSC	6.80
10	*HX Fast	1.00×	10	bzip3	6.05
11	gzip -9	1.00×	11	7-Zip PPMd (-mx9 -m0=PPMd)	2.58
12	zstd -19	1.00×	12	precomp (lzma)	1.95
13	Brotli 11	1.00×	13	*H-Deflate + HX Complete + image ro	1.59
14	LZ4 (default)	1.00×	14	rprecomp -cn + HX Complete	1.41
15	7-Zip PPMd (-mx9 -m0=PPMd)	0.99×	15	rprecomp deflate-only + HX Complete	1.33

[JPEG photos] JPEG photos - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	packJPG	1.28×	1	LZ4 (default)	250.8
2	*HX Complete + specialists	1.28×	2	zstd -19	231.0
3	Lepton	1.28×	3	Brotli 11	166.2
4	rprecomp -cn + HX Complete	1.27×	4	*HX Fast	152.9
5	precomp (lzma)	1.27×	5	gzip -9	95.41
6	*HX Complete	1.05×	6	xz -9e	35.83
7	bzip3	1.05×	7	7-Zip -mx9 (LZMA2)	31.15
8	BSC	1.05×	8	Lepton	7.53
9	7-Zip PPMd (-mx9 -m0=PPMd)	1.03×	9	rprecomp -cn + HX Complete	7.35
10	*HX Fast	1.01×	10	precomp (lzma)	7.23
11	zstd -19	1.01×	11	*HX Complete	5.24
12	gzip -9	1.01×	12	bzip3	4.91
13	7-Zip -mx9 (LZMA2)	1.01×	13	BSC	4.30
14	Brotli 11	1.01×	14	7-Zip PPMd (-mx9 -m0=PPMd)	2.78
15	xz -9e	1.01×	15	packJPG	2.31
16	LZ4 (default)	1.00×	16	*HX Complete + specialists	2.27

[GIF and TIFF files] GIF and TIFF files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	precomp (lzma)	1.39×	1	LZ4 (default)	113.2
2	rprecomp -cn + HX Complete	1.38×	2	zstd -19	89.18
3	*HX Complete	1.30×	3	*HX Fast	75.46
4	bzip3	1.30×	4	gzip -9	62.62
5	Brotli 11	1.29×	5	Brotli 11	60.06
6	xz -9e	1.28×	6	xz -9e	16.53
7	7-Zip -mx9 (LZMA2)	1.28×	7	7-Zip -mx9 (LZMA2)	14.94
8	BSC	1.27×	8	precomp (lzma)	14.43
9	7-Zip PPMd (-mx9 -m0=PPMd)	1.26×	9	*HX Complete	6.07
10	*HX Fast	1.24×	10	bzip3	5.67
11	zstd -19	1.19×	11	rprecomp -cn + HX Complete	5.12
12	gzip -9	1.17×	12	BSC	4.80
13	LZ4 (default)	1.04×	13	7-Zip PPMd (-mx9 -m0=PPMd)	3.56

[Raw video (Y4M; frames decoded from H.264 sources + one raw test clip)] Raw video (Y4M; frames decoded from H.264 sources + one raw test clip) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	x264 lossless (veryslow)	16.86×	1	LZ4 (default)	691.4
2	rHX Complete + specialists + x264	16.86×	2	zstd -19	390.9
3	*HX Complete + specialists	11.58×	3	MagicYUV -pred gradient (FFmpeg)	293.1
4	FFV1 (level 3)	11.58×	4	Ut Video	282.0
5	*HX Fast	10.47×	5	Brotli 11	245.9
6	*HX Complete	10.47×	6	MagicYUV -pred median (FFmpeg)	192.8
7	bzip3	8.11×	7	gzip -9	157.0
8	BSC	8.03×	8	*HX Complete	128.9
9	xz -9e	7.94×	9	*HX Fast	128.2
10	7-Zip PPMd (-mx9 -m0=PPMd)	7.75×	10	x264 lossless (veryslow)	110.7
11	7-Zip -mx9 (LZMA2)	7.72×	11	7-Zip -mx9 (LZMA2)	92.65
12	Brotli 11	7.37×	12	rHX Complete + specialists + x264	87.30
13	zstd -19	6.83×	13	xz -9e	83.07
14	MagicYUV -pred median (FFmpeg)	5.64×	14	FFV1 (level 3)	79.97
15	MagicYUV -pred gradient (FFmpeg)	5.62×	15	*HX Complete + specialists	66.39
16	gzip -9	5.12×	16	BSC	25.44
17	Ut Video	4.48×	17	bzip3	13.47
18	LZ4 (default)	3.22×	18	7-Zip PPMd (-mx9 -m0=PPMd)	7.99

[Compressed video files (H.264, HEVC, VP9)] Compressed video files (H.264, HEVC, VP9) -

same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	Brotli 11	1.04×	*HX Fast	122.3
2	kanzi -l 9	1.04×	*HX Complete	122.3
3	zstd -19	1.04×	LZ4 (default)	71.95
4	*HX Fast	1.04×	Brotli 11	55.60
5	*HX Complete	1.04×	gzip -9	55.60
6	gzip -9	1.04×	zstd -19	53.18
7	BSC	1.04×	xz -9e	17.99
8	bzip3	1.04×	7-Zip -mx9 (LZMA2)	15.89
9	xz -9e	1.03×	bzip3	5.94
10	7-Zip -mx9 (LZMA2)	1.03×	BSC	5.05
11	LZ4 (default)	1.03×	7-Zip PPMd (-mx9 -m0=PPMd)	2.74
12	7-Zip PPMd (-mx9 -m0=PPMd)	1.01×	kanzi -l 9	0.600

[PDF files from the pdf.js test set] PDF files from the pdf.js test set - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*H-Deflate + HX Complete + image ro	1.69×	*HX Fast	177.2
2	rprecomp deflate-only + HX Complete	1.68×	zstd -19	129.4
3	rprecomp -cn + HX Complete	1.66×	LZ4 (default)	129.4
4	precomp (lzma)	1.66×	gzip -9	78.38
5	*HX Complete	1.35×	Brotli 11	74.78
6	bzip3	1.34×	xz -9e	29.53
7	Brotli 11	1.34×	7-Zip -mx9 (LZMA2)	27.05
8	BSC	1.34×	*HX Complete	20.43
9	7-Zip -mx9 (LZMA2)	1.34×	precomp (lzma)	8.28
10	xz -9e	1.34×	rprecomp -cn + HX Complete	6.41
11	*HX Fast	1.33×	*H-Deflate + HX Complete + image ro	6.29
12	zstd -19	1.33×	BSC	6.25
13	7-Zip PPMd (-mx9 -m0=PPMd)	1.32×	bzip3	6.05
14	gzip -9	1.28×	rprecomp deflate-only + HX Complete	5.50
15	LZ4 (default)	1.22×	7-Zip PPMd (-mx9 -m0=PPMd)	3.25

[Packet captures, public pcapng files] Packet captures, public pcapng files - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	kanzi -l 9	1.42×	*HX Fast	103.5
2	Brotli 11	1.40×	LZ4 (default)	90.39
3	xz -9e	1.39×	zstd -19	73.42
4	7-Zip -mx9 (LZMA2)	1.38×	gzip -9	58.64
5	*HX Complete	1.38×	Brotli 11	57.96
6	*HX Fast	1.37×	xz -9e	15.72
7	zstd -19	1.37×	7-Zip -mx9 (LZMA2)	15.18
8	bzip3	1.36×	*HX Complete	9.57
9	BSC	1.36×	BSC	4.98
10	7-Zip PPMd (-mx9 -m0=PPMd)	1.34×	bzip3	4.70
11	gzip -9	1.29×	7-Zip PPMd (-mx9 -m0=PPMd)	2.51
12	LZ4 (default)	1.21×	kanzi -l 9	0.525

[Packet capture, one 113 MB merged pcap] Packet capture, one 113 MB merged pcap - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	7-Zip -mx9 (LZMA2)	1.39×	LZ4 (default)	855.4
2	*HX Fast	1.38×	*HX Complete	464.7
3	*HX Complete	1.38×	*HX Fast	418.2
4	BSC	1.30×	zstd -19	358.5
5	bzip3	1.27×	gzip -9	116.7
6	Brotli 11	1.26×	Brotli 11	98.88
7	xz -9e	1.26×	7-Zip -mx9 (LZMA2)	19.40
8	zstd -19	1.24×	xz -9e	18.34
9	7-Zip PPMd (-mx9 -m0=PPMd)	1.22×	BSC	5.41
10	gzip -9	1.20×	bzip3	4.54
11	LZ4 (default)	1.15×	7-Zip PPMd (-mx9 -m0=PPMd)	1.67

[Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data)] Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*H-Deflate + HX Complete + image ro	2.06×	*HX Fast	56.29
2	rprecomp -cn + HX Complete	1.94×	zstd -19	37.02
3	precomp (lzma)	1.92×	LZ4 (default)	35.30
4	rprecomp deflate-only + HX Complete	1.92×	Brotli 11	32.33
5	*HX Complete	1.53×	gzip -9	32.33
6	Brotli 11	1.53×	xz -9e	19.18

7	xz -9e	1.53×	7	7-Zip -mx9 (LZMA2)	14.46
8	7-Zip -mx9 (LZMA2)	1.53×	8	*HX Complete	14.32
9	*HX Fast	1.52×	9	precomp (lzma)	6.42
10	zstd -19	1.51×	10	bzip3	5.98
11	bzip3	1.51×	11	BSC	5.13
12	BSC	1.49×	12	*H-Deflate + HX Complete + image ro	4.50
13	7-Zip PPMd (-mx9 -m0=PPMd)	1.45×	13	rprecomp -cn + HX Complete	4.27
14	gzip -9	1.43×	14	rprecomp deflate-only + HX Complete	4.10
15	LZ4 (default)	1.36×	15	7-Zip PPMd (-mx9 -m0=PPMd)	3.17

[SQLite database (Chinook)] SQLite database (Chinook) - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	kanzi -l 9	5.28×	1	*HX Fast	201.5
2	7-Zip -mx9 (LZMA2)	4.64×	2	zstd -19	167.9
3	xz -9e	4.55×	3	LZ4 (default)	167.9
4	precomp (lzma)	4.54×	4	Brotli 11	83.97
5	Brotli 11	4.35×	5	gzip -9	67.17
6	bzip3	3.68×	6	7-Zip -mx9 (LZMA2)	40.30
7	*HX Complete	3.68×	7	xz -9e	33.59
8	rprecomp -cn + HX Complete	3.68×	8	precomp (lzma)	30.53
9	BSC	3.48×	9	BSC	11.72
10	zstd -19	3.37×	10	rprecomp -cn + HX Complete	9.08
11	*HX Fast	3.37×	11	bzip3	8.33
12	7-Zip PPMd (-mx9 -m0=PPMd)	3.16×	12	*HX Complete	8.26
13	gzip -9	2.52×	13	7-Zip PPMd (-mx9 -m0=PPMd)	6.81
14	LZ4 (default)	1.85×	14	kanzi -l 9	1.04

[Archives: ZIP, TAR, TAR.GZ] Archives: ZIP, TAR, TAR.GZ - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	rprecomp -cn + HX Complete	1.65×	1	*HX Fast	372.1
2	rprecomp deflate-only + HX Complete	1.65×	2	LZ4 (default)	372.1
3	*H-Deflate + HX Complete + image ro	1.64×	3	zstd -19	316.6
4	precomp (lzma)	1.52×	4	Brotli 11	216.1
5	*HX Complete	1.28×	5	gzip -9	131.3
6	BSC	1.28×	6	xz -9e	100.8
7	Brotli 11	1.28×	7	7-Zip -mx9 (LZMA2)	86.15
8	bzip3	1.28×	8	*HX Complete	10.41
9	xz -9e	1.28×	9	BSC	9.15
10	7-Zip -mx9 (LZMA2)	1.28×	10	bzip3	4.89
11	*HX Fast	1.28×	11	precomp (lzma)	2.62
12	zstd -19	1.28×	12	7-Zip PPMd (-mx9 -m0=PPMd)	2.51
13	7-Zip PPMd (-mx9 -m0=PPMd)	1.27×	13	*H-Deflate + HX Complete + image ro	2.36
14	gzip -9	1.25×	14	rprecomp deflate-only + HX Complete	1.66
15	LZ4 (default)	1.21×	15	rprecomp -cn + HX Complete	1.65

[Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.coli), FITS, NumPy] Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.coli), FITS, NumPy - same-machine campaign 3-4 Nov 2026

#	by ratio	ratio	#	by decode speed	dec MB/s
1	rprecomp -cn + HX Complete	2.72×	1	LZ4 (default)	454.8
2	*HX Complete + audio/colour lines	2.72×	2	zstd -19	353.0
3	bzip3	2.68×	3	*HX Fast + audio/colour lines	250.1
4	BSC	2.65×	4	Brotli 11	135.6
5	7-Zip -mx9 (LZMA2)	2.64×	5	gzip -9	125.0
6	xz -9e	2.62×	6	xz -9e	40.04
7	precomp (lzma)	2.62×	7	precomp (lzma)	39.60
8	Brotli 11	2.61×	8	7-Zip -mx9 (LZMA2)	37.95
9	7-Zip PPMd (-mx9 -m0=PPMd)	2.58×	9	*HX Complete + audio/colour lines	11.11
10	*HX Fast + audio/colour lines	2.56×	10	rprecomp -cn + HX Complete	10.98
11	zstd -19	2.49×	11	BSC	10.03
12	gzip -9	2.36×	12	bzip3	8.44
13	LZ4 (default)	1.92×	13	7-Zip PPMd (-mx9 -m0=PPMd)	3.69

[Silesia 4 GB] Silesia repeated to 4 GB (4,026,833,020 B) - hs_zstd race (1 Nov 2026)

#	by ratio	ratio	#	by decode speed	dec MB/s
1	zstd -3 --long=27 -T12	3.21×	1	*hs_zstd plain-blocks control 4 MB	1,830
2	zstd -3, one thread	3.18×	2	*hs_zstd 1.3 (Sets, raw off, safe),	1,774
3	zstd -3 -T12	3.18×	3	*hs_zstd plain-blocks control 16 MB	1,685
4	*hs_zstd plain-blocks control 16 MB	3.18×	4	*hs_zstd 1.2 rule, 16 MB	1,685
5	*hs_zstd 1.2 rule, 16 MB	3.18×	5	*hs_zstd 1.3 (Sets, raw off, safe),	1,604
6	*hs_zstd 1.3 (Sets, raw off, safe),	3.18×	6	pzstd -3 -p 12	1,413
7	pzstd -3 -p 12	3.17×	7	zstd -3, one thread	1,080
8	*hs_zstd plain-blocks control 4 MB	3.17×	8	zstd -3 --long=27 -T12	1,062
9	*hs_zstd 1.3 (Sets, raw off, safe),	3.17×	9	zstd -3 -T12	947.5

[Mixed 3.2 GB tar] Mixed public tar 3.2 GB (3,179,306,496 B: Linux, Blender, pdf.js, NYT CSV) - hs_zstd race

#	by ratio	ratio	by decode speed	dec MB/s
1	zstd -3 --long=27 -T12	4.29×	*hs_zstd plain-blocks control 16 MB	1,904
2	*hs_zstd 1.3 (Sets, raw off, safe),	4.09×	*hs_zstd 1.2 rule, 16 MB	1,848
3	zstd -3, one thread	4.08×	*hs_zstd 1.3 (Sets, raw off, safe),	1,817
4	zstd -3 -T12	4.08×	*hs_zstd 1.3 (Sets, raw off, safe),	1,817
5	*hs_zstd plain-blocks control 16 MB	4.08×	*hs_zstd plain-blocks control 4 MB	1,786
6	*hs_zstd 1.2 rule, 16 MB	4.06×	pzstd -3 -p 12	1,656
7	pzstd -3 -p 12	4.05×	zstd -3 -T12	1,452
8	*hs_zstd 1.3 (Sets, raw off, safe),	4.05×	zstd -3, one thread	1,426
9	*hs_zstd plain-blocks control 4 MB	4.04×	zstd -3 --long=27 -T12	454.2

[Plant data] Industrial plant-data exports, tar (163,870,720 B, 7 public UCI/NAB data sets) - 1 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6.1 Harrington's Sets 16 MB	9.27×	zstd -19 -T2	744.9
2	*HCM v3.6.1 fast (one core)	9.11×	*hs_zstd -19 Harrington's Sets 4 MB	682.8
3	xz -9	7.48×	*hs_zstd -19 plain blocks 4 MB	606.9
4	zstd -19	7.13×	*hs_zstd -19 Harrington's Sets 16 M	585.3
5	zstd -19 -T2	7.13×	*hs_zstd -19 plain blocks 16 MB	565.1
6	*hs_zstd -19 plain blocks 16 MB	7.12×	zstd -19	482.0
7	*hs_zstd -19 Harrington's Sets 16 M	7.11×	gzip -9	79.94
8	*hs_zstd -19 plain blocks 4 MB	7.01×	xz -9	74.15
9	*hs_zstd -19 Harrington's Sets 4 MB	7.01×	*HCM v3.6.1 Harrington's Sets 16 MB	2.64
10	gzip -9	5.36×	*HCM v3.6.1 fast (one core)	1.36

[Seismic] Seismic miniSEED, 8 EarthScope station-days (132,250,680 B as int32 samples) - 1 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6.1 on raw samples	4.63×	Steim2 in miniSEED (archive standa	339.1
2	difference+shuffle, xz -9	4.25×	zstd -19 on raw samples	228.0
3	*difference+shuffle, HCM v3.6.1	3.95×	difference+shuffle, zstd -19	176.3
4	difference+shuffle, zstd -19	3.80×	rrecord delta + zstd -19	124.8
5	xz -9 on raw samples	3.47×	difference+shuffle, xz -9	43.65
6	Steim2 in miniSEED (archive standa	3.12×	xz -9 on raw samples	33.65
7	zstd -19 on raw samples	3.06×	rrecord delta + xz -9	31.19
8	rrecord delta + xz -9	2.65×	*difference+shuffle, HCM v3.6.1	1.30
9	rrecord delta + zstd -19	2.28×	*HCM v3.6.1 on raw samples	0.912

[Bank data] Bank data: ACH files tar (64,143,360 B, synthetic NACHA) - 1 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6.1 fast	6.38×	zstd -19	400.9
2	xz -9	4.98×	rrecord delta + zstd -19	160.4
3	zstd -19	4.72×	gzip -9	83.30
4	rrecord delta + xz -9	3.90×	xz -9	44.86
5	rrecord delta + zstd -19	3.62×	rrecord delta + xz -9	35.05
6	gzip -9	3.49×	*HCM v3.6.1 fast	1.08

[Bank data] Bank data: ATM electronic journals tar (55,552,000 B, synthetic) - 1 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6.1 fast	28.55×	zstd -19	617.2
2	zstd -19	15.28×	rrecord delta + zstd -19	205.7
3	xz -9	14.92×	gzip -9	173.6
4	gzip -9	11.31×	xz -9	118.2
5	rrecord delta + xz -9	11.29×	rrecord delta + xz -9	79.36
6	rrecord delta + zstd -19	9.75×	*HCM v3.6.1 fast	1.35

[Bank data] Bank data: Mixed bank backup tar (119,685,120 B, synthetic) - 1 Nov 2026

#	by ratio	ratio	by decode speed	dec MB/s
1	*HCM v3.6.1 fast	9.87×	zstd -19	569.9
2	xz -9	7.06×	rrecord delta + zstd -19	168.6
3	zstd -19	6.87×	gzip -9	142.5
4	rrecord delta + xz -9	5.26×	xz -9	64.00
5	gzip -9	5.12×	rrecord delta + xz -9	44.99
6	rrecord delta + zstd -19	4.72×	*HCM v3.6.1 fast	1.17

[Silesia 10 GB] Silesia repeated to 10 GB (10,000,000,000 B) - gzip -6 alone vs hs_wrap 1.4 (31 Oct 2026)

#	by ratio	ratio	by decode speed	dec MB/s
1	gzip -6 (zlib 1.3.1 minigzip)	3.10×	*hs_wrap 1.4 + gzip -6	925.9
2	*hs_wrap 1.4 + gzip -6	3.10×	gzip -6 (zlib 1.3.1 minigzip)	285.7

[Silesia 0.5 GB] Silesia repeated to 0.5 GB (500,000,000 B) - Tier-1 check (31 Oct 2026)

# by ratio	ratio	by decode speed	dec MB/s
1 zstd 1.5.6 default	3.14×	LZ4 1.10.0 default	1,190
2 *hs_wrap 1.4 + gzip -6	3.06×	zstd 1.5.6 default	1,042
3 LZ4 1.10.0 default	2.07×	*hs_wrap 1.4 + gzip -6	819.7

[Silesia 2 GB] Silesia repeated to 2 GB - hs_zstd vs pzstd (31 Oct 2026)

# by ratio	ratio	by decode speed	dec MB/s
1 zstd -3 (1 thread)	3.19×	*hs_zstd 1.1, 4 MB blocks	3,175
2 *hs_zstd 1.1, 16 MB blocks	3.19×	*hs_zstd 1.1, 16 MB blocks	2,899
3 pzstd -3 -p 12	3.18×	pzstd -3 -p 12	2,128
4 *hs_zstd 1.1, 4 MB blocks	3.18×	zstd -3 (1 thread)	1,198

ABSOLUTE BEST PER STANDARD CORPUS - everything we built or modified vs the best original program

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(across all tables of a corpus; machines differ between tables, so each line names its machine)

Silesia

best ratio, ours	rkanzi -l 9 + Sets reorder		5.13×	(cloud)
best ratio, other	ZPAQ -m5		5.42×	(cloud)
fastest ours	*hs_zstd 1.1, 4 MB blocks	440.6 MB/s	3.18×	(desktop)

Silesia (silesia.tar)

best ratio, ours	*HCM v3.6.1 max		4.93×	(cloud)
best ratio, other	zpaq 5		5.24×	(cloud)
fastest ours	*hs_zstd -19 plain blocks 16 MB, 2 jo	642.3 MB/s	3.96×	(cloud)

Silesia 1 MB sample

best ratio, ours	*HCM v3.4 max		3.55×	(cloud)
best ratio, other	ZPAQ -m5		3.68×	(cloud)
fastest ours	mxz-hi	39.37 MB/s	3.14×	(cloud)

Silesia 3 MB heads

best ratio, ours	*hs_wrap+paq8px -6 1 MB 6 jobs		4.56×	(desktop)
best ratio, other	paq8px -1 single stream		4.50×	(desktop)
fastest ours	*hs_wrap+paq8px -1 1 MB 12 jobs	0.017 MB/s	4.39×	(desktop)

Silesia x-ray

best ratio, ours	*HX Complete, every route (router build)		2.39×	(mixed)
best ratio, other	JPEG-LS (CharLS) (cloud)		2.36×	(mixed)
fastest ours	*HX Fast (desktop 12 thr, in memory)	642.0 MB/s	2.32×	(mixed)

Silesia mr

best ratio, ours	*HX Complete, every route (router build)		5.61×	(mixed)
best ratio, other	JPEG XL -d 0 -e 9 + xz header (cloud)		5.61×	(mixed)
fastest ours	*HX Fast (desktop 12 thr, in memory)	722.5 MB/s	4.82×	(mixed)

Calgary

best ratio, ours	*hcmx max		4.60×	(desktop)
best ratio, other	ZPAQ 7.15 -m5		4.53×	(desktop)
fastest ours	*HX Fast	40.64 MB/s	3.56×	(cloud)

Canterbury

best ratio, ours	rpaq8px lite model set (lab)		9.28×	(cloud)
best ratio, other	paq8px v217 -8 (full)		9.31×	(cloud)
fastest ours	mxz-hi	113.3 MB/s	6.07×	(cloud)

kennedy.xls

best ratio, ours	mzpaq-hi -m5		64.35×	(cloud)
best ratio, other	ZPAQ -m5		61.21×	(cloud)
fastest ours	rzstd -22 + record delta 13	1,980 MB/s	39.01×	(cloud)

Canterbury large

best ratio, ours	mzpaq-hi -m5		5.19×	(desktop)
best ratio, other	ZPAQ 7.15 -m5		5.18×	(desktop)
fastest ours	*HX Fast	328.2 MB/s	4.38×	(cloud)

enwik8

best ratio, ours	*HCM v3.3 max		4.61×	(cloud)
best ratio, other	ZPAQ -m5		5.10×	(cloud)
fastest ours	*HCM v3.3 fast	1.01 MB/s	4.51×	(cloud)

enwik9

```
  best ratio, ours : mzpaq-hi -m5                                  5.94×  (desktop)
  best ratio, other: lpaq1 9                                       6.08×  (desktop)
  fastest ours     : *HCM Sets fast                          10.87 MB/s    5.10×  (desktop)

GCB (uiq2)
  best ratio, ours : rlpaq1-h 9                                    2.14×  (desktop)
  best ratio, other: ZPAQ 7.15 -m5                                 2.14×  (desktop)
  fastest ours     : *HX Fast                                238.4 MB/s    1.92×  (cloud)

Synthetic (ours)
  best ratio, ours : *HX Complete                                  1.66×  (cloud)
  best ratio, other: 7-Zip -mx9 (LZMA2)                            1.65×  (cloud)
  fastest ours     : *HX Fast                                330.2 MB/s    1.64×  (cloud)

Goodwin synthetic
  best ratio, ours : rlpaq1-h                                      2.22×  (cloud)
  best ratio, other: ZPAQ -m5                                      2.19×  (cloud)
  fastest ours     : mxz-hi                                  31.62 MB/s    1.96×  (cloud)

Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC
  best ratio, ours : rprecomp -cn + HX Complete                    1.03×  (cloud)
  best ratio, other: precomp (lzma)                                1.03×  (cloud)
  fastest ours     : *HX Fast                                265.3 MB/s    1.01×  (cloud)

Archives: ZIP, TAR, TAR.GZ
  best ratio, ours : rprecomp -cn + HX Complete                    1.65×  (cloud)
  best ratio, other: precomp (lzma)                                1.52×  (cloud)
  fastest ours     : *HX Fast                                372.1 MB/s    1.28×  (cloud)

Bank data
  best ratio, ours : *HCM v3.6.1 fast                             28.55×  (cloud)
  best ratio, other: zstd -19                                     15.28×  (cloud)
  fastest ours     : rrecord delta + zstd -19                205.7 MB/s    9.75×  (cloud)

Biased test files, 8 MB each
  best ratio, ours : *HX Fast                                      2.02×  (cloud)
  best ratio, other: lpaq1 -9                                      2.02×  (cloud)
  fastest ours     : *HX lines                               159.8 MB/s    1.93×  (cloud)

Colour photos, raw RGB (Kodak 24, PPM)
  best ratio, ours : *HX Complete + specialists                    2.83×  (cloud)
  best ratio, other: JPEG XL lossless (e9)                         2.83×  (cloud)
  fastest ours     : *HX Fast                                83.27 MB/s    1.95×  (cloud)

Compressed video files (H.264, HEVC, VP9)
  best ratio, ours : *HX Fast                                      1.04×  (cloud)
  best ratio, other: Brotli 11                                     1.04×  (cloud)
  fastest ours     : *HX Fast                                122.3 MB/s    1.04×  (cloud)

Environmental recordings, WAVs, 44.1 kHz mono (ESC-50)
  best ratio, ours : *HX Fast + audio/colour lines                 2.35×  (cloud)
  best ratio, other: WavPack -hhx6 (official, max)                 2.34×  (cloud)
  fastest ours     : *HX Fast                                53.33 MB/s    2.14×  (cloud)

GIF and TIFF files
  best ratio, ours : rprecomp -cn + HX Complete                    1.38×  (cloud)
  best ratio, other: precomp (lzma)                                1.39×  (cloud)
  fastest ours     : *HX Fast                                75.46 MB/s    1.24×  (cloud)

JPEG photos
  best ratio, ours : *HX Complete + specialists                    1.28×  (cloud)
  best ratio, other: packJPG                                       1.28×  (cloud)
  fastest ours     : *HX Fast                                152.9 MB/s    1.01×  (cloud)

Medical DICOM files (imageio test images)
  best ratio, ours : *HX Fast                                      3.15×  (cloud)
  best ratio, other: bzip3                                         3.10×  (cloud)
  fastest ours     : *HX Fast                                70.93 MB/s    3.15×  (cloud)

Mixed 3.2 GB tar
  best ratio, ours : *hs_zstd 1.3 (Sets, raw off, safe), 16 MB     4.09×  (desktop)
  best ratio, other: zstd -3 --long=27 -T12                        4.29×  (desktop)
  fastest ours     : *hs_zstd plain-blocks control 16 MB   1,904 MB/s    4.08×  (desktop)
```

```
Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data)
  best ratio, ours : *HX Complete + audio/colour lines              2.59×  (cloud)
  best ratio, other: Monkey's Audio -c5000                          2.85×  (cloud)
  fastest ours     : *HX Fast                                 118.2 MB/s    2.22×  (cloud)

NIST SP 800-90B real noise samples
  best ratio, ours : *HX Complete                                   2.49×  (cloud)
  best ratio, other: lpaq1 -9                                       2.49×  (cloud)
  fastest ours     : *HX lines + Sets                         304.3 MB/s    2.49×  (cloud)

Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data)
  best ratio, ours : *H-Deflate + HX Complete + image routes        2.06×  (cloud)
  best ratio, other: precomp (lzma)                                 1.92×  (cloud)
  fastest ours     : *HX Fast                                 56.29 MB/s    1.52×  (cloud)

PDF files from the pdf.js test set
  best ratio, ours : *H-Deflate + HX Complete + image routes        1.69×  (cloud)
  best ratio, other: precomp (lzma)                                 1.66×  (cloud)
  fastest ours     : *HX Fast                                 177.2 MB/s    1.33×  (cloud)

PNG files (Kodak 24)
  best ratio, ours : *H-Deflate + HX Complete + image routes        1.54×  (cloud)
  best ratio, other: precomp (lzma)                                 1.08×  (cloud)
  fastest ours     : *HX Fast                                 149.5 MB/s    1.00×  (cloud)

Packet capture, one 113 MB merged pcap
  best ratio, ours : *HX Fast                                       1.38×  (cloud)
  best ratio, other: 7-Zip -mx9 (LZMA2)                             1.39×  (cloud)
  fastest ours     : *HX Complete                             464.7 MB/s    1.38×  (cloud)

Packet captures, public pcapng files
  best ratio, ours : *HX Complete                                   1.38×  (cloud)
  best ratio, other: kanzi -l 9                                     1.42×  (cloud)
  fastest ours     : *HX Fast                                 103.5 MB/s    1.37×  (cloud)

Plant data
  best ratio, ours : *HCM v3.6.1 Harrington's Sets 16 MB, 2 cores   9.27×  (cloud)
  best ratio, other: xz -9                                          7.48×  (cloud)
  fastest ours     : *hs_zstd -19 Harrington's Sets 4 MB      682.8 MB/s    7.01×  (cloud)

Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc,
    libstdc++, cc1plus, git)
  best ratio, ours : *HX Fast + audio/colour lines                  3.04×  (cloud)
  best ratio, other: kanzi -l 9                                     3.96×  (cloud)
  fastest ours     : *HX Fast + audio/colour lines            201.6 MB/s    3.04×  (cloud)

Raw video (Y4M; frames decoded from H.264 sources + one raw test clip)
  best ratio, ours : rHX Complete + specialists + x264             16.86×  (cloud)
  best ratio, other: x264 lossless (veryslow)                      16.86×  (cloud)
  fastest ours     : *HX Complete                             128.9 MB/s   10.47×  (cloud)

Rendered document pages / screenshots, raw (PGM/PPM)
  best ratio, ours : *HX Complete + specialists                    34.83×  (cloud)
  best ratio, other: 7-Zip PPMd (-mx9 -m0=PPMd)                    36.22×  (cloud)
  fastest ours     : *HX Fast + audio/colour lines            479.6 MB/s   30.48×  (cloud)

SQLite database (Chinook)
  best ratio, ours : *HX Complete                                   3.68×  (cloud)
  best ratio, other: kanzi -l 9                                     5.28×  (cloud)
  fastest ours     : *HX Fast                                 201.5 MB/s    3.37×  (cloud)

Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA,
    E.coli), FITS, NumPy
  best ratio, ours : rprecomp -cn + HX Complete                     2.72×  (cloud)
  best ratio, other: bzip3                                          2.68×  (cloud)
  fastest ours     : *HX Fast + audio/colour lines            250.1 MB/s    2.56×  (cloud)

Seismic
  best ratio, ours : *HCM v3.6.1 on raw samples                     4.63×  (cloud)
  best ratio, other: difference+shuffle, xz -9                      4.25×  (cloud)
  fastest ours     : rrecord delta + zstd -19                 124.8 MB/s    2.28×  (cloud)

Silesia 0.5 GB
  best ratio, ours : *hs_wrap 1.4 + gzip -6                         3.06×  (desktop)
```

```
  best ratio, other: zstd 1.5.6 default                          3.14×  (desktop)
  fastest ours     : *hs_wrap 1.4 + gzip -6                819.7 MB/s    3.06×  (desktop)

Silesia 10 GB
  best ratio, ours : *hs_wrap 1.4 + gzip -6                       3.10×  (desktop)
  best ratio, other: gzip -6 (zlib 1.3.1 minigzip)                3.10×  (desktop)
  fastest ours     : *hs_wrap 1.4 + gzip -6                925.9 MB/s    3.10×  (desktop)

Silesia 2 GB
  best ratio, ours : *hs_zstd 1.1, 16 MB blocks                   3.19×  (desktop)
  best ratio, other: zstd -3 (1 thread)                           3.19×  (desktop)
  fastest ours     : *hs_zstd 1.1, 4 MB blocks             3,175 MB/s    3.18×  (desktop)

Silesia 4 GB
  best ratio, ours : *hs_zstd plain-blocks control 16 MB          3.18×  (desktop)
  best ratio, other: zstd -3 --long=27 -T12                       3.21×  (desktop)
  fastest ours     : *hs_zstd plain-blocks control 4 MB    1,830 MB/s    3.17×  (desktop)

Speech, WAVs, 8 kHz mono (FSDD)
  best ratio, ours : *HX Complete + audio/colour lines            1.92×  (cloud)
  best ratio, other: OptimFROG --preset max                       2.07×  (cloud)
  fastest ours     : *HX Fast                              108.2 MB/s    1.91×  (cloud)

Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR)
  best ratio, ours : *HX Complete + audio/colour lines            8.62×  (cloud)
  best ratio, other: kanzi -l 9                                  10.01×  (cloud)
  fastest ours     : *HX Fast + audio/colour lines         106.2 MB/s    7.46×  (cloud)

  Showcase - exceptional results: HX Fast decodes x-ray at 642 MB/s on 12 threads (0.0132 s) at
  2.32×, smaller than mzip, bzip3, BSC, kanzi and ZPAQ -m5; HX Complete is the smallest of every
  tool run on Calgary, Canterbury, Canterbury large and our synthetic set in the 3-4 Nov campaign
  while decoding faster than the runner-up; hcmx max is the smallest on Calgary in the Mahoney
  race (4.60×), zpaq-hi on Canterbury large (5.19×), lpaq1-h on the GCB (2.14×) and on Goodwin's
  synthetic set; HCM v3.4 max is the smallest of 14 tools on Canterbury (8.02×, decode 3.4 s vs
  ZPAQ -m5 11.5 s); HCM v3.6.1 is the smallest on the seismic, bank and plant data (plant: Sets
  mode 9.27×, 19% smaller than xz -9); the Sets reorder takes the 113 MB packet capture to 1.40×,
  the smallest of every tool run on it; HX lines sit within 0.06-0.31% of the entropy floor on
  structured biased data at 2-10 GB/s.

================================================================================================
9b. TYPE x SPEED-TIER MATRIX - HX vs the best other tool in every tier
================================================================================================
  Cells: 29 data types x 10 speed tiers = 290 cells. Filled (at least one HX or rival result
  measured inside the tier): 159; not tested: 131. HX measured inside the tier: 62 cells; a rival
  measured inside the tier: 155; both: 58. Cells where HX only qualifies from a faster tier (rule
  3d) show its entry with "^" and count as contested only if a rival is in the tier. The type list
  is the one master_report.py produces now: 29 types (its 3 Nov output had 26, the source of the
  26 x 10 = 260 count). The tiers are the Compression Tier Standard v2.0 1-2-5 series
  (plans\Compression-Tier-Standard-v2.0.txt; master_report.py itself defines no tiers): M = decode
  time / LZ4 default decode time on the same set; T1 M<=2, T2 <=5, T3 <=10, T4 <=20, T5 <=50, T6
  <=100, T7 <=200, T8 <=500, T9 <=2,000, T10 >2,000.
  Rule used (as in Results-Summary-Kings-2026-11-04.txt): the HX entry is the smallest HX result
  whose M is in that tier or a faster one; the rival is the smallest original third-party tool
  whose M lands in that tier. Only full-coverage exact results count; HX research combinations
  (precomp, mzip, x264 routes) are left out. Data: the 3-4 Nov cloud campaign (section 2), the
  same data behind the HX-vs-best-other radar PNGs. Caveat from the Tier Standard: LZ4 was timed
  through its command line in this campaign, so slow tools' M is too low; a re-measurement will
  move some tools to slower tiers. Speeds are decode MB/s. Matrix CSV: tier-matrix-2026-11-06.csv.

  Biased test files, 8 MB each
   tier        HX entry                  ratio    MB/s | best other in tier       ratio    MB/s
   T1 M<=2     none                                    | zstd -19                1.87×   364.7
   T2 M<=5     HX Fast                   2.02×   152.5 | Brotli 11               1.95×   101.7
   T3 M<=10    HX Fast^                  2.02×   152.5 | no rival measured in tier
   T4 M<=20    HX Fast^                  2.02×   152.5 | xz -9e                  1.90×   23.29
   T5 M<=50    HX Fast^                  2.02×   152.5 | adaptive arithmetic (a   1.79×   16.83
   T6 M<=100   HX Fast^                  2.02×   152.5 | bzip3                    2.00×    4.51
   T7 M<=200   HX Fast^                  2.02×   152.5 | 7-Zip PPMd (-mx9 -m0=P   1.85×    2.21
   T8 M<=500   HX Fast^                  2.02×   152.5 | no rival measured in tier
   T9 M<=2,000 HX Fast^                  2.02×   152.5 | lpaq1 -9                2.02×   0.372
   T10 M>2,000 HX Fast^                  2.02×   152.5 | no rival measured in tier

  NIST SP 800-90B real noise samples
   tier        HX entry                  ratio    MB/s | best other in tier       ratio    MB/s
```

T1 M<=2	HX lines	2.49×	241.4	rANS 1 MB	2.45×	166.7
T2 M<=5	HX Complete	2.49×	54.26	xz -9e	2.40×	36.46
T3 M<=10	HX Complete^	2.49×	54.26	no rival measured in tier		
T4 M<=20	HX Complete^	2.49×	54.26	adaptive arithmetic (a	2.45×	13.21
T5 M<=50	HX Complete^	2.49×	54.26	bzip3	2.48×	4.85
T6 M<=100	HX Complete^	2.49×	54.26	no rival measured in tier		
T7 M<=200	HX Complete^	2.49×	54.26	no rival measured in tier		
T8 M<=500	HX Complete^	2.49×	54.26	lpaq1 -9	2.49×	0.360
T9 M<=2,000	HX Complete^	2.49×	54.26	ZPAQ -m5	2.48×	0.239
T10 M>2,000	HX Complete^	2.49×	54.26	no rival measured in tier		

Random Turing machine output (uiq2, Generic Compression Benchmark data), 3 seeds

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.92×	238.4	LZ4 (default)	1.21×	291.8
T2 M<=5	HX Fast^	1.92×	238.4	Brotli 11	1.94×	91.37
T3 M<=10	HX Fast^	1.92×	238.4	7-Zip -mx9 (LZMA2)	1.92×	30.55
T4 M<=20	HX Fast^	1.92×	238.4	no rival measured in tier		
T5 M<=50	HX Complete	2.13×	7.22	bzip2 -9	1.77×	12.00
T6 M<=100	HX Complete^	2.13×	7.22	no rival measured in tier		
T7 M<=200	HX Complete^	2.13×	7.22	7-Zip PPMd (-mx9 -m0=P	2.08×	2.27
T8 M<=500	HX Complete^	2.13×	7.22	kanzi -l 9	2.08×	1.10
T9 M<=2,000	HX Complete^	2.13×	7.22	lpaq1 8	2.13×	0.578
T10 M>2,000	HX Complete^	2.13×	7.22	no rival measured in tier		

Text: server logs (16 systems), Chinese poetry JSON, CSV data, C++ source tree (TAR)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast + audio/colour	7.46×	106.2	Brotli 11	7.94×	61.17
T2 M<=5	HX Complete + audio/col	8.62×	19.27	BSC	8.58×	18.98
T3 M<=10	HX Complete + audio/col^	8.62×	19.27	7-Zip PPMd (-mx9 -m0=P	8.69×	7.67
T4 M<=20	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		
T5 M<=50	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		
T6 M<=100	HX Complete + audio/col^	8.62×	19.27	kanzi -l 9	10.01×	0.697
T7 M<=200	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		
T8 M<=500	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		
T9 M<=2,000	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		
T10 M>2,000	HX Complete + audio/col^	8.62×	19.27	no rival measured in tier		

Programs: Windows x64 EXE (zstd, pzstd, bgzip, distlib) and Linux x86-64 (python, libc, libstdc+

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast + audio/colour	3.04×	201.6	zstd -19	2.83×	225.4
T2 M<=5	HX Fast + audio/colour ^	3.04×	201.6	Brotli 11	2.98×	123.3
T3 M<=10	HX Fast + audio/colour ^	3.04×	201.6	7-Zip -mx9 (LZMA2)	3.50×	41.29
T4 M<=20	HX Fast + audio/colour ^	3.04×	201.6	no rival measured in tier		
T5 M<=50	HX Fast + audio/colour	3.04×	201.6	BSC	2.72×	9.73
T6 M<=100	HX Fast + audio/colour ^	3.04×	201.6	7-Zip PPMd (-mx9 -m0=P	3.28×	4.53
T7 M<=200	HX Fast + audio/colour ^	3.04×	201.6	no rival measured in tier		
T8 M<=500	HX Fast + audio/colour ^	3.04×	201.6	kanzi -l 9	3.96×	1.30
T9 M<=2,000	HX Fast + audio/colour ^	3.04×	201.6	no rival measured in tier		
T10 M>2,000	HX Fast + audio/colour ^	3.04×	201.6	no rival measured in tier		

Silesia, 12 files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	4.26×	308.9	LZ4 (default)	2.11×	482.8
T2 M<=5	HX Fast^	4.26×	308.9	Brotli 11	4.28×	176.2
T3 M<=10	HX Fast^	4.26×	308.9	7-Zip -mx9 (LZMA2)	4.35×	62.19
T4 M<=20	HX Complete	4.72×	24.58	no rival measured in tier		
T5 M<=50	HX Complete^	4.72×	24.58	BSC	4.55×	15.52
T6 M<=100	HX Complete^	4.72×	24.58	mzip	4.92×	8.56
T7 M<=200	HX Complete^	4.72×	24.58	7-Zip PPMd (-mx9 -m0=P	4.44×	4.26
T8 M<=500	HX Complete^	4.72×	24.58	kanzi -l 9	5.20×	2.19
T9 M<=2,000	HX Complete^	4.72×	24.58	no rival measured in tier		
T10 M>2,000	HX Complete^	4.72×	24.58	no rival measured in tier		

Calgary corpus, 18 files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	3.56×	40.64	Brotli 11	3.79×	41.16
T2 M<=5	HX Complete	4.08×	10.32	BSC	4.04×	9.65
T3 M<=10	HX Complete^	4.08×	10.32	bzip3	4.05×	7.42
T4 M<=20	HX Complete^	4.08×	10.32	no rival measured in tier		
T5 M<=50	HX Complete^	4.08×	10.32	no rival measured in tier		
T6 M<=100	HX Complete^	4.08×	10.32	kanzi -l 9	4.44×	0.433
T7 M<=200	HX Complete^	4.08×	10.32	no rival measured in tier		
T8 M<=500	HX Complete^	4.08×	10.32	no rival measured in tier		
T9 M<=2,000	HX Complete^	4.08×	10.32	no rival measured in tier		

T10 M>2,000	HX Complete^	4.08×	10.32	no rival measured in tier			

Canterbury corpus, 11 files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	5.44×	57.36	Brotli 11	5.73×	55.11
T2 M<=5	HX Complete	6.33×	16.06	BSC	6.14×	12.78
T3 M<=10	HX Complete^	6.33×	16.06	bzip3	6.12×	9.07
T4 M<=20	HX Complete^	6.33×	16.06	no rival measured in tier		
T5 M<=50	HX Complete^	6.33×	16.06	no rival measured in tier		
T6 M<=100	HX Complete^	6.33×	16.06	kanzi -l 9	6.81×	0.556
T7 M<=200	HX Complete^	6.33×	16.06	no rival measured in tier		
T8 M<=500	HX Complete^	6.33×	16.06	no rival measured in tier		
T9 M<=2,000	HX Complete^	6.33×	16.06	no rival measured in tier		
T10 M>2,000	HX Complete^	6.33×	16.06	no rival measured in tier		

Canterbury large corpus, 3 files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	4.38×	328.2	Brotli 11	4.46×	146.8
T2 M<=5	HX Fast^	4.38×	328.2	7-Zip -mx9 (LZMA2)	4.36×	61.32
T3 M<=10	HX Fast^	4.38×	328.2	no rival measured in tier		
T4 M<=20	HX Fast^	4.38×	328.2	BSC	4.94×	17.01
T5 M<=50	HX Complete	4.96×	7.95	bzip3	4.95×	6.95
T6 M<=100	HX Complete^	4.96×	7.95	7-Zip PPMd (-mx9 -m0=P	4.81×	4.57
T7 M<=200	HX Complete^	4.96×	7.95	kanzi -l 9	5.16×	2.17
T8 M<=500	HX Complete^	4.96×	7.95	no rival measured in tier		
T9 M<=2,000	HX Complete^	4.96×	7.95	no rival measured in tier		
T10 M>2,000	HX Complete^	4.96×	7.95	no rival measured in tier		

Synthetic test files, 5 files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.64×	330.2	LZ4 (default)	1.11×	539.0
T2 M<=5	HX Fast^	1.64×	330.2	Brotli 11	1.58×	128.6
T3 M<=10	HX Fast^	1.64×	330.2	gzip -9	1.22×	92.17
T4 M<=20	HX Fast^	1.64×	330.2	7-Zip -mx9 (LZMA2)	1.65×	44.66
T5 M<=50	HX Complete	1.66×	22.21	no rival measured in tier		
T6 M<=100	HX Complete^	1.66×	22.21	BSC	1.52×	7.15
T7 M<=200	HX Complete^	1.66×	22.21	no rival measured in tier		
T8 M<=500	HX Complete^	1.66×	22.21	kanzi -l 9	1.61×	1.41
T9 M<=2,000	HX Complete^	1.66×	22.21	no rival measured in tier		
T10 M>2,000	HX Complete^	1.66×	22.21	no rival measured in tier		

Speech, WAVs, 8 kHz mono (FSDD)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	none			zstd -19	1.43×	196.2
T2 M<=5	HX Fast + audio/colour	1.91×	101.9	WavPack -hhx6 (officia	1.98×	63.24
T3 M<=10	HX Complete + audio/col	1.92×	36.39	WavPack -hh (official)	1.94×	50.84
T4 M<=20	HX Complete + audio/col^	1.92×	36.39	OptimFROG --preset 2	2.04×	19.88
T5 M<=50	HX Complete + audio/col^	1.92×	36.39	bzip3	1.75×	6.61
T6 M<=100	HX Complete + audio/col^	1.92×	36.39	no rival measured in tier		
T7 M<=200	HX Complete + audio/col^	1.92×	36.39	OptimFROG --preset max	2.07×	2.61
T8 M<=500	HX Complete + audio/col^	1.92×	36.39	no rival measured in tier		
T9 M<=2,000	HX Complete + audio/col^	1.92×	36.39	no rival measured in tier		
T10 M>2,000	HX Complete + audio/col^	1.92×	36.39	no rival measured in tier		

Music and sounds, WAVs, 44.1 kHz stereo, decoded from Ogg Vorbis (librosa data)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	none			zstd -19	1.21×	264.8
T2 M<=5	HX Complete	2.22×	106.5	flac -8 -e -p (max)	2.61×	128.0
T3 M<=10	HX Complete + audio/col	2.59×	72.46	WavPack -hhx6 (officia	2.62×	63.17
T4 M<=20	HX Complete + audio/col^	2.59×	72.46	Monkey's Audio -c4000	2.85×	42.28
T5 M<=50	HX Complete + audio/col^	2.59×	72.46	Monkey's Audio -c5000	2.85×	18.93
T6 M<=100	HX Complete + audio/col^	2.59×	72.46	bzip3	1.34×	5.05
T7 M<=200	HX Complete + audio/col^	2.59×	72.46	no rival measured in tier		
T8 M<=500	HX Complete + audio/col^	2.59×	72.46	7-Zip PPMd (-mx9 -m0=P	1.60×	2.23
T9 M<=2,000	HX Complete + audio/col^	2.59×	72.46	no rival measured in tier		
T10 M>2,000	HX Complete + audio/col^	2.59×	72.46	no rival measured in tier		

Environmental recordings, WAVs, 44.1 kHz mono (ESC-50)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	2.14×	53.33	flac -8 -e -p (max)	2.22×	78.20
T2 M<=5	HX Fast + audio/colour	2.35×	38.52	WavPack -hhx6 (officia	2.34×	43.07
T3 M<=10	HX Fast + audio/colour ^	2.35×	38.52	7-Zip -mx9 (LZMA2)	1.80×	17.24
T4 M<=20	HX Fast + audio/colour ^	2.35×	38.52	TTA (ffmpeg)	2.22×	7.47
T5 M<=50	HX Fast + audio/colour ^	2.35×	38.52	7-Zip PPMd (-mx9 -m0=P	1.80×	4.52

T6 M<=100	HX Fast + audio/colour ^	2.35×	38.52	no rival measured in tier			
T7 M<=200	HX Fast + audio/colour ^	2.35×	38.52	kanzi -l 9	1.76×	0.664	
T8 M<=500	HX Fast + audio/colour ^	2.35×	38.52	no rival measured in tier			
T9 M<=2,000	HX Fast + audio/colour ^	2.35×	38.52	no rival measured in tier			
T10 M>2,000	HX Fast + audio/colour ^	2.35×	38.52	no rival measured in tier			

Already-compressed audio files: MP3, AAC, Opus, Ogg, FLAC

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.01×	265.3	Brotli 11	1.01×	105.1
T2 M<=5	HX Fast^	1.01×	265.3	gzip -9	1.01×	95.86
T3 M<=10	HX Fast^	1.01×	265.3	no rival measured in tier		
T4 M<=20	HX Fast^	1.01×	265.3	precomp (lzma)	1.03×	13.50
T5 M<=50	HX Complete	1.01×	7.93	bzip3	1.01×	5.29
T6 M<=100	HX Complete^	1.01×	7.93	7-Zip PPMd (-mx9 -m0=P	0.99×	2.68
T7 M<=200	HX Complete^	1.01×	7.93	no rival measured in tier		
T8 M<=500	HX Complete^	1.01×	7.93	no rival measured in tier		
T9 M<=2,000	HX Complete^	1.01×	7.93	no rival measured in tier		
T10 M>2,000	HX Complete^	1.01×	7.93	no rival measured in tier		

Colour photos, raw RGB (Kodak 24, PPM)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	none			zstd -19	1.86×	153.0
T2 M<=5	HX Fast	1.95×	83.27	WebP lossless (z9)	2.53×	78.64
T3 M<=10	HX Fast^	1.95×	83.27	7-Zip -mx9 (LZMA2)	1.96×	26.14
T4 M<=20	HX Complete + specialis	2.83×	13.64	JPEG XL lossless (e9)	2.83×	13.82
T5 M<=50	HX Complete + specialis	2.83×	13.64	bzip3	2.32×	7.66
T6 M<=100	HX Complete + specialis^	2.83×	13.64	no rival measured in tier		
T7 M<=200	HX Complete + specialis^	2.83×	13.64	no rival measured in tier		
T8 M<=500	HX Complete + specialis^	2.83×	13.64	no rival measured in tier		
T9 M<=2,000	HX Complete + specialis^	2.83×	13.64	no rival measured in tier		
T10 M>2,000	HX Complete + specialis^	2.83×	13.64	no rival measured in tier		

Rendered document pages / screenshots, raw (PGM/PPM)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	30.48×	459.6	xz -9e	33.13×	171.0
T2 M<=5	HX Complete + specialis	34.83×	68.51	BSC	34.46×	75.04
T3 M<=10	HX Complete + specialis	34.83×	68.51	bzip3	33.35×	46.74
T4 M<=20	HX Complete + specialis^	34.83×	68.51	7-Zip PPMd (-mx9 -m0=P	36.22×	29.81
T5 M<=50	HX Complete + specialis^	34.83×	68.51	no rival measured in tier		
T6 M<=100	HX Complete + specialis^	34.83×	68.51	kanzi -l 9	35.33×	3.99
T7 M<=200	HX Complete + specialis^	34.83×	68.51	no rival measured in tier		
T8 M<=500	HX Complete + specialis^	34.83×	68.51	no rival measured in tier		
T9 M<=2,000	HX Complete + specialis^	34.83×	68.51	no rival measured in tier		
T10 M>2,000	HX Complete + specialis^	34.83×	68.51	no rival measured in tier		

Medical DICOM files (imageio test images)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	3.15×	70.93	Brotli 11	2.56×	67.09
T2 M<=5	HX Fast^	3.15×	70.93	7-Zip -mx9 (LZMA2)	2.64×	24.10
T3 M<=10	HX Fast^	3.15×	70.93	no rival measured in tier		
T4 M<=20	HX Fast^	3.15×	70.93	bzip3	3.10×	7.55
T5 M<=50	HX Fast^	3.15×	70.93	no rival measured in tier		
T6 M<=100	HX Fast^	3.15×	70.93	no rival measured in tier		
T7 M<=200	HX Fast^	3.15×	70.93	no rival measured in tier		
T8 M<=500	HX Fast^	3.15×	70.93	no rival measured in tier		
T9 M<=2,000	HX Fast^	3.15×	70.93	no rival measured in tier		
T10 M>2,000	HX Fast^	3.15×	70.93	no rival measured in tier		

PNG files (Kodak 24)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.00×	149.5	gzip -9	1.00×	81.45
T2 M<=5	HX Fast^	1.00×	149.5	7-Zip -mx9 (LZMA2)	1.00×	33.18
T3 M<=10	HX Fast^	1.00×	149.5	no rival measured in tier		
T4 M<=20	HX Complete	1.01×	9.25	BSC	1.01×	6.80
T5 M<=50	HX Complete^	1.01×	9.25	bzip3	1.01×	6.05
T6 M<=100	H-Deflate + HX Complete	1.54×	1.59	precomp (lzma)	1.08×	1.95
T7 M<=200	H-Deflate + HX Complete^	1.54×	1.59	no rival measured in tier		
T8 M<=500	H-Deflate + HX Complete^	1.54×	1.59	no rival measured in tier		
T9 M<=2,000	H-Deflate + HX Complete^	1.54×	1.59	no rival measured in tier		
T10 M>2,000	H-Deflate + HX Complete^	1.54×	1.59	no rival measured in tier		

JPEG photos

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.01×	152.9	zstd -19	1.01×	231.0

T2 M<=5	HX Fast^	1.01×	152.9	gzip -9	1.01×	95.41
T3 M<=10	HX Fast^	1.01×	152.9	7-Zip -mx9 (LZMA2)	1.01×	31.15
T4 M<=20	HX Fast^	1.01×	152.9	no rival measured in tier		
T5 M<=50	HX Complete	1.05×	5.24	Lepton	1.28×	7.53
T6 M<=100	HX Complete^	1.05×	5.24	bzip3	1.05×	4.91
T7 M<=200	HX Complete + specialis	1.28×	2.27	packJPG	1.28×	2.31
T8 M<=500	HX Complete + specialis^	1.28×	2.27	no rival measured in tier		
T9 M<=2,000	HX Complete + specialis^	1.28×	2.27	no rival measured in tier		
T10 M>2,000	HX Complete + specialis^	1.28×	2.27	no rival measured in tier		

GIF and TIFF files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.24×	75.46	Brotli 11	1.29×	60.06
T2 M<=5	HX Fast^	1.24×	75.46	no rival measured in tier		
T3 M<=10	HX Fast^	1.24×	75.46	precomp (lzma)	1.39×	14.43
T4 M<=20	HX Complete	1.30×	6.07	bzip3	1.30×	5.67
T5 M<=50	HX Complete^	1.30×	6.07	BSC	1.27×	4.80
T6 M<=100	HX Complete^	1.30×	6.07	no rival measured in tier		
T7 M<=200	HX Complete^	1.30×	6.07	no rival measured in tier		
T8 M<=500	HX Complete^	1.30×	6.07	no rival measured in tier		
T9 M<=2,000	HX Complete^	1.30×	6.07	no rival measured in tier		
T10 M>2,000	HX Complete^	1.30×	6.07	no rival measured in tier		

Raw video (Y4M; frames decoded from H.264 sources + one raw test clip)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	none			zstd -19	6.83×	390.9
T2 M<=5	none			Brotli 11	7.37×	245.9
T3 M<=10	HX Fast	10.47×	128.2	x264 lossless (veryslo	16.86×	110.7
T4 M<=20	HX Complete + specialis	11.58×	66.39	no rival measured in tier		
T5 M<=50	HX Complete + specialis^	11.58×	66.39	BSC	8.03×	25.44
T6 M<=100	HX Complete + specialis^	11.58×	66.39	bzip3	8.11×	13.47
T7 M<=200	HX Complete + specialis^	11.58×	66.39	no rival measured in tier		
T8 M<=500	HX Complete + specialis^	11.58×	66.39	no rival measured in tier		
T9 M<=2,000	HX Complete + specialis^	11.58×	66.39	no rival measured in tier		
T10 M>2,000	HX Complete + specialis^	11.58×	66.39	no rival measured in tier		

Compressed video files (H.264, HEVC, VP9)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.04×	122.3	Brotli 11	1.04×	55.60
T2 M<=5	HX Fast^	1.04×	122.3	xz -9e	1.03×	17.99
T3 M<=10	HX Fast^	1.04×	122.3	no rival measured in tier		
T4 M<=20	HX Fast^	1.04×	122.3	BSC	1.04×	5.05
T5 M<=50	HX Fast^	1.04×	122.3	7-Zip PPMd (-mx9 -m0=P	1.01×	2.74
T6 M<=100	HX Fast^	1.04×	122.3	no rival measured in tier		
T7 M<=200	HX Fast^	1.04×	122.3	kanzi -l 9	1.04×	0.600
T8 M<=500	HX Fast^	1.04×	122.3	no rival measured in tier		
T9 M<=2,000	HX Fast^	1.04×	122.3	no rival measured in tier		
T10 M>2,000	HX Fast^	1.04×	122.3	no rival measured in tier		

PDF files from the pdf.js test set

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.33×	177.2	Brotli 11	1.34×	74.78
T2 M<=5	HX Fast^	1.33×	177.2	7-Zip -mx9 (LZMA2)	1.34×	27.05
T3 M<=10	HX Complete	1.35×	20.43	no rival measured in tier		
T4 M<=20	HX Complete^	1.35×	20.43	precomp (lzma)	1.66×	8.28
T5 M<=50	H-Deflate + HX Complete	1.69×	6.29	bzip3	1.34×	6.05
T6 M<=100	H-Deflate + HX Complete^	1.69×	6.29	no rival measured in tier		
T7 M<=200	H-Deflate + HX Complete^	1.69×	6.29	no rival measured in tier		
T8 M<=500	H-Deflate + HX Complete^	1.69×	6.29	no rival measured in tier		
T9 M<=2,000	H-Deflate + HX Complete^	1.69×	6.29	no rival measured in tier		
T10 M>2,000	H-Deflate + HX Complete^	1.69×	6.29	no rival measured in tier		

Packet captures, public pcapng files

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.37×	103.5	Brotli 11	1.40×	57.96
T2 M<=5	HX Fast^	1.37×	103.5	no rival measured in tier		
T3 M<=10	HX Complete	1.38×	9.57	xz -9e	1.39×	15.72
T4 M<=20	HX Complete^	1.38×	9.57	bzip3	1.36×	4.70
T5 M<=50	HX Complete^	1.38×	9.57	7-Zip PPMd (-mx9 -m0=P	1.34×	2.51
T6 M<=100	HX Complete^	1.38×	9.57	no rival measured in tier		
T7 M<=200	HX Complete^	1.38×	9.57	kanzi -l 9	1.42×	0.525
T8 M<=500	HX Complete^	1.38×	9.57	no rival measured in tier		
T9 M<=2,000	HX Complete^	1.38×	9.57	no rival measured in tier		
T10 M>2,000	HX Complete^	1.38×	9.57	no rival measured in tier		

Packet capture, one 113 MB merged pcap

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Complete	1.38×	464.7	LZ4 (default)	1.15×	855.4
T2 M<=5	HX Fast	1.38×	418.2	zstd -19	1.24×	358.5
T3 M<=10	HX Fast^	1.38×	418.2	Brotli 11	1.26×	98.88
T4 M<=20	HX Fast^	1.38×	418.2	no rival measured in tier		
T5 M<=50	HX Fast^	1.38×	418.2	7-Zip -mx9 (LZMA2)	1.39×	19.40
T6 M<=100	HX Fast^	1.38×	418.2	no rival measured in tier		
T7 M<=200	HX Fast^	1.38×	418.2	BSC	1.30×	5.41
T8 M<=500	HX Fast^	1.38×	418.2	no rival measured in tier		
T9 M<=2,000	HX Fast^	1.38×	418.2	7-Zip PPMd (-mx9 -m0=P	1.22×	1.67
T10 M>2,000	HX Fast^	1.38×	418.2	no rival measured in tier		

Office files: DOCX/XLSX/PPTX/DOC/XLS/PPT (Apache POI test data)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.52×	56.29	Brotli 11	1.53×	32.33
T2 M<=5	HX Complete	1.53×	14.32	7-Zip -mx9 (LZMA2)	1.53×	14.46
T3 M<=10	H-Deflate + HX Complete	2.06×	4.50	precomp (lzma)	1.92×	6.42
T4 M<=20	H-Deflate + HX Complete^	2.06×	4.50	7-Zip PPMd (-mx9 -m0=P	1.45×	3.17
T5 M<=50	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		
T6 M<=100	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		
T7 M<=200	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		
T8 M<=500	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		
T9 M<=2,000	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		
T10 M>2,000	H-Deflate + HX Complete^	2.06×	4.50	no rival measured in tier		

SQLite database (Chinook)

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	3.37×	201.5	Brotli 11	4.35×	83.97
T2 M<=5	HX Fast^	3.37×	201.5	7-Zip -mx9 (LZMA2)	4.64×	40.30
T3 M<=10	HX Fast^	3.37×	201.5	precomp (lzma)	4.54×	30.53
T4 M<=20	HX Fast^	3.37×	201.5	BSC	3.48×	11.72
T5 M<=50	HX Complete	3.68×	8.26	bzip3	3.68×	8.33
T6 M<=100	HX Complete^	3.68×	8.26	no rival measured in tier		
T7 M<=200	HX Complete^	3.68×	8.26	kanzi -l 9	5.28×	1.04
T8 M<=500	HX Complete^	3.68×	8.26	no rival measured in tier		
T9 M<=2,000	HX Complete^	3.68×	8.26	no rival measured in tier		
T10 M>2,000	HX Complete^	3.68×	8.26	no rival measured in tier		

Archives: ZIP, TAR, TAR.GZ

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast	1.28×	372.1	Brotli 11	1.28×	216.1
T2 M<=5	HX Fast^	1.28×	372.1	xz -9e	1.28×	100.8
T3 M<=10	HX Fast^	1.28×	372.1	no rival measured in tier		
T4 M<=20	HX Fast^	1.28×	372.1	no rival measured in tier		
T5 M<=50	HX Complete	1.28×	10.41	BSC	1.28×	9.15
T6 M<=100	HX Complete^	1.28×	10.41	bzip3	1.28×	4.89
T7 M<=200	H-Deflate + HX Complete	1.64×	2.36	precomp (lzma)	1.52×	2.62
T8 M<=500	H-Deflate + HX Complete^	1.64×	2.36	no rival measured in tier		
T9 M<=2,000	H-Deflate + HX Complete^	1.64×	2.36	no rival measured in tier		
T10 M>2,000	H-Deflate + HX Complete^	1.64×	2.36	no rival measured in tier		

Scientific: climate NetCDF/GRIB, radar, ice velocity, 3D models (OBJ), genomics (SAM, FASTA, E.c

tier	HX entry	ratio	MB/s	best other in tier	ratio	MB/s
T1 M<=2	HX Fast + audio/colour	2.56×	250.1	zstd -19	2.49×	353.0
T2 M<=5	HX Fast + audio/colour ^	2.56×	250.1	Brotli 11	2.61×	135.6
T3 M<=10	HX Fast + audio/colour ^	2.56×	250.1	no rival measured in tier		
T4 M<=20	HX Fast + audio/colour ^	2.56×	250.1	7-Zip -mx9 (LZMA2)	2.64×	37.95
T5 M<=50	HX Complete + audio/col	2.72×	11.11	BSC	2.65×	10.03
T6 M<=100	HX Complete + audio/col^	2.72×	11.11	bzip3	2.68×	8.44
T7 M<=200	HX Complete + audio/col^	2.72×	11.11	7-Zip PPMd (-mx9 -m0=P	2.58×	3.69
T8 M<=500	HX Complete + audio/col^	2.72×	11.11	no rival measured in tier		
T9 M<=2,000	HX Complete + audio/col^	2.72×	11.11	no rival measured in tier		
T10 M>2,000	HX Complete + audio/col^	2.72×	11.11	no rival measured in tier		

10. GAPS - results with ratio but no speed, or speed but no ratio

Ratio recorded, decode speed not recorded:
7-Zip -mx9 [Silesia mr]; 7-Zip -mx9 [Silesia x-ray]; 7-Zip Ultra -mx9 [Canterbury]; 7z-hi (Delta 2) [Silesia mr]; 7z-hi (Delta 2) [Silesia x-ray]; 7z-hi [Canterbury]; Brotli 11 [Canterbury]; Brotli 5 [Canterbury]; Brotli 8 [Canterbury]; Brotli-hi 11 [Canterbury]; Brotli-hi 5 [Canterbury]; Brotli-hi 8 [Canterbury]; HCM v2 [Canterbury]; HCM v3 fast

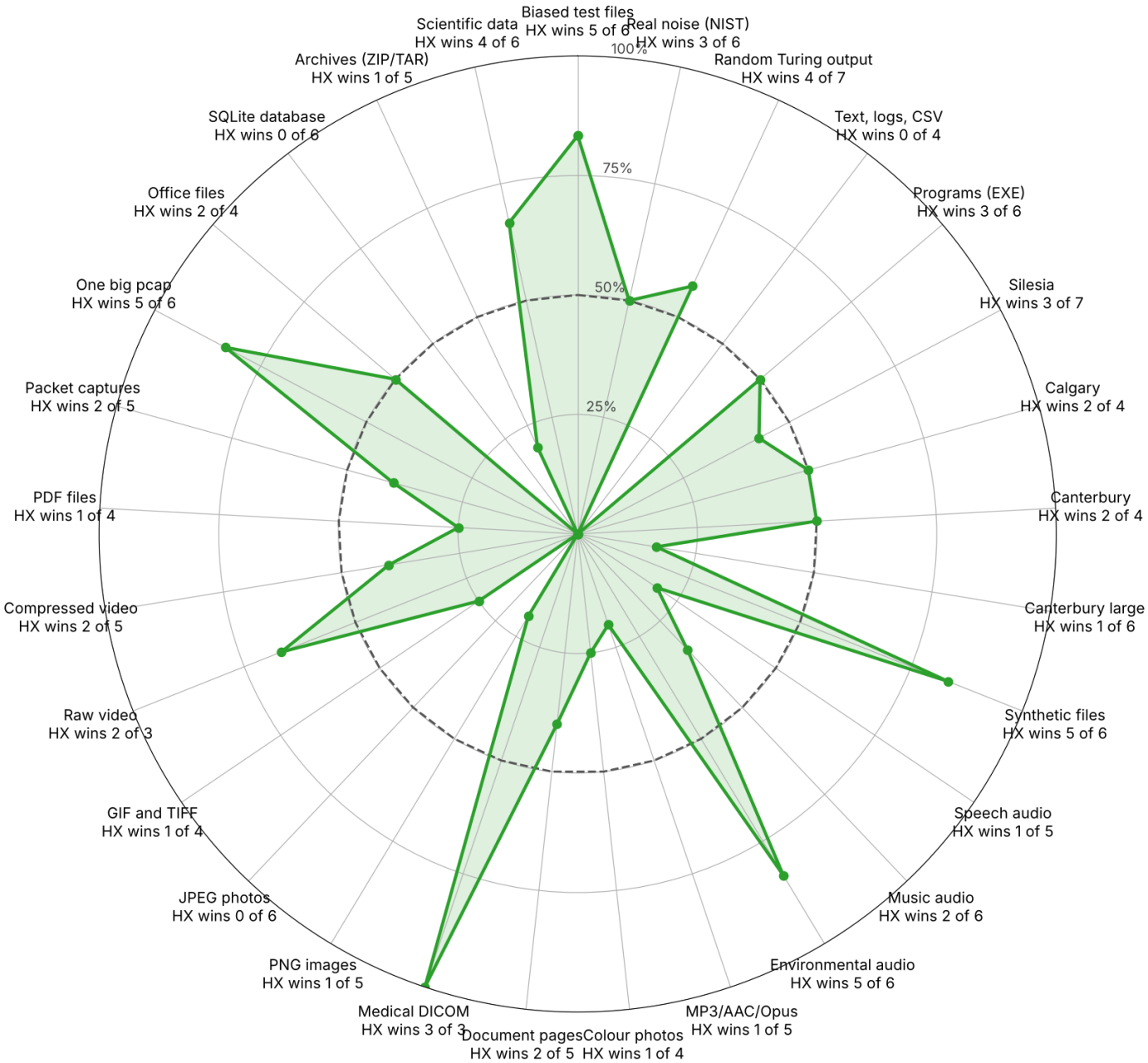
[Canterbury]; HCM v3 fast [kennedy.xls]; HCM v3 max [Canterbury]; HCM v3.1 fast [Canterbury]; HCM v3.1 max [Canterbury]; HCM v3.2 fast [Canterbury]; HCM v3.2 max [Canterbury]; HCM v3.3 fast + cmix English dictionary [Canterbury]; HCM v3.3 fast + x86 E8/E9 filter [Silesia]; HCM v3.3 fast [Canterbury]; HCM v3.3 fast [kennedy.xls]; HCM v3.3 max + x86 E8/E9 filter [Silesia]; HCM v3.3 max [Canterbury]; HCM v3.7 max, reorder kept only if smaller [Silesia]; HCM-Bin v1 (overall) [Canterbury]; HCM-Bin v1 [kennedy.xls]; HCM-Bin v1 max (hcmx) [kennedy.xls]; HX Complete, every route (router build) [Silesia mr]; HX Complete, every route (router build) [Silesia x-ray]; ZPAQ -m5 [Canterbury]; bzip2 -9 [kennedy.xls]; bzip2 rebuild + candidate escape flag [Canterbury]; bzip2-hi (100k blocks) [kennedy.xls]; bzip2-style rebuild + arithmetic coding [Canterbury]; cmix v20 (published) [Silesia mr]; cmix v20 (published) [Silesia x-ray]; gzip -1 + record delta [Silesia]; gzip -1 [Silesia]; gzip -9 + change map [kennedy.xls]; gzip -9 + record delta [Silesia]; gzip -9 [Silesia]; hcm-trit v1 [Canterbury]; hcm-trit v1.1 [Canterbury]; hcm-trit v2 (pointer) 6-order [Canterbury]; hcm-trit v2 max + cmix dictionary [Canterbury]; hcm-trit v2 max [Canterbury]; hcmx fast best-of [Canterbury]; hcmx max best-of [Canterbury]; hcmx-s fast (x86 inside format) [Silesia]; hcmx-s trit2 lite [Silesia]; lpaq1 -6 [kennedy.xls]; lpaq1-h [kennedy.xls]; paq8px lite model set (lab) [Canterbury]; paq8px v217 -8 (full) [Canterbury]; xz -0 + record delta [Silesia]; xz -0 [Silesia]; xz -9e + record delta [Silesia]; xz -9e [Silesia]; xz -9e [kennedy.xls]; xz-hi (delta 13) [kennedy.xls]; zpaq-hi (one file per archive) [Canterbury]; zstd -1 + record delta [Silesia]; zstd -1 [Canterbury]; zstd -1 [Silesia]; zstd -19 + record delta [Silesia]; zstd -19 [Canterbury]; zstd -19 [Silesia]; zstd -22 [Canterbury]; zstd -3 [Canterbury]; zstd -9 [Canterbury]; zstd-hi -1 [Canterbury]; zstd-hi -19 [Canterbury]; zstd-hi -22 [Canterbury]; zstd-hi -3 [Canterbury]; zstd-hi -9 [Canterbury]

Speed recorded, ratio not recorded:
none

Compression time is not recorded for: the 27 Oct Tier-4 race, the enwik8 race, the 26-27 Oct record-delta and king-race totals, all Canterbury rows recorded only as ratios, zstd-hi / Brotli-hi, jxl-hi, the pzstd race summary and the Steim2 archive files. The per-slice build has no results yet. hx15 live run covers 10 of 30 listed unseen files.

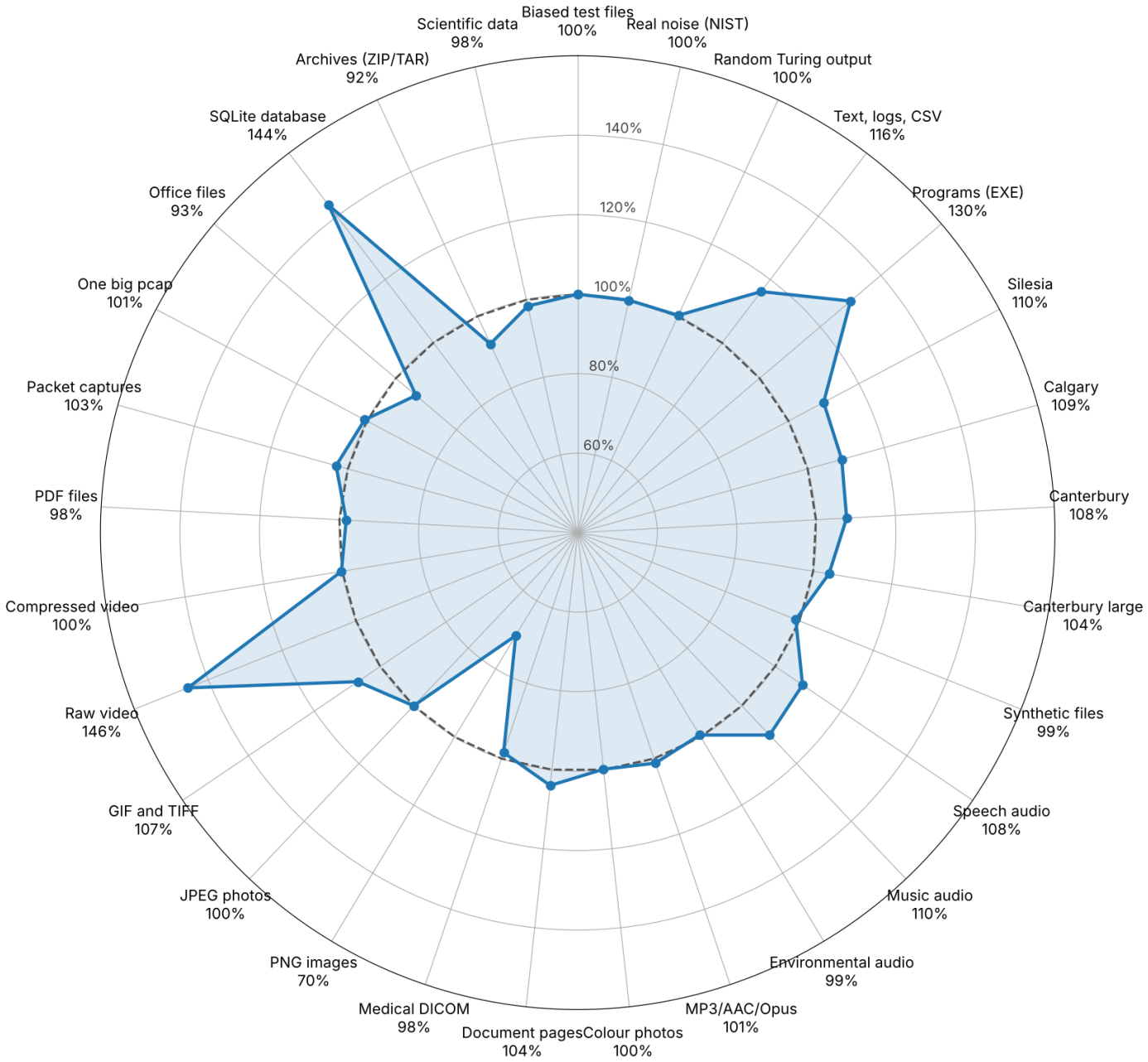
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END
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HX vs the best other tool, every data type
% of speed tiers where HX is smaller



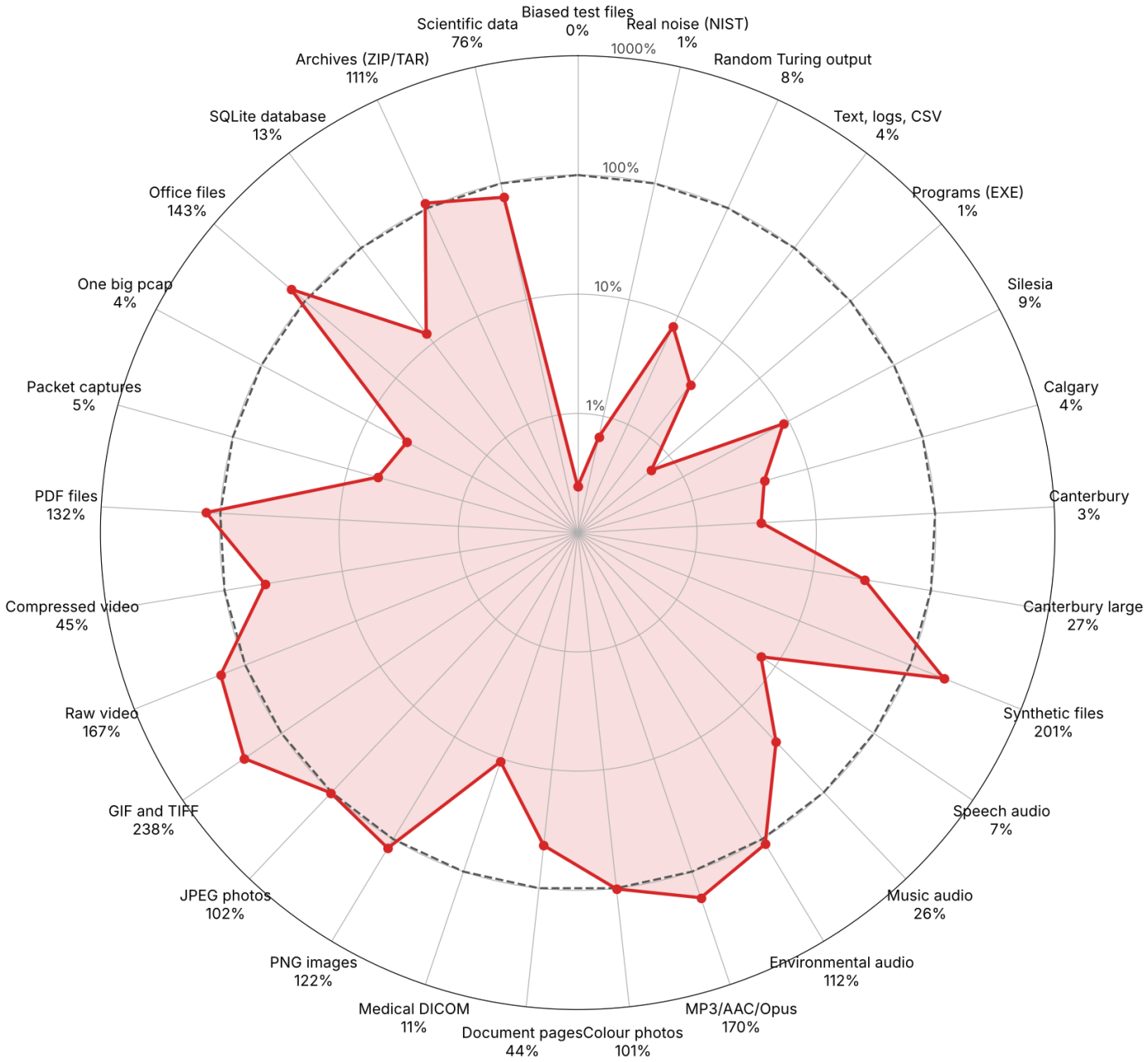
29 data types x 10 speed tiers. HX smaller in 64, tied in 36, of the 149 type-and-tier cells where both were measured.
Outside the dashed circle = HX wins more than half of that type's tiers. Patent Pending (U.S. App. 64/192,843)

Compressed size: HX's best vs the best other tool
(HX size as % of theirs, any speed)



Inside the dashed circle = HX is smaller. HX smaller on 8, tied on 6, larger on 15 of 29 types. Patent Pending (U.S. App. 64/192,843)

Decode time: HX vs that same best other tool
(HX time as % of theirs, log scale)



Inside the dashed circle = HX decodes faster. HX faster on 18 of 29 types. Patent Pending (U.S. App. 64/192,843)